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Detection assays

Abstract

Presented is a nucleic acid-guided nuclease cascade assay that can detect one or more target nucleic acids of interest without the need for amplifying the target nucleic acids of interest. The nucleic acid-guided nuclease cascade assays utilize signal amplification mechanisms comprising various components including nucleic acid-guided nucleases and guide RNAs (gRNAs) forming ribonucleoprotein complexes, blocked primer molecules, template molecules, polymerases and reporter moieties.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. Ser. No. 18/372,098, filed 24 Sep. 2023; which is a continuation of U.S. Ser. No. 18/208,272, filed 10 Jun. 2023; which is a continuation of U.S. Ser. No. 18/204,337, filed 31 May 2023, now U.S. Pat. No. 11,821,025; which is a continuation of U.S. Ser. No. 18/106,420, filed 6 Feb. 2023, now U.S. Pat. No. 11,702,686, which is a continuation of U.S. Ser. No. 17/861,208, filed 9 Jul. 2022, now U.S. Pat. No. 11,639,520, which claims priority to U.S. Ser. No. 63/220,987, filed 12 Jul. 2021, and U.S. Ser. No. 63/289,112, filed 13 Dec. 2021.

FIELD OF THE INVENTION

(1) The present disclosure relates to methods, compositions of matter and assay systems used to

detect one or more target nucleic acids of interest in a sample. The assay systems provide signal amplification upon detection of a target nucleic acids of interest without amplification of the target nucleic acids.

INCORPORATION BY REFERENCE

(2) Submitted on 4 Aug. 2022 is an electronically filed sequence listing via EFS-Web a Sequence Listing XML, entitled “VB002US2C5_SeqList_2_21_2024”, created 4 Aug. 2022, which is 6,799,000 bytes in size. The sequence listing is part of this specification and is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

(3) In the following discussion certain articles and methods will be described for background and introductory purposes. Nothing contained herein is to be construed as an “admission” of prior art. Applicant expressly reserves the right to demonstrate, where appropriate, that the articles and methods referenced herein do not constitute prior art under the applicable statutory provisions.

(4) Rapid and accurate identification of infectious agents is important in order to select correct treatment and prevent further spreading of viral infections and pandemic diseases. For example, viral pathogens, such as SARS-CoV-2, and the associated COVID-19 disease require immediate detection and response to decrease mortality, morbidity and transmission.

(5) Classic nucleic acid-guided nuclease or CRISPR (clustered regularly interspaced short palindromic repeats) detection methods usually rely on pre-amplification of target nucleic acids of interest to enhance detection sensitivity. However, amplification increases time to detection and may cause changes to the relative proportion of nucleic acids in samples that, in turn, lead to artifacts or inaccurate results. Improved technologies that allow very rapid and accurate detection of pathogens are therefore needed for timely diagnosis, prevention and treatment of disease, as well as in other applications.

SUMMARY OF THE INVENTION

(6) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other features, details, utilities, and advantages of the claimed subject matter will be apparent from the following written Detailed Description including those aspects illustrated in the accompanying drawings and defined in the appended claims. Further, all of the functionalities described in connection with one embodiment of the compositions and methods described herein are intended to be applicable to the additional embodiments of the compositions and methods described herein except where expressly stated or where the feature or function is incompatible with the additional embodiments. For example, where a given feature or function is expressly described in connection with one embodiment but not expressly mentioned in connection with an alternative embodiment, it should be understood that the feature or function may be deployed, utilized, or implemented in connection with the alternative embodiment unless the feature or function is incompatible with the alternative embodiment.

(7) The present disclosure provides compositions of matter, methods, and cascade assays to detect target nucleic acids of interest. The cascade assays described herein comprise two different ribonucleoprotein complexes and either blocked nucleic acid molecules or blocked primer molecules. The blocked nucleic acid molecules or blocked primer molecules keep one of the ribonucleoprotein complexes “locked” unless and until a target nucleic acid of interest activates the other ribonucleoprotein complex. The present nucleic acid-guided nuclease cascade assay can detect one or more target nucleic acids of interest (e.g., DNA, RNA and/or cDNA) at attomolar (aM) (or lower) limits in about 10 minutes or less without the need for amplifying the target nucleic acid(s) of interest, thereby avoiding the drawbacks of multiplex amplification, such as primer-dimerization. A particularly advantageous feature of the cascade assay is that, with the exception of the gRNA in RNP1, the cascade assay components stay the same no matter what target nucleic

acid(s) of interest are being detected. In this sense, the cascade assay is modular.

(8) There is provided herein in one embodiment of the disclosure a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; and (iii) a plurality of blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecules cannot activate the RNP1 or the RNP2.

(9) There is provided in a second embodiment of the disclosure, a reaction mixture comprising: (i) a first complex comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; and (iii) a plurality of blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecule cannot activate the first or second complex.

(10) Provided in a third embodiment is a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) (RNP1) complex comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both sequence-specific activity and non-sequence-specific activity; (ii) a second ribonucleoprotein (RNP2) complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both sequence-specific activity and non-sequence-specific activity; and (iii) a plurality of blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecules do not bind to the RNP1 complex or the RNP2 complex. In yet another fourth embodiment of the disclosure there is provided a reaction mixture comprising: (i) a first complex comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both sequence-specific activity and non-sequence-specific activity; (ii) a second complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both sequence-specific activity and non-sequence-specific activity; and (iii) a plurality of blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecules are not recognized by the RNP1s or RNP2s.

(11) A fifth embodiment provides a cascade assay method for detecting a target nucleic acid of interest in a sample comprising the steps of: (a) providing a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; and (iii) a plurality of blocked nucleic acid

molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecules cannot activate the RNP1 or the RNP2; (b) contacting the reaction mixture with the sample under conditions that allow the target nucleic acid of interest in the sample to bind to RNP1; wherein upon binding of the target nucleic acid of interest RNP1 becomes active initiating trans-cleavage of at least one of the blocked nucleic acid molecules thereby producing at least one unblocked nucleic acid molecule and the at least one unblocked nucleic acid molecule binds to RNP2 initiating cleavage of at least one further blocked nucleic acid molecule; and (c) detecting products of the cleavage, thereby detecting the target nucleic acid of interest in the sample.

(12) In a sixth embodiment there is provided a kit for detecting a target nucleic acid of interest in a sample comprising: (i) a first ribonucleoprotein (RNP1) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first gRNA, wherein the first gRNA comprises a sequence complementary to the target nucleic acid of interest; and wherein binding of the RNP1 complex to the target nucleic acid of interest activates cis-cleavage and trans-cleavage activity of the first nucleic acid-guided nuclease; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; a (iii) plurality of blocked nucleic acid molecules comprising a sequence corresponding to the second gRNA, wherein trans-cleavage activity of the blocked nucleic acid molecules results in at least one unblocked nucleic acid molecule; and wherein the unblocked nucleic acid molecule activates trans-cleavage activity of the second nucleic acid-guided nuclease in at least one RNP2 initiating cleavage of more blocked nucleic acid molecules; and (iv) a reporter moiety, wherein the reporter molecule comprises a nucleic acid molecule and/or is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon trans-cleavage activity by the RNP1 or the RNP2, to identify the presence of the target nucleic acid of interest in the sample.

(13) In some aspects of any one of the aforementioned embodiments, the first and/or second nucleic acid-guided nuclease is a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the first nucleic acid-guided nuclease can be a different nucleic acid-guided nuclease than the second nucleic acid-guided nuclease; in some aspects, the first and/or second nucleic acid-guided nuclease is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease and/or in some aspects, the first and/or second nucleic acid-guided nuclease comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(14) In some aspects of any one of the aforementioned embodiments, the blocked nucleic acid molecules comprise a structure represented by any one of Formulas I-IV, wherein Formulas I-IV comprise in the 5'-to-3' direction:

(a) A-(B-L).sub.J-C-M-T-D (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L).sub.J-C and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length and comprises at least 50% sequence complementarity to B and C; and D is 0-10 nucleotides in length and comprises at least 50% sequence complementarity to A;

(b) D-T-T'-C-(L-B).sub.J-A (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length; T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D;

(c) T-D-M-A-(B-L).sub.J-C (Formula III); wherein T is 17-135 nucleotides in length; D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D

and A-(B-L).sub.J-C are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length; or

(d) T-D-M-A-L.sub.p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T. And in some aspects, (a) T of Formula I comprises at least 80% sequence complementarity to B and C; (b) D of Formula I comprises at least 80% sequence complementarity to A; (c) C of Formula II comprises at least 80% sequence complementarity to T; (d) B of Formula II comprises at least 80% sequence complementarity to T; (e) A of Formula II comprises at least 80% sequence complementarity to D; (f) A of Formula III comprises at least 80% sequence complementarity to D; (g) B of Formula III comprises at least 80% sequence complementarity to T; (h) A of Formula IV comprises at least 80% sequence complementarity to D; and/or (i) C of Formula IV comprises at least 80% sequence complementarity to T.

(15) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecules comprise a first sequence complementary to the second gRNA and a second sequence not complementary to the second gRNA, wherein the second sequence at least partially hybridizes to the first sequence resulting in at least one loop.

(16) In some aspects of the aforementioned embodiments, the reaction mixture comprises about 1 fM to about 10 μ M of the RNP1 and in some aspects the reaction mixture comprises about 1 fM to about 1 mM of the RNP2.

(17) In some aspects of the aforementioned embodiments, the reaction mixture comprises at least two different RNP1s, wherein different RNP1s comprise different gRNA sequences, and in some aspects the reaction mixture comprises 2 to 10000 different RNP1s, or 2 to 1000 different RNP1s, or 2 to 100 different RNP1s, or 2 to 10 different RNP1s.

(18) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecules include the sequence of any one of SEQ ID NOs: 14-1421.

(19) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecules are circular, and in some aspects the blocked nucleic acid molecules are linear.

(20) In some aspects the K.sub.d of the blocked nucleic acid molecules to the RNP2 is about 10.sup.5-fold greater, 10.sup.6-fold greater, 10.sup.7-fold greater, 10.sup.8-fold greater, 10.sup.9-fold greater, 10.sup.10-fold greater or more than the K.sub.d of unblocked nucleic acid molecules.

(21) In some aspects of the aforementioned embodiments, the target nucleic acid of interest is of bacterial, viral, fungal, mammalian or plant origin, and in some aspects, the sample may include peripheral blood, serum, plasma, ascites, urine, cerebrospinal fluid (CSF), sputum, saliva, bone marrow, synovial fluid, aqueous humor, amniotic fluid, cerumen, breast milk, bronchoalveolar lavage fluid, semen, prostatic fluid, cowper's fluid or pre-ejaculatory fluid, sweat, fecal matter, hair, tears, cyst fluid, pleural and peritoneal fluid, pericardial fluid, lymph, chyme, chyle, bile, interstitial fluid, menses, pus, sebum, vomit, vaginal secretions, mucosal secretion, stool water, pancreatic juice, lavage fluids from sinus cavities, bronchopulmonary aspirates, blastocyst cavity fluid, and umbilical cord blood; food; agricultural products; pharmaceuticals; cosmetics, nutraceuticals; personal care products; environmental substances such as soil, water, or air; industrial sites and products; or manufactured or natural chemicals and compounds.

(22) In some aspects of the aforementioned embodiments, the reaction mixture further comprises a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid

molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(23) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecules do not comprise a PAM sequence, yet in other aspects, the blocked nucleic acid molecules comprise a PAM sequence, and in some aspects the PAM sequence is disposed between the first and second sequences, wherein the first sequence is 5' to the PAM sequence.

(24) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecule is a blocked primer molecule.

(25) In a seventh embodiment of the disclosure, there is provided a blocked nucleic acid molecule comprising: a first region recognized by a ribonucleoprotein (RNP) complex; one or more second regions not complementary to the first region; and one or more third regions complementary and hybridized to the first region, wherein cleavage of the one or more second regions results in dehybridization of the third region from the first region, resulting in an unblocked nucleic acid molecule.

(26) An eighth embodiment provides a method of unblocking a blocked nucleic acid comprising: (a) providing a blocked nucleic acid molecule comprising: a first region recognized by a ribonucleoprotein (RNP) complex; one or more second regions not complementary to the first region; and one or more third regions complementary and hybridized to the first region, wherein cleavage of the one or more second regions results in dehybridization of the third region from the first region, resulting in an unblocked nucleic acid molecule; and (b) initiating cleavage of the one or more second regions, wherein the blocked nucleic acid molecule becomes an unblocked nucleic acid molecule.

(27) A ninth embodiment provides a composition of matter comprising: a first region recognized by a ribonucleoprotein (RNP) complex; one or more second regions of not complementary to the first region; and one or more third regions complementary and hybridized to the first region, wherein cleavage of the one or more second regions results in dehybridization of the third region from the first region, resulting in an unblocked nucleic acid molecule; and the RNP complex comprising a gRNA that is complementary to the first region and a nucleic acid-guided nuclease, wherein the nucleic acid-guided nuclease exhibits both sequence-specific and non-sequence-specific nuclease activity.

(28) Additionally, a tenth embodiment of the disclosure provides a cascade assay method of detecting a target nucleic acid of interest in a sample comprising the steps of: (a) providing a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; and (iii) a plurality of linear blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the linear blocked nucleic acid molecules cannot activate the RNP1 or the RNP2; (b) contacting the reaction mixture with the sample under conditions that allow the target nucleic acid of interest in the sample to bind to RNP1; wherein upon binding of the target nucleic acid of interest RNP1 becomes active initiating trans-cleavage of at least one of the linear blocked nucleic acid molecules thereby producing at least one linear unblocked nucleic acid

molecule and the at least one linear unblocked nucleic acid molecule to RNP2 initiating cleavage of at least one further linear blocked nucleic acid molecule; and (c) detecting the cleavage products, thereby detecting the target nucleic acid of interest in the sample.

(29) In some aspects of any one of the aforementioned embodiments, the first and/or second nucleic acid-guided nuclease is a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the first nucleic acid-guided nuclease can be a different nucleic acid-guided nuclease than the second nucleic acid-guided nuclease; in some aspects, the first and/or second nucleic acid-guided nuclease is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease and/or in some aspects, the first and/or second nucleic acid-guided nuclease comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(30) In some aspects, the blocked nucleic acid molecule comprises a structure represented by any one of Formulas I-IV, wherein Formulas I-IV are in the 5'-to-3' direction:

(a) A-(B-L).sub.J-C-M-T-D (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L).sub.J-C and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length and comprises at least 50% sequence complementarity to B and C; and D is 0-10 nucleotides in length and comprises at least 50% sequence complementarity to A;

(b) D-T-T'-C-(L-B).sub.J-A (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length; T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D;

(c) T-D-M-A-(B-L).sub.J-C (Formula III); wherein T is 17-135 nucleotides in length; D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D and A-(B-L).sub.J-C are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length; or

(d) T-D-M-A-L.sub.p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T.

Further (a) T of Formula I comprises at least 80% sequence complementarity to B and C; (b) D of Formula I comprises at least 80% sequence complementarity to A; (c) C of Formula II comprises at least 80% sequence complementarity to T; (d) B of Formula II comprises at least 80% sequence complementarity to T; (e) A of Formula II comprises at least 80% sequence complementarity to D; (f) A of Formula III comprises at least 80% sequence complementarity to D; (g) B of Formula III comprises at least 80% sequence complementarity to T; (h) A of Formula IV comprises at least 80% sequence complementarity to D; and/or (i) C of Formula IV comprises at least 80% sequence complementarity to T.

(31) In some aspects of the aforementioned embodiments, the blocked nucleic acid molecule comprises a modified nucleoside or nucleotide, including but not limited to a locked nucleic acid (LNA), peptide nucleic acid (PNA), 2'-O-methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bond. In some aspects, the blocked nucleic acid molecule includes the sequence of any one of SEQ ID NOs: 14-1421; the blocked nucleic acid molecule is a blocked primer molecule; the blocked nucleic acid molecule does not comprise a PAM sequence; and/or in some aspects the blocked nucleic acid molecule comprises a PAM

sequence, and the PAM sequence is disposed between the first and second sequences, wherein the first sequence is 5' to the PAM sequence.

(32) In some aspects of the aforementioned embodiments, the reaction mixture further comprises a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(33) In some aspects, the K.sub.d of the blocked nucleic acid molecules to the RNP2 is about 10.sup.5-fold greater, 10.sup.6-fold greater, 10.sup.7-fold greater, 10.sup.8-fold greater, 10.sup.9-fold greater, 10.sup.10-fold greater or more than the K.sub.d of unblocked nucleic acid molecules.

(34) In an eleventh embodiment, there is provided composition of matter comprising a ribonucleoprotein (RNP) complex and a blocked nucleic acid molecule, wherein the blocked nucleic acid molecule is represented by any one of Formula I-IV, wherein Formulas I-IV comprise in the 5'-to-3' direction comprises:

(a) A-(B-L).sub.J-C-M-T-D (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L).sub.J-C and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length and comprises at least 50% sequence complementarity to B and C; and D is 0-10 nucleotides in length and comprises at least 50% sequence complementarity to A;

(b) D-T-T'-C-(L-B).sub.J-A (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length; T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D;

(c) T-D-M-A-(B-L).sub.J-C (Formula III); wherein T is 17-135 nucleotides in length; D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D and A-(B-L).sub.J-C are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length; or

(d) T-D-M-A-L.sub.p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T.

(35) In some aspects of this embodiment,

(36) T of Formula I comprises at least 80% sequence complementarity to B and C;

(37) (a) D of Formula I comprises at least 80% sequence complementarity to A; (b) C of Formula II comprises at least 80% sequence complementarity to T; (c) B of Formula II comprises at least 80% sequence complementarity to T; (d) A of Formula II comprises at least 80% sequence complementarity to D; (e) A of Formula III comprises at least 80% sequence complementarity to D; (f) B of Formula III comprises at least 80% sequence complementarity to T; (g) A of Formula IV

comprises at least 80% sequence complementarity to D; and/or (h) C of Formula IV comprises at least 80% sequence complementarity to T.

(38) In some aspects of the aforementioned embodiment, the blocked primer molecules include the sequence of any one of SEQ ID NOs: 14-1421.

(39) In some aspects of the aforementioned embodiment, the RNP complex comprises a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the RNP complex comprises a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease and/or in some aspects, the RNP complex comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(40) In some aspects of the aforementioned embodiment, the blocked nucleic acid molecule comprises a modified nucleoside or nucleotide comprises a locked nucleic acid (LNA), peptide nucleic acid (PNA), 2'-O-methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bond.

(41) In some aspects, the blocked nucleic acid molecule does not comprise a PAM sequence, and in other aspects, the blocked nucleic acid molecule comprises a PAM sequence where the PAM sequence is disposed between the first and second sequences, wherein the first sequence is 5' to the PAM sequence. In some aspects, the blocked nucleic acid molecule is a blocked primer molecule.

(42) In some aspects of the aforementioned embodiment(s), the composition of matter further comprises a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(43) Yet another embodiment provides a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (iii) a plurality of template molecules comprising a sequence corresponding to the second gRNA; (iv) a plurality of blocked primer molecules comprising a sequence complementary to the template molecules, wherein the blocked nucleic acid molecules cannot be extended by a polymerase; and (v) a polymerase and a plurality of nucleotides.

(44) Another embodiment provides a cascade assay method for detecting a target nucleic acid of interest in a sample comprising: (a) providing a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; wherein the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; (iii) a plurality of template molecules comprising a sequence corresponding to the second gRNA; (iv) a plurality of blocked primer molecules

comprising a sequence complementary to the template molecules, wherein the blocked nucleic acid molecules cannot be extended by a polymerase; and (v) a polymerase and a plurality of nucleotides; (b) contacting the reaction mixture with the sample under conditions that allow target nucleic acids of interest in the sample to bind to the first gRNA, wherein: upon binding of the target nucleic acid of interest, the RNP1 active cleaving at least one of the blocked primer molecules, thereby producing at least one unblocked primer molecule that can be extended by the polymerase; at least one unblocked primer molecule binds to one of the template molecules and is extended by the polymerase and nucleotides to form at least one synthesized activating molecule having a sequence complementary to the second gRNA; at least one synthesized activating molecule binds to the second gRNA, and RNP2 becomes active cleaving at least one further blocked primer molecule; and detecting the cleavage products of step (b), thereby detecting the target nucleic acid of interest in the sample.

(45) In some aspects the K.sub.d of the blocked nucleic acid molecules to the RNP2 is about 10.sup.5-fold greater, 10.sup.6-fold greater, 10.sup.7-fold greater, 10.sup.8-fold greater, 10.sup.9-fold greater, 10.sup.10-fold greater or more than the K.sub.d of unblocked nucleic acid molecules.

(46) A further embodiment provides a kit for detecting a target nucleic acid of interest in a sample comprising: (i) a first ribonucleoprotein complex (RNP1) comprising a first nucleic acid-guided nuclease and a first gRNA, wherein the first gRNA comprises a sequence complementary to the target nucleic acid of interest; and wherein binding of the RNP1 complex to the target nucleic acid of interest activates cis-cleavage and trans-cleavage activity of the first nucleic acid-guided nuclease; (ii) a second ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; (iii) a plurality of template molecules comprising a non-target sequence to the second gRNA; (iv) a polymerase and nucleotides; (v) a plurality of blocked primer molecules comprising a sequence complementary to the template molecules, wherein the blocked primer molecule cannot be extended by the polymerase, trans-cleavage of the blocked primer molecules results in at least one unblocked primer molecule; and wherein the unblocked primer molecule can bind to at least one the template molecule and be extended by the polymerase to form a synthesized activating molecule; and (vi) a reporter moiety, wherein the reporter moiety comprises a nucleic acid molecule and/or is operably linked to the blocked primer molecule and produces a detectable signal upon trans-cleavage activity of the blocked primer molecule by the RNP1 or the RNP2, to identify the presence of the target nucleic acid of interest in the sample.

(47) In any of these embodiments, the first and/or second nucleic acid-guided nuclease in the reaction mixture is a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the first nucleic acid-guided nuclease is a different nucleic acid-guided nuclease than the second nucleic acid-guided nuclease; in some aspects the first and/or second nucleic acid-guided nuclease is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease; and in some aspects, the first and/or second nucleic acid-guided nuclease comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(48) In some aspects the blocked primer molecules comprise a first sequence complementary to the second gRNA and a second sequence not complementary to the second gRNA, wherein the second sequence at least partially hybridizes to the first sequence resulting in at least one loop; and in some aspects, the blocked primer molecules comprise a structure represented by any one of Formulas I-IV, wherein Formulas I-IV are in the 5'-to-3' direction:

(a) A-(B-L).sub.J-C-M-T-D (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L).sub.J-C and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length and comprises at least 50% sequence complementarity to B and C; and D is 0-10 nucleotides in length

and comprises at least 50% sequence complementarity to A;

(b) D-T-T'-C-(L-B).sub.J-A (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length; T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D;

(c) T-D-M-A-(B-L).sub.J-C (Formula III); wherein T is 17-135 nucleotides in length; D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D and A-(B-L).sub.J-C are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length; or

(d) T-D-M-A-L.sub.p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T.

In some aspects, (a) T of Formula I comprises at least 80% sequence complementarity to B and C;

(b) D of Formula I comprises at least 80% sequence complementarity to A; (c) C of Formula II

comprises at least 80% sequence complementarity to T; (d) B of Formula II comprises at least 80% sequence complementarity to T; (c) A of Formula II comprises at least 80% sequence

complementarity to D; (f) A of Formula III comprises at least 80% sequence complementarity to D;

(g) B of Formula III comprises at least 80% sequence complementarity to T; (h) A of Formula IV

comprises at least 80% sequence complementarity to D; and/or (i) C of Formula IV comprises at least 80% sequence complementarity to T.

(49) In some aspects the reaction mixture comprises about 1 fM to about 10 μ M of the RNP1, and in some aspects, the reaction mixture of claim 1, wherein the reaction mixture comprises about 1 fM to about 1 mM of the RNP2.

(50) In some aspects of these embodiments, the reaction mixture comprises at least two different RNP1s, wherein different RNP1s comprise different gRNA sequences, and in some aspects, the reaction mixture comprises 2 to 10000 different RNP1s, 2 to 1000 different RNP1s, 2 to 100 different RNP1s, or 2 to 10 different RNP1s.

(51) In some aspects the blocked primer molecules include the sequence of any one of SEQ ID NOs: 14-1421. In some aspects the K.sub.d of the blocked primer molecules to the RNP2 is about 10.sup.5-fold greater, 10.sup.6-fold greater, 10.sup.7-fold greater, 10.sup.8-fold greater, 10.sup.9-fold greater, 10.sup.10-fold greater or more than the K.sub.d of unblocked primer molecules.

(52) In some aspects of the aforementioned embodiments, the reaction mixture further comprises a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(53) In some aspects of the aforementioned embodiments, the template molecules do not comprise a complement of a PAM sequence, and in some aspects, the template molecules comprise a

complement of a PAM sequence. In some aspects, the template molecules are single-stranded. In some aspects, the template molecules are linear; in yet other aspects the template molecules are circularized. In some aspects comprising circular blocked nucleic acid molecules, at least one of the plurality of circular high Kd blocked nucleic acid molecules comprises a first region comprising a sequence specific to the second guide RNA and a second region comprising a nuclease-cleavable sequence; where at least one circular high Kd blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA sequence in the second region; at least one circular high Kd blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; at least one circular high Kd blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; or at least one circular high Kd blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable RNA sequence in the second region.

(54) In some aspects the polymerase comprises strand displacement activity and/or 3'-to-5' exonuclease activity, and in some aspects, the polymerase is Phi29 polymerase.

(55) Yet another embodiment provides a composition of matter comprising a circular high Kd blocked nucleic acid molecule comprising: a region recognized by a gRNA in an RNP complex; a region comprising a sequence cleavable by a nucleic acid-guided nuclease in the RNP complex; and wherein the circular high Kd blocked nucleic acid molecule cannot activate the RNP complex, and wherein the circular high Kd blocked nucleic molecules are high Kd in relation to binding to the RNP complex.

(56) A further embodiment provides a method of unblocking a circular high Kd blocked nucleic acid molecule comprising the steps of: (a) providing a circular high Kd blocked nucleic acid molecule comprising: a first region recognized by a gRNA in an RNP complex; a second region comprising a sequence cleavable by a nucleic acid-guided nuclease in the RNP complex, wherein the circular high Kd blocked nucleic acid molecule cannot substantially activate the RNP complex; and (b) initiating cleavage of the second region by the nucleic acid-guided nuclease in the RNP complex, wherein the circular high Kd blocked nucleic acid molecule becomes a linear low Kd unblocked nucleic acid molecule, and wherein the circular high Kd blocked nucleic acid molecules are high Kd and linear low K_{sub.d} unblocked nucleic acid molecules are high K_{sub.d} and low K_{sub.d} in relation to binding the RNP complex.

(57) Also provided as an embodiment is a cascade assay method of detecting a target nucleic acid of interest in a sample comprising the steps of: (a) providing a reaction mixture comprising: (i) a first ribonucleoprotein (RNP) complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest; (ii) a second ribonucleoprotein (RNP2) complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid molecule; and (iii) a plurality of circular blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the circular blocked nucleic acid molecules cannot activate the RNP1 complex or the RNP2 complex; (b) contacting the reaction mixture with the sample under conditions that allow the target nucleic acid of interest in the sample to bind to RNP1; wherein upon binding of the target nucleic acid of interest, RNP1 becomes active initiating trans-cleavage of at least one of the circular blocked nucleic acid molecules thereby producing at least one linear unblocked nucleic acid molecule; the at least one linear unblocked nucleic acid molecule binds to RNP2 initiating cleavage of at least one further circular blocked nucleic acid molecule; and (c) detecting the cleavage products, thereby detecting the target nucleic acid of interest in the sample.

(58) In some aspects, the RNP complex (either RNP1 or RNP2) comprises a nucleic acid-guided nuclease selected from Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i,

Cas12j, Cas13a, or Cas13b, and in some aspects, the RNP complex comprises a nucleic acid-guided nuclease that is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease; the RNP complex comprises a nucleic acid-guided nuclease that exhibits both cis-cleavage and trans-cleavage activity; and/or the RNP complex comprises a nucleic acid-guided nuclease comprising a RuvC nuclease domain or a RuvC-like nuclease domain but lacks an HNH nuclease domain.

(59) In some aspects of any embodiments comprising circular high K.sub.d blocked nucleic acid molecules, the circular high K.sub.d blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA sequence in the second region; the circular high K.sub.d blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; the circular high K.sub.d blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; or the circular high K.sub.d blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable RNA sequence in the second region.

(60) In some aspects of these embodiments, the circular high K.sub.d blocked nucleic acid molecule comprises 5' and 3' ends hybridized to each other and DNA, RNA, LNA or PNA bases having a high T.sub.m; and in some aspects, the K.sub.d of the circular high K.sub.d blocked nucleic acid molecules to the RNP complex or RNP2 is about 10⁵-fold greater, 10⁶-fold greater, 10⁷-fold greater, 10⁸-fold greater, 10⁹-fold greater, 10¹⁰-fold greater or more than the K.sub.d of unblocked circular high K.sub.d blocked nucleic acid molecules.

(61) In some aspects the circular high K.sub.d blocked nucleic acid molecule comprises a modified nucleoside or nucleotide, including but not limited to a locked nucleic acid (LNA), a peptide nucleic acid (PNA), a 2'-O-methyl (2'-O-Me) modified nucleoside, a 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bond.

(62) In some aspects the circular high K.sub.d blocked nucleic acid molecule is a single-stranded, double-stranded, or partially double-stranded molecule comprising one or more different combinations of DNA-DNA, DNA-RNA or RNA-RNA hybrid molecules. In some aspects the circular high K.sub.d blocked nucleic acid molecule is a circular high K.sub.d primer molecule. In some aspects the circular high K.sub.d blocked nucleic acid molecule does not comprise a PAM sequence or the circular high K.sub.d blocked nucleic acid molecule comprises a PAM sequence.

(63) In some aspects of the aforementioned embodiments, the compositions of matter or reaction further comprises a reporter moiety wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(64) Yet another embodiment provides a composition of matter comprising: (a) a first preassembled ribonucleoprotein complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA that is specific to a target nucleic acid of interest, wherein the first nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; (b) a second preassembled ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second guide RNA, wherein the second nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; and (c) a plurality of circular high K.sub.d blocked nucleic acid molecules

comprising a sequence complementary to the second gRNA, wherein the circular high K.sub.d blocked nucleic acid molecules are not recognized by the RNP1 or RNP2, and wherein the circular high K.sub.d blocked nucleic acid molecules are high K.sub.d in relation to binding to RNP2.

(65) Another embodiment provides a composition of matter comprising: (a) a first preassembled ribonucleoprotein complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA that is specific to a target nucleic acid of interest, wherein the first nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; (b) a second preassembled ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second guide RNA, wherein the second nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; and (c) a plurality of engineered linear high K.sub.d blocked nucleic acid molecules comprising a first sequence complementary to the second gRNA, wherein the linear high K.sub.d blocked nucleic acid molecules are not recognized by the RNP1 and RNP2, and wherein the linear high K.sub.d blocked nucleic acid molecules are high K.sub.d in relation to binding to the RNP2.

(66) Yet another embodiment provides a composition of matter comprising: (a) a first preassembled ribonucleoprotein complex (RNP1) comprising a first nucleic acid-guided nuclease and a first guide RNA that is specific to a target nucleic acid of interest, wherein the first nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; (b) a second preassembled ribonucleoprotein complex (RNP2) comprising a second nucleic acid-guided nuclease and a second guide RNA, wherein the second nucleic acid-guided nuclease exhibits cis-cleavage activity and trans-cleavage activity; and (c) a plurality of engineered high K.sub.d blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the high K.sub.d blocked nucleic acid molecules are not recognized by the RNP1 and RNP2, and wherein the high K.sub.d blocked nucleic acid molecules are high K.sub.d in relation to binding to the RNP complex.

(67) In aspects of any one of the foregoing embodiments, the high K.sub.d blocked nucleic acid molecule comprises DNA, RNA, LNA or PNA bases having a high T.sub.m; the 5' and 3' ends of the high K.sub.d blocked nucleic acid molecule comprise phosphorothioate bonds (PS); high K.sub.d blocked nucleic acid molecule comprises one or more different combinations of DNA-DNA, DNA-RNA or RNA-RNA hybrid molecules; and/or the high K.sub.d blocked nucleic acid molecule comprises a nucleic acid region comprising nanoparticles attached thereto, wherein the nanoparticles provide steric hindrance to internalization in RNP2 and block RNP2 activation until cleavage and removal of the nucleic acid region comprising the nanoparticles.

(68) In other aspects, the first and/or second nucleic acid-guided nuclease is a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the first nucleic acid-guided nuclease can be a different nucleic acid-guided nuclease than the second nucleic acid-guided nuclease; in some aspects, the first and/or second nucleic acid-guided nuclease is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease and/or in some aspects, the first and/or second nucleic acid-guided nuclease comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(69) Aspects also include the composition of matter comprises about 1 fM to about 10 μ M of the RNP1; and/or the composition of matter comprises about 1 fM to about 1 mM of the RNP2.

(70) In some aspects the composition of matter comprises at least two different RNP1 complex species, wherein different RNP1s comprise different gRNA sequences; and in some aspects the composition comprises 2 to 10000 different RNP1s, 2 to 1000 different RNP1s, 2 to 100 different RNP1s, or 2 to 10 different RNP1s.

(71) In some aspects the RNP2 recognizes a PAM sequence, and in other aspects the RNP2 complex does not recognize a PAM sequence.

(72) In some aspects of the aforementioned embodiments, the composition of matter further comprises a reporter moiety, wherein the reporter moiety comprises a DNA, RNA or chimeric

nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds.

(73) In some aspects the high K_d blocked nucleic acid molecule is a high K_d blocked primer molecule.

(74) In some aspects the linear high K_{sub.d} blocked nucleic acid molecule is converted to a linear low K_{sub.d} blocked nucleic acid molecule upon trans-cleavage by RNP1 and/or RNP2. In some aspects the K_{sub.d} of the blocked nucleic acid molecules to the RNP2 is about 10^{sup.5}-fold greater, 10^{sup.6}-fold greater, 10^{sup.7}-fold greater, 10^{sup.8}-fold greater, 10^{sup.9}-fold greater, 10^{sup.10}-fold greater or more than the K_{sub.d} of unblocked nucleic acid molecules.

(75) In some aspects of the compositions of matter comprising circular blocked nucleic acid molecules, at least one of the plurality of circular high K_d blocked nucleic acid molecules comprises a first region comprising a sequence specific to the second guide RNA and a second region comprising a nuclease-cleavable sequence; where at least one circular high K_d blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA sequence in the second region; at least one circular high K_d blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; at least one circular high K_d blocked nucleic acid molecule comprises a nuclease-resistant DNA sequence in the first region and a nuclease-cleavable DNA and RNA sequence in the second region; or at least one circular high K_d blocked nucleic acid molecule comprises a nuclease-resistant RNA sequence in the first region and a nuclease-cleavable RNA sequence in the second region.

(76) In some aspects of the compositions of matter comprising linear blocked nucleic acid molecules, the linear high K_{sub.d} nucleic acid molecules comprise a structure represented by any one of Formulas I-IV, where Formulas I-IV comprise in the 5'-to-3' direction:

(a) A-(B-L)_{sub.J-C-M-T-D} (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L)_{sub.J-C} and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length and comprises at least 50% sequence complementarity to B and C; and D is 0-10 nucleotides in length and comprises at least 50% sequence complementarity to A;

(b) D-T-T'-C-(L-B)_{sub.J-A} (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length; T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D;

(c) T-D-M-A-(B-L)_{sub.J-C} (Formula III); wherein T is 17-135 nucleotides in length; D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D and A-(B-L)_{sub.J-C} are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises at least 50% sequence complementarity to D; B is 4-12 nucleotides in length and comprises at least 50% sequence complementarity to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length; or

(d) T-D-M-A-L.sub.p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T. And in some aspects, (a) T of Formula I comprises at least 80% sequence complementarity to B and C; (b) D of Formula I comprises at least 80% sequence complementarity to A; (c) C of Formula II comprises at least 80% sequence complementarity to T; (d) B of Formula II comprises at least 80% sequence complementarity to T; (e) A of Formula II comprises at least 80% sequence complementarity to D; (f) A of Formula III comprises at least 80% sequence complementarity to D; (g) B of Formula III comprises at least 80% sequence complementarity to T; (h) A of Formula IV comprises at least 80% sequence complementarity to D; and/or (i) C of Formula IV comprises at least 80% sequence complementarity to T.

(77) In some aspects, at least one of the linear blocked nucleic acid molecules include the sequence of any one of SEQ ID NOs: 14-1421.

(78) In another embodiment, there is provided a method for syndromic testing comprising: (a) providing a reaction mixture comprising: (i) a plurality of first ribonucleoprotein complexes (RNP1s), each RNP1 comprising a nucleic acid-guided nuclease exhibiting both cis-cleavage and trans-cleavage activity and a first guide RNA (gRNA), wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the reaction mixture comprises at least two different RNP1s, wherein different RNP1s comprise different first gRNAs; (ii) a second ribonucleoprotein complex (RNP2), wherein the RNP2 comprises a second nucleic acid-guided nuclease and a second gRNA that does not recognize any of the target nucleic acids of interest; and (iii) a plurality of blocked nucleic acid molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecule cannot substantially activate the plurality of RNP1s or the RNP2; (b) contacting the reaction mixture with a sample under conditions that allow target nucleic acids of interest in the sample to bind to the RNP1s, wherein: upon binding of any one of the target nucleic acids of interest, the RNP1 becomes active, cleaving at least one of the blocked nucleic acid molecules, thereby producing at least one unblocked nucleic acid molecule; and at least one unblocked nucleic acid molecule binds to the second gRNA thereby activating RNP2 and initiating trans-cleavage of at least one further blocked nucleic acid molecule; and (c) detecting products of the cleavage of step (b), thus identifying at least one target nucleic acid of interest in the sample.

(79) In some aspects of this embodiment, the first and/or second nucleic acid-guided nuclease is a Cas3, Cas12a, Cas12b, Cas12c, Cas12d, Cas12e, Cas14, Cas12h, Cas12i, Cas12j, Cas13a, Cas13b nuclease; in some aspects, the first nucleic acid-guided nuclease can be a different nucleic acid-guided nuclease than the second nucleic acid-guided nuclease; in some aspects, the first and/or second nucleic acid-guided nuclease is a Type V nucleic acid-guided nuclease or a Type VI nucleic acid-guided nuclease and/or in some aspects, the first and/or second nucleic acid-guided nuclease comprises a RuvC nuclease domain or a RuvC-like nuclease domain and lacks an HNH nuclease domain.

(80) Aspects also include the reaction mixture comprises about 1 fM to about 10 UM of the RNP1; and/or the reaction mixture comprises about 1 fM to about 1 mM of the RNP2. In some aspects the reaction mixture comprises at least two different RNP1 complex species, wherein different RNP1s comprise different gRNA sequences; and in some aspects the composition comprises 2 to 10000 different RNP1s, 2 to 1000 different RNP1s, 2 to 100 different RNP1s, or 2 to 10 different RNP1s.

(81) In some aspects the K.sub.d of the plurality of blocked nucleic acid molecules to the RNP2 is about 10^{sup.5}-fold greater, 10^{sup.6}-fold greater, 10^{sup.7}-fold greater, 10^{sup.8}-fold greater, 10^{sup.9}-fold greater, 10^{sup.10}-fold greater or more than the K.sub.d of unblocked nucleic acid molecules.

(82) In some aspects of the aforementioned embodiment, the target nucleic acid of interest is of

bacterial, viral, fungal, or mammalian origin, and in some aspects, the sample may include peripheral blood, serum, plasma, ascites, urine, cerebrospinal fluid (CSF), sputum, saliva, bone marrow, synovial fluid, aqueous humor, amniotic fluid, cerumen, breast milk, bronchoalveolar lavage fluid, semen, prostatic fluid, cowper's fluid or pre-ejaculatory fluid, sweat, fecal matter, hair, tears, cyst fluid, pleural and peritoneal fluid, pericardial fluid, lymph, chyme, chyle, bile, interstitial fluid, menses, pus, sebum, vomit, vaginal secretions, mucosal secretion, stool water, pancreatic juice, lavage fluids from sinus cavities, bronchopulmonary aspirates, blastocyst cavity fluid, and/or umbilical cord blood.

(83) In some aspects of the aforementioned embodiments, the reaction mixture further comprises a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule that is operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2; or wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and is not operably linked to the blocked nucleic acid molecule and produces a detectable signal upon cleavage by RNP1 and/or RNP2. In some aspects, the detectable signal is produced within about 1-10 minutes upon binding of the target nucleic acid of interest to RNP1; in some aspects, the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or optical signal; and/or in some aspects, the reporter moiety comprises a modified nucleoside or nucleotide including but not limited to locked nucleic acids (LNAs), peptide nucleic acids (PNAs), 2'-O-methyl (2'-O-Me) modified nucleosides, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bonds. In some aspects the detectable signal is produced within about 1-10 minutes upon the target nucleic acid of interest activating RNP1.

(84) In some aspects the blocked nucleic acid molecules comprise a PAM sequence and in other aspects, the blocked nucleic acid molecules do not comprise a PAM sequence. In some aspects the blocked nucleic acid molecules are linear and in some aspects, the blocked nucleic acids are circular and in yet other aspects, the blocked nucleic acid molecules are a mixture of circular and linear blocked nucleic acid molecules.

(85) In some aspects the blocked nucleic acid molecules are blocked primer molecules and wherein the reaction mixture further comprises a polymerase and nucleotides.

(86) In some aspects, the syndromic testing is for any two or more of common flu (e.g., influenza A, influenza A/H1, influenza A/H3, influenza A/H1-2009, and influenza B); one of the multiple strains of respiratory syncytial virus (RSV), such as RSV-A and RSV-B; at least one variant of SARS-CoV-2 (e.g., B.1.1.7, B.1.351, P.1, B.1.617.2, BA.1, BA.2, BA.2.12.1, BA.4, and BA.5); and at least one of other pathogens of interest (e.g., parainfluenza virus 1-4, human metapneumovirus, human rhinovirus, human enterovirus, adenovirus, coronavirus HKU1, coronavirus NL63, coronavirus 229E, coronavirus OC43, MERS).

(87) Yet other embodiments provide: a method of detecting a target nucleic acid of interest in a sample comprising the steps of: providing a reaction mixture comprising a first RNP complex comprising a first nucleic acid-guided nuclease and a first guide RNA (gRNA), wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest; and a second RNP complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest; and contacting the reaction mixture with the sample under conditions that allow the target nucleic acid of interest in the sample to bind to the first gRNA, wherein upon binding of the target nucleic acid of interest, the first RNP complex becomes active which catalyzes activation of the second RNP complex via one or more blocked nucleic acids to produce a detectable signal from a reporter moiety.

(88) A further embodiment provides a modular cascade assay comprising: a first nucleic acid-guided nuclease, wherein the first nucleic acid-guided nuclease will form a first ribonucleoprotein complex with a first gRNA that is complementary to a target nucleic acid of interest; a second RNP2 complex comprising a second nucleic acid-guided nuclease and a second gRNA that is not complementary to a target nucleic acid of interest; and a plurality of blocked nucleic acid

molecules comprising a sequence complementary to the second gRNA, wherein the blocked nucleic acid molecules cannot activate the RNP1 complex or the RNP2 complex; wherein by changing the sequence of the first gRNA, the modular cascade assay is changed to detect different target nucleic acids of interest.

(89) These aspects and other features and advantages of the invention are described below in more detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing and other features and advantages of the present invention will be more fully understood from the following detailed description of illustrative embodiments taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1A is an overview of a prior art assay where target nucleic acids of interest from a sample must be amplified before performing a detection assay.
- (3) FIG. 1B is an overview of the general principles underlying the nucleic acid-guided nuclease cascade assay described in detail herein where target nucleic acids of interest from a sample do not need to be amplified before detection.
- (4) FIG. 2A is a diagram showing the sequence of steps in an exemplary cascade assay utilizing blocked nucleic acids.
- (5) FIG. 2B is a diagram showing an exemplary blocked nucleic acid molecule and a method for unblocking the blocked nucleic acid molecules of the disclosure.
- (6) FIG. 2C shows schematics of several exemplary blocked nucleic acid molecules containing the structure of Formula I, as described herein.
- (7) FIG. 2D shows schematics of several exemplary blocked nucleic acid molecules containing the structure of Formula II, as described herein.
- (8) FIG. 2E shows schematics of several exemplary blocked nucleic acid molecules containing the structure of Formula III, as described herein.
- (9) FIG. 2F shows schematics of several exemplary blocked nucleic acid molecules containing the structure of Formula IV, as described herein.
- (10) FIG. 2G shows an exemplary single-stranded blocked nucleic acid molecule with a design able to block R-loop formation with an RNP complex, thereby blocking activation of the trans-nuclease activity of an RNP complex (i.e., RNP2).
- (11) FIG. 2H shows schematics of exemplary circularized blocked nucleic acid molecules.
- (12) FIG. 3A is a diagram showing the sequence of steps in an exemplary cascade assay involving circular blocked primer molecules and linear template molecules.
- (13) FIG. 3B is a diagram showing the sequence of steps in an exemplary cascade assay involving circular blocked primer molecules and circular template molecules.
- (14) FIG. 4 illustrates three embodiments of reporter moieties.
- (15) FIG. 5A shows a lateral flow assay that can be used to detect the cleavage and separation of a signal from a reporter moiety.
- (16) FIG. 5B shows a schematic of a lateral flow assay device illustrating the results of an exemplary syndromic test.
- (17) FIG. 6 shows a titrated quantification of a synthesized nucleocapsid gene (N-gene) using the nucleic acid detection methods described herein. As described in Example VI, a cascade assay was initiated using the detection methods described in Examples II-V above.
- (18) FIG. 7 shows titrated quantification of an inactivated SARS-CoV-2 virus using the nucleic acid detection methods described in Examples II-V above.
- (19) FIG. 8 shows titrated quantification of DNA from Methicillin-resistant *Staphylococcus*

(MRSA) using the nucleic acid detection methods described in Examples II-V.

(20) FIG. 9 shows titrated quantification of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using the nucleic acid detection methods described in Examples II-V.

(21) FIG. 10 shows the detection of 3 copies of a molecule of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using Molecule C5 as the blocked nucleic acid molecule.

(22) FIG. 11 shows the detection of 3 copies of a molecule of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using Molecule C6 as the blocked nucleic acid molecule.

(23) FIG. 12 shows the detection of 3 copies of a molecule of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using Molecule C7 as the blocked nucleic acid molecule.

(24) FIG. 13 shows the detection of 3 copies of a molecule of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using Molecule C8 as the blocked nucleic acid molecule.

(25) FIG. 14 shows the detection of 3 copies of a molecule of DNA from Methicillin-resistant *Staphylococcus* (MRSA) using Molecule C9 as the blocked nucleic acid molecule.

(26) It should be understood that the drawings are not necessarily to scale, and that like reference numbers refer to like features.

Definitions

(27) All of the functionalities described in connection with one embodiment of the compositions and methods described herein are intended to be applicable to the additional embodiments of the compositions and methods described herein except where expressly stated or where the feature or function is incompatible with the additional embodiments. For example, where a given feature or function is expressly described in connection with one embodiment but not expressly mentioned in connection with an alternative embodiment, it should be understood that the feature or function may be deployed, utilized, or implemented in connection with the alternative embodiment unless the feature or function is incompatible with the alternative embodiment.

(28) Note that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a cell” refers to one or more cells, and reference to “a system” includes reference to equivalent steps, methods and devices known to those skilled in the art, and so forth. Additionally, it is to be understood that terms such as “left,” “right,” “top,” “bottom,” “front,” “rear,” “side,” “height,” “length,” “width,” “upper,” “lower,” “interior,” “exterior,” “inner,” “outer” that may be used herein merely describe points of reference and do not necessarily limit embodiments of the present disclosure to any particular orientation or configuration. Furthermore, terms such as “first,” “second,” “third,” etc., merely identify one of a number of portions, components, steps, operations, functions, and/or points of reference as disclosed herein, and likewise do not necessarily limit embodiments of the present disclosure to any particular configuration or orientation.

(29) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All publications mentioned herein are incorporated by reference for the purpose of describing and disclosing devices, formulations and methodologies that may be used in connection with the presently described invention. Conventional methods are used for the procedures described herein, such as those provided in the art, and demonstrated in the Examples and various general references. Unless otherwise stated, nucleic acid sequences described herein are given, when read from left to right, in the 5' to 3' direction. Nucleic acid sequences may be provided as DNA, as RNA, or a combination of DNA and RNA (e.g., a chimeric nucleic acid).

(30) Where a range of values is provided, it is understood that each intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in smaller ranges, and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in

the invention.

(31) The term “and/or” where used herein is to be taken as specific disclosure of each of the multiple specified features or components with or without another. Thus, the term “and/or” as used in a phrase such as “A and/or B” herein is intended to include “A and B,” “A or B,” “A” (alone), and “B” (alone). Likewise, the term “and/or” as used in a phrase such as “A, B, and/or C” is intended to encompass each of the following embodiments: A, B, and C; A, B, or C; A or C; A or B; B or C; A and C; A and B; B and C; A (alone); B (alone); and C (alone).

(32) In the following description, numerous specific details are set forth to provide a more thorough understanding of the present invention. However, it will be apparent to one of skill in the art that the present invention may be practiced without one or more of these specific details. In other instances, features and procedures well known to those skilled in the art have not been described in order to avoid obscuring the invention. The terms used herein are intended to have the plain and ordinary meaning as understood by those of ordinary skill in the art.

(33) As used herein, the term “about,” as applied to one or more values of interest, refers to a value that falls within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of a stated reference value, unless otherwise stated or otherwise evident from the context (except where such number would exceed 100% of a possible value).

(34) As used herein, the terms “binding affinity” or “dissociation constant” or “K_{sub.d}” refer to the tendency of a molecule to bind (covalently or non-covalently) to a different molecule. A high K_{sub.d} (which in the context of the present disclosure refers to blocked nucleic acid molecules or blocked primer molecules binding to RNP2) indicates the presence of more unbound molecules, and a low K_{sub.d} (which in the context of the present disclosure refers to unblocked nucleic acid molecules or unblocked primer molecules binding to RNP2) indicates the presence of more bound molecules. In the context of the present disclosure and the binding of blocked or unblocked nucleic acid molecules or blocked or unblocked primer molecules to RNP2, low K_{sub.d} values are in a range from about 100 fM to about 1 aM or lower (e.g., 100 zM) and high K_{sub.d} values are in the range of 100 nM-100 μM (10 mM) and thus are about 10^{sup.5}- to 10^{sup.10}-fold or higher as compared to low K_{sub.d} values.

(35) As used herein, the terms “binding domain” or “binding site” refer to a region on a protein, DNA, or RNA, to which specific molecules and/or ions (ligands) may form a covalent or non-covalent bond. By way of example, a polynucleotide sequence present on a nucleic acid molecule (e.g., a primer binding domain) may serve as a binding domain for a different nucleic acid molecule (e.g., an unblocked primer nucleic acid molecule). Characteristics of binding sites are chemical specificity, a measure of the types of ligands that will bond, and affinity, which is a measure of the strength of the chemical bond.

(36) As used herein, the term “blocked nucleic acid molecule” refers to nucleic acid molecules that cannot bind to the first or second RNP complex to activate cis- or trans-cleavage. “Unblocked nucleic acid molecule” refers to a formerly blocked nucleic acid molecule that can bind to the second RNP complex (RNP2) to activate trans-cleavage of additional blocked nucleic acid molecules.

(37) The terms “Cas RNA-guided endonuclease” or “CRISPR nuclease” or “nucleic acid-guided nuclease” refer to a CRISPR-associated protein that is an RNA-guided endonuclease suitable for assembly with a sequence-specific gRNA to form a ribonucleoprotein (RNP) complex.

(38) As used herein, the terms “cis-cleavage”, “cis-endonuclease activity”, “cis-mediated endonuclease activity”, “cis-nuclease activity”, “cis-mediated nuclease activity”, and variations thereof refer to sequence-specific cleavage of a target nucleic acid of interest, including an unblocked nucleic acid molecule or synthesized activating molecule, by a nucleic acid-guided nuclease in an RNP complex. Cis-cleavage is a single turn-over cleavage event in that only one substrate molecule is cleaved per event.

(39) The term “complementary” as used herein refers to Watson-Crick base pairing between

nucleotides and specifically refers to nucleotides hydrogen-bonded to one another with thymine or uracil residues linked to adenine residues by two hydrogen bonds and cytosine and guanine residues linked by three hydrogen bonds. In general, a nucleic acid includes a nucleotide sequence described as having a “percent complementarity” or “percent homology” to a specified second nucleotide sequence. For example, a nucleotide sequence may have 80%, 90%, or 100% complementarity to a specified second nucleotide sequence, indicating that 8 of 10, 9 of 10, or 10 of 10 nucleotides of a sequence are complementary to the specified second nucleotide sequence. For instance, the nucleotide sequence 3'-TCGA-5' is 100% complementary to the nucleotide sequence 5'-AGCT-3'; and the nucleotide sequence 3'-TCGATCGATCGA-5' [SEQ ID NO: 1] is 100% complementary to a region of the nucleotide sequence 5'-GCTAGCTAGC-3' [SEQ ID NO: 2].

(40) As used herein, the term “contacting” refers to placement of two moieties in direct physical association, including in solid or liquid form. Contacting can occur in vitro with isolated cells (for example in a tissue culture dish or other vessel) or in vivo by administering an agent to a subject.

(41) A “control” is a reference standard of a known value or range of values.

(42) The terms “guide nucleic acid” or “guide RNA” or “gRNA” refer to a polynucleotide comprising 1) a crRNA region or guide sequence capable of hybridizing to the target strand of a target nucleic acid of interest, and 2) a scaffold sequence capable of interacting or complexing with a nucleic acid-guided nuclease. The crRNA region of the gRNA is a customizable component that enables specificity in every nucleic acid-guided nuclease reaction. A gRNA can include any polynucleotide sequence having sufficient complementarity with a target nucleic acid of interest to hybridize with the target nucleic acid of interest and to direct sequence-specific binding of a ribonucleoprotein (RNP) complex containing the gRNA and nucleic acid-guided nuclease to the target nucleic acid. Target nucleic acids of interest may include a protospacer adjacent motif (PAM), and, following gRNA binding, the nucleic acid-guided nuclease induces a double-stranded break either inside or outside the protospacer region on the target nucleic acid of interest, including on an unblocked nucleic acid molecule or synthesized activating molecule. A gRNA may contain a spacer sequence including a plurality of bases complementary to a protospacer sequence in the target nucleic acid. For example, a spacer can contain about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, or more bases. The gRNA spacer may be 50%, 60%, 75%, 80%, 85%, 90%, 95%, 97.5%, 98%, 99%, or more complementary to its corresponding target nucleic acid of interest. Optimal alignment may be determined with the use of any suitable algorithm for aligning sequences. A guide RNA may be from about 20 nucleotides to about 300 nucleotides long. Guide RNAs may be produced synthetically or generated from a DNA template.

(43) “Modified” refers to a changed state or structure of a molecule. Molecules may be modified in many ways including chemically, structurally, and functionally. In one embodiment, a nucleic acid molecule (for example, a blocked nucleic acid molecule) may be modified by the introduction of non-natural nucleosides, nucleotides, and/or internucleoside linkages. In another embodiment, a modified protein (e.g., a nucleic acid-guided nuclease) may refer to any polypeptide sequence alteration which is different from the wildtype.

(44) The terms “percent sequence identity”, “percent identity”, or “sequence identity” refer to percent (%) sequence identity with respect to a reference polynucleotide or polypeptide sequence following alignment by standard techniques. Alignment for purposes of determining percent sequence identity can be achieved in various ways that are within the capabilities of one of skill in the art, for example, using publicly available computer software such as BLAST, BLAST-2, PSI-BLAST, or MEGALIGN™ software. In some embodiments, the software is MUSCLE (Edgar, *Nucleic Acids Res.*, 32 (5): 1792-1797 (2004)). Those skilled in the art can determine appropriate parameters for aligning sequences, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. For example, in embodiments, percent sequence identity values are generated using the sequence comparison computer program BLAST

(Altschul et al., J. Mol. Biol., 215:403-410 (1990)).

(45) As used herein, the terms “preassembled ribonucleoprotein complex”, “ribonucleoprotein complex”, “RNP complex”, or “RNP” refer to a complex containing a guide RNA (gRNA) and a nucleic acid-guided nuclease, where the gRNA is integrated with the nucleic acid-guided nuclease. The gRNA, which includes a sequence complementary to a target nucleic acid of interest, guides the RNP to the target nucleic acid of interest and hybridizes to it. The hybridized target nucleic acid-gRNA units are cleaved by the nucleic acid-guided nuclease. In the cascade assays described herein, a first ribonucleoprotein complex (RNP1) includes a first guide RNA (gRNA) specific to a nucleic acid target nucleic acid of interest, and a first nucleic acid-guided nuclease, such as, for example, cas12a or cas14a for a DNA target nucleic acid, or cas13a for an RNA target nucleic acid. A second ribonucleoprotein complex (RNP2) for signal amplification includes a second guide RNA specific to an unblocked nucleic acid or synthesized activating molecule, and a second nucleic acid-guided nuclease, which may be different from or the same as the first nucleic acid-guided nuclease.

(46) As used herein, the terms “protein” and “polypeptide” are used interchangeably. Proteins may or may not be made up entirely of amino acids.

(47) As used herein, the term “sample” refers to tissues; cells or component parts; body fluids, including but not limited to peripheral blood, serum, plasma, ascites, urine, cerebrospinal fluid (CSF), sputum, saliva, bone marrow, synovial fluid, aqueous humor, amniotic fluid, cerumen, breast milk, bronchoalveolar lavage fluid, semen, prostatic fluid, cowper's fluid or pre-ejaculatory fluid, sweat, fecal matter, hair, tears, cyst fluid, pleural and peritoneal fluid, pericardial fluid, lymph, chyme, chyle, bile, interstitial fluid, menses, pus, sebum, vomit, vaginal secretions, mucosal secretion, stool water, pancreatic juice, lavage fluids from sinus cavities, bronchopulmonary aspirates, blastocyst cavity fluid, and umbilical cord blood; food; agricultural products; pharmaceuticals; cosmetics, nutraceuticals; personal care products; environmental substances such as soil, water, or air; industrial sites and products; and chemicals and compounds. A sample further may include a homogenate, lysate or extract. A sample further refers to a medium, such as a nutrient broth or gel, which may contain cellular components, such as proteins or nucleic acid molecules.

(48) The terms “target DNA sequence”, “target sequence”, “target nucleic acid of interest”, “target molecule of interest”, “target nucleic acid”, or “target of interest” refer to any locus that is recognized by a gRNA sequence in vitro or in vivo. The “target strand” of a target nucleic acid of interest is the strand of the double-stranded target nucleic acid that is complementary to a gRNA. The spacer sequence of a gRNA may be 50%, 60%, 75%, 80%, 85%, 90%, 95%, 97.5%, 98%, 99% or more complementary to the target nucleic acid of interest. Optimal alignment can be determined with the use of any suitable algorithm for aligning sequences. Full complementarity is not necessarily required provided there is sufficient complementarity to cause hybridization and trans-cleavage activation of an RNP complex. A target nucleic acid of interest can include any polynucleotide, such as DNA (ssDNA or dsDNA) or RNA polynucleotides. A target nucleic acid of interest may be located in the nucleus or cytoplasm of a cell such as, for example, within an organelle of a eukaryotic cell, such as a mitochondrion or a chloroplast, or it can be exogenous to a host cell, such as a eukaryotic cell or a prokaryotic cell. The target nucleic acid of interest may be present in a sample, such as a biological or environmental sample, and it can be a viral nucleic acid molecule, a bacterial nucleic acid molecule, a fungal nucleic acid molecule, or a polynucleotide of another organism, such as a coding or a non-coding sequence, and it may include single-stranded or double-stranded DNA molecules, such as a cDNA or genomic DNA, or RNA molecules, such as mRNA, RNA, and rRNA. The target nucleic acid may be associated with a protospacer adjacent motif (PAM) sequence, which may include a 2-5 base pair sequence adjacent to the protospacer. In some embodiments 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more target nucleic acids can be detected by the disclosed method.

(49) As used herein, the terms “trans-cleavage”, “trans-endonuclease activity”, “trans-mediated endonuclease activity”, “trans-nuclease activity”, “trans-mediated nuclease activity”, and variations thereof, refer to indiscriminate, non-sequence-specific cleavage of a nucleic acid molecule by an endonuclease (such as by a Cas12, Cas13, and Cas14) which is triggered by cis- (sequence-specific) cleavage. Trans-cleavage is a “multiple turn-over” event, in that more than one substrate molecule is cleaved after initiation by a single turn-over cis-cleavage event.

(50) Type V CRISPR/Cas nucleic acid-guided nucleases are a subtype of Class 2 CRISPR/Cas effector nucleases such as, but not limited to, engineered Cas12a, Cas12b, Cas12c, C2c4, C2c8, C2c5, C2c10, C2c9, CasX (Cas12c), CasY (Cas12d), Cas 13a nucleases or naturally-occurring proteins, such as a Cas12a isolated from, for example, *Francisella tularensis* subsp. *novicida* (Gene ID: 60806594), *Candidatus methanoplasma termitum* (Gene ID: 24818655), *Candidatus methanomethylophilus alvus* (Gene ID: 15139718), and [*Eubacterium*] *eligens* ATCC 27750 (Gene ID: 41356122), and an artificial polypeptide, such as a chimeric protein.

(51) The term “variant” refers to a polypeptide or polynucleotide that differs from a reference polypeptide or polynucleotide but retains essential properties. A typical variant of a polypeptide differs in amino acid sequence from another reference polypeptide. Generally, differences are limited so that the sequences of the reference polypeptide and the variant are closely similar overall and, in many if not most regions, identical. A variant and reference polypeptide may differ in amino acid sequence by one or more modifications (e.g., substitutions, additions, and/or deletions). A variant of a polypeptide may be a conservatively modified variant. A substituted or inserted amino acid residue may or may not be one encoded by the genetic code (e.g., a non-natural amino acid). A variant of a polypeptide may be naturally occurring, such as an allelic variant, or it may be a variant that is not known to occur naturally. Variants include modifications—including chemical modifications—to one or more amino acids that do not involve amino acid substitutions, additions or deletions.

(52) A “vector” is any of a variety of nucleic acids that comprise a desired sequence or sequences to be delivered to and/or expressed in a cell. Vectors are typically composed of DNA, although RNA vectors are also available. Vectors include, but are not limited to, plasmids, fosmids, phagemids, virus genomes, synthetic chromosomes, and the like.

DETAILED DESCRIPTION

(53) The present disclosure provides compositions of matter, methods, and cascade assays for detecting nucleic acids. The cascade assays described herein comprise first and second ribonucleoprotein complexes and either blocked nucleic acid molecules or blocked primer molecules. The blocked nucleic acid molecules or blocked primer molecules keep the second ribonucleoprotein complexes “locked” unless and until a target nucleic acid of interest activates the first ribonucleoprotein complex. The methods comprise the steps of providing cascade assay components, contacting the cascade assay components with a sample, and detecting a signal that is generated only when a target nucleic acid of interest is present in the sample.

(54) Early and accurate detection and determination of infections and diseases is crucial for appropriate prevention strategies, accurate testing, confirmation, and further diagnosis and treatment. Nucleic acid-guided nucleases, such as the Cas12a endonuclease, can be utilized as diagnostic tools for the detection of target nucleic acids of interest associated with diseases. However, currently available state-of-the-art CRISPR Cas12a-based nucleic acid detection relies on DNA amplification before using Cas12a enzymes, which significantly hinders the ability to perform rapid point-of-care testing. The lack of rapidity is due to the fact that target-specific activation of Cas12a enzymes, referred herein as cis-cleavage, is a single turnover event in which the number of activated enzyme complexes is, at most, equal to the number of copies of the target nucleic acids of interest in the sample. Once a ribonucleoprotein (RNP) complex is activated after completion of cis-cleavage, the RNP complex initiates rapid non-specific trans-endonuclease activity, which is a multi-turnover event. Some currently available methods use trans-cleavage to

cleave fluorescent reporters that are initially quenched to generate a signal, thereby indicating the presence of a cis-cleavage event—the target nucleic acid. However, the $K_{sub.cat}$ of activated Cas12a complex is 17/sec and 3/sec for dsDNA and ssDNA targets, respectively. Therefore, for less than 10,000 target copies, the number of reporters cleaved is not sufficient to generate a signal in less than 60 minutes. Hence, all current technologies rely on DNA amplification to first generate billions of target copies to activate a proportional number of ribonucleoprotein complexes to generate a detectable signal in 30-60 minutes.

(55) The present disclosure describes a nucleic acid-guided nuclease cascade assay that can detect one or more target nucleic acids of interest (e.g., DNA, RNA and/or cDNA) at attomolar (aM) (or lower) limits in about 10 minutes or less without the need for amplifying the target nucleic acid(s) of interest, thereby avoiding the drawbacks of multiplex amplification, such as primer-dimerization. As described in detail below, the nucleic acid-guided nuclease cascade assays utilize signal amplification mechanisms comprising various components including nucleic acid-guided nucleases, guide RNAs (gRNAs), blocked nucleic acid molecules or blocked primer molecules, reporter moieties, and, in some embodiments, polymerases. A particularly advantageous feature of the cascade assay is that, with the exception of the gRNA (gRNA1) in RNP1, the cascade assay components stay the same no matter what target nucleic acid(s) of interest are being detected. In this sense, the cascade assay is modular.

(56) FIG. 1A provides a simplified diagram demonstrating a prior art method (1) of a nucleic acid-guided nuclease detection assay where target nucleic acids of interest from a sample must be amplified in order to be detected. First, assuming the presence of a target nucleic acid of interest in a sample, the target nucleic acid of interest (2) is amplified to produce many copies of the target nucleic acid of interest (4). The detection assay is initiated (step 2) when the target nucleic acid of interest (4) is combined with and binds to a pre-assembled ribonucleoprotein complex (6), which is part of a reaction mix. The ribonucleoprotein complex (6) comprises a guide RNA (gRNA) and a nucleic acid-guided nuclease, where the gRNA is integrated with the nucleic acid-guided nuclease. The gRNA, which includes a sequence complementary to the target nucleic acid of interest, guides the RNP complex to the target nucleic acid of interest and hybridizes to it thereby activating the ribonucleoprotein complex (6). The nucleic acid-guided nuclease exhibits (i.e., possesses) both cis- and trans-cleavage activity, where trans-cleavage activity is initiated by cis-cleavage activity. Cis-cleavage activity occurs as the target nucleic acid of interest binds to the gRNA and is cleaved by the nucleic acid guided nuclease (i.e., activation). Once cis-cleavage of the target nucleic acid of interest is initiated, trans-cleavage activity is triggered, where trans-cleavage activity is indiscriminate, non-sequence-specific cleavage of nucleic acid molecules in the sample and is a multi-turnover event.

(57) In step 3, the trans-cleavage activity triggers activation of reporter moieties (12) that are present in the reaction mix. The reporter moieties (12) may be a synthetic molecule linked or conjugated to a quencher (14) and a fluorophore (16) such as, for example, a probe with a dye label (e.g., FAM or FITC) on the 5' end and a quencher on the 3' end. The quencher (14) and fluorophore (16) typically are about 20-30 bases apart or less for effective quenching via fluorescence resonance energy transfer (FRET). Reporter moieties (12) are described in greater detail below. As more activated ribonucleoprotein complexes (6) are activated (6.fwdarw.8), more trans-cleavage activity of the nucleic acid-guided nuclease in the ribonucleoprotein complex is activated and more reporter moieties are activated (where here, “activated” means unquenched); thus, the binding of the target nucleic acid of interest (4).

(58) As noted above, the downside to the prior art, currently available state-of-the-art nucleic acid-guided nuclease detection assays is that these detection assays rely on DNA amplification, which, in addition to issues with multiplexing, significantly hinders the ability to perform rapid point-of-care testing. The lack of rapidity is due to cis-cleavage of a target nucleic acid of interest being a single turnover event in which the number of activated enzyme complexes is, at most, equal to the

number of copies of the target nucleic acids of interest in the sample. Once the ribonucleoprotein complex is activated after completion of cis-cleavage, trans-cleavage activity of the reporter moieties that are initially quenched is generated. However, the k_{cat} of, e.g., activated Cas12a complex is 17/sec and 3/sec for dsDNA and ssDNA targets, respectively. Therefore, for less than 10,000 target copies, the number of reporters cleaved is not sufficient to generate a signal in less than 30-60 minutes.

(59) FIG. 1B provides a simplified diagram demonstrating a method (100) of a nucleic acid-guided nuclease cascade assay. The cascade assay is initiated when the target nucleic acid of interest (104) binds to and activates a first pre-assembled ribonucleoprotein complex (RNP1) (102). A ribonucleoprotein complex comprises a guide RNA (gRNA) and a nucleic acid-guided nuclease, where the gRNA is integrated with the nucleic acid-guided nuclease. The gRNA, which includes a sequence complementary to the target nucleic acid of interest, guides an RNP complex to the target nucleic acid of interest and hybridizes to it. Typically, preassembled RNP complexes are employed in the reaction mix—as opposed to separate nucleic acid-guided nucleases and gRNAs—to facilitate rapid detection of the target nucleic acid(s) of interest.

(60) “Activation” of RNP1 refers to activating trans-cleavage activity of the nucleic acid-guided nuclease in RNP1 (106) by first initiating cis-cleavage where the target nucleic acid of interest is cut by the nucleic acid-guided nuclease. The cis-cleavage activity initiates trans-cleavage activity (i.e., multi-turnover activity) of the nucleic acid-guided nuclease, where trans-cleavage is indiscriminate, non-sequence-specific cutting of nucleic acid molecules by the nucleic acid-guided nuclease of RNP1 (102). This trans-cleavage activity triggers activation of blocked ribonucleoprotein complexes (RNP2s) (108) in various ways, which are described in detail below. Each newly activated RNP2 (110) activates more RNP2 (108.fwdarw.110), which in turn cleave reporter moieties (112). The reporter moieties (112) may be a synthetic molecule linked or conjugated to a quencher (114) and a fluorophore (116) such as, for example, a probe with a dye label (e.g., FAM or FITC) on the 5′ end and a quencher on the 3′ end. The quencher (114) and fluorophore (116) can be about 20-30 bases apart or less for effective quenching via fluorescence resonance energy transfer (FRET). Reporter moieties also are described in greater detail below. As more RNP2s are activated (108.fwdarw.110), more trans-cleavage activity is activated and more reporter moieties are activated (where here, “activated” means unquenched); thus, the binding of the target nucleic acid of interest (104) to RNP1 (102) initiates what becomes a cascade of signal production (120), which increases exponentially. The cascade assay thus comprises a single turnover event that triggers a multi-turnover event that then triggers another multi-turnover event. As described below in relation to FIG. 4, the reporter moieties (112) may be provided as molecules that are separate from the other components of the nucleic acid-guided nuclease cascade assay, or the reporter moieties may be covalently or non-covalently linked to the blocked nucleic acid molecules or synthesized activating molecules (i.e., the target molecules for the RNP2). The various components common to the embodiments of the cascade assay and methods described herein are described below.

(61) Target Nucleic Acids of Interest

(62) The target nucleic acid of interest may be a DNA, RNA, or cDNA molecule. Target nucleic acids of interest may be isolated from a sample or organism by standard laboratory techniques or may be synthesized by standard laboratory techniques (e.g., RT-PCR). In some embodiments, the target nucleic acids of interest are identified in a sample, such as a biological sample from a subject or an environmental sample (e.g., water or soil). Non-limiting examples of biological samples include blood, serum, plasma, saliva, mucus, a nasal swab, a buccal swab, a cell, a cell culture, and tissue. The source of the sample could be any mammal, such as, but not limited to, a human, primate, monkey, cat, dog, mouse, pig, cow, horse, sheep, and bat. Samples may also be obtained from any other source, such as air, water, soil, surfaces, food, beverages, nutraceuticals, clinical sites or products, industrial sites and products, cosmetics, personal care products, pharmaceuticals,

medical devices, agricultural equipment and sites, and commercial samples.

(63) In some embodiments, the target nucleic acid of interest is from an infectious agent (e.g., a bacteria, protozoan, insect, worm, virus, or fungus). As a non-limiting example, the target nucleic acid of interest could be one or more nucleic acid molecules from bacteria, such as *Bordetella parapertussis*, *Bordetella pertussis*, *Chlamydia pneumoniae*, *Legionella pneumophila*, *Mycoplasma pneumoniae*, *Acinetobacter calcoaceticus-baumannii* complex, *Bacteroides fragilis*, *Enterobacter cloacae* complex, *Escherichia coli*, *Klebsiella aerogenes*, *Klebsiella oxytoca*, *Klebsiella pneumoniae* group, *Moraxella catarrhalis*, *Proteus* spp., *Salmonella enterica*, *Serratia marcescens*, *Haemophilus influenzae*, *Neisseria meningitidis*, *Pseudomonas aeruginosa*, *Stenotrophomonas maltophilia*, *Enterococcus faecalis*, *Enterococcus faecium*, *Listeria monocytogenes*, *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Staphylococcus lugdunensis*, *Streptococcus agalactiae*, *Streptococcus pneumoniae*, *Streptococcus pyogenes*, *Chlamydia trachomatis*, *Neisseria gonorrhoeae*, Syphilis (*Treponema pallidum*), *Ureaplasma urealyticum*, *Mycoplasma genitalium*, and/or *Gardnerella vaginalis*. As a non-limiting example, the target nucleic acid of interest could be one or more nucleic acid molecules from a virus, such as adenovirus, coronavirus HKU1, coronavirus NL63, coronavirus 229E, coronavirus OC43, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), human metapneumovirus, human rhinovirus, enterovirus, influenza A, influenza A/H1, influenza A/H3, influenza A/H1-2009, influenza B, parainfluenza virus 1, parainfluenza virus 2, parainfluenza virus 3, parainfluenza virus 4, respiratory syncytial virus, herpes simplex virus 1, herpes simplex virus 2, human immunodeficiency virus (HIV), human papillomavirus, hepatitis A virus (HAV), hepatitis B virus (HBV), hepatitis C virus (HCV), and/or human parvovirus B19 (B19V). Also, as a non-limiting example, the target nucleic acid of interest could be one or more nucleic acid molecules from a fungus, such as *Candida albicans*, *Candida auris*, *Candida glabrata*, *Candida krusei*, *Candida parapsilosis*, *Candida tropicalis*, *Cryptococcus neoformans*, and/or *Cryptococcus gattii*. As another non-limiting example, the target nucleic acid of interest could be one or more nucleic acid molecules from a protozoan, such as *Trichomonas vaginalis*. In some embodiments, other target nucleic acids of interest may be for non-infectious conditions, e.g., to be used for genotyping. Other target nucleic acids of interest and samples are described herein.

(64) The cascade assays described herein are particularly well-suited for syndromic testing. Syndromic testing allows simultaneous testing for multiple causative agents that cause similar symptoms. Syndromic testing allows rapid triage of patients, such as those needing emergency care, those amenable to treatment with pharmaceutical agents, those needing to be quarantined, etc. In syndrome testing, multiple target nucleic acids of interest are pooled into a single reaction, and this process may be repeated in multiple, separate reactions. A positive result in one of the reactions indicates that one of the target nucleic acids of interest in that pool is present. Pools of two to 10,000 target nucleic acids of interest may be employed, e.g., 2-1000, 2-100, 2-50, or 2-10. Further testing may be used to identify the specific member of the pool, if warranted. Syndromic testing allows the rapid triage of patients with the ability to focus further care rapidly.

(65) While the methods described herein do not require the target nucleic acid of interest to be DNA (and in fact it is specifically contemplated that the target nucleic acid of interest may be RNA), it is understood by those in the field that a reverse transcription step to convert target RNA to cDNA may be performed prior to or while contacting the biological sample with the composition.

(66) Nucleic Acid-Guided Nucleases

(67) The cascade assays comprise nucleic acid-guided nucleases in the reaction mix, either provided as a protein, a coding sequence for the protein, or in a ribonucleoprotein (RNP) complex. In some embodiments, the one or more nucleic acid-guided nucleases in the reaction mix may be, for example, a Cas endonuclease. Any nucleic acid-guided nuclease having both cis- and trans-endonuclease activity may be employed, and the same nucleic acid-guided nuclease may be used

for both RNPs or different nucleic acid-guided nucleases may be used in RNP1 and RNP2. Note that trans-cleavage activity is not triggered unless and until cis-cleavage activity (i.e., sequence specific activity) is initiated. Nucleic acid-guided nucleases include Type V and Type VI nucleic acid-guided nucleases, as well as nucleic acid-guided nucleases that comprise a RuvC nuclease domain or a RuvC-like nuclease domain but lack an HNH nuclease domain. Nucleic acid-guided nucleases with these properties are reviewed in Makarova and Koonin, *Methods Mol. Biol.*, 1311:47-75 (2015) and Koonin, et al., *Current Opinion in Microbiology*, 37:67-78 (2020) and updated databases of nucleic acid-guided nucleases and nuclease systems that include newly-discovered systems include BioGRID ORCS ([orcs:thebiogrid.org](http://orcs.thebiogrid.org)); GenomeCRISPR (genomecrispr.org); Plant Genome Editing Database (plantcrispr.org) and CRISPRCasFinder (crisprcas.i2bc.paris-saclay.fr).

(68) The type of nucleic acid-guided nuclease utilized in the method of detection depends on the type of target nucleic acid of interest to be detected. For example, a DNA nucleic acid-guided nuclease (e.g., a Cas12a, Cas14a, or Cas3) should be utilized if the target nucleic acid of interest is a DNA molecule, and an RNA nucleic acid-guided nuclease (e.g., Cas13a or Cas12g) should be utilized if the target nucleic acid of interest is an RNA molecule. Exemplary nucleic acid-guided nucleases include, but are not limited to, Cas RNA-guided DNA endonucleases, such as Cas3, Cas12a (e.g., AsCas12a, LbCas12a), Cas12b, Cas12c, Cas12d, Cas12e, Cas12f, Cas12g, Cas12h, Cas12i, and Cas12j; Cas RNA-guided RNA endonucleases, such as Cas13a (LbaCas13, LbuCas13, LwaCas13), Cas13b (e.g., CccaCas13b, PsmCas13b), and Cas12g; and any other nucleic acid (DNA, RNA, or cDNA) targeting nucleic acid-guided nuclease with cis-cleavage activity and collateral trans-cleavage activity. In some embodiments, the nucleic acid-guided nuclease is a Type V CRISPR-Cas nuclease, such as a Cas12a, Cas13a, or Cas14a. In some embodiments, the nucleic acid-guided nuclease is a Type I CRISPR-Cas nuclease, such as Cas3. Type II and Type VI nucleic acid-guided nucleases may also be employed.

(69) Guide RNA (gRNA)

(70) The present disclosure detects a target nucleic acid of interest via a reaction mixture containing at least two gRNAs. Suitable guide RNAs include at least one crRNA region to enable specificity in every reaction. The gRNA of RNP1 is specific to a target nucleic acid of interest, and the gRNA of RNP2 is specific to an unblocked nucleic acid or a synthesized activating molecule (both described in detail herein). As will be clear given the description below, an advantageous feature of the cascade assay is that, with the exception of the gRNA in the RNP1 (i.e., the gRNA specific to the target nucleic acid of interest), the cascade assay components can stay the same no matter what target nucleic acid(s) of interest are being detected. In this sense, the cascade assay is modular.

(71) Like the nucleic acid-guided nuclease, the gRNA may be provided in the cascade assay reaction mix in a preassembled RNP, as an RNA molecule, or may also be provided as a DNA sequence to be transcribed, in, e.g., a vector backbone. If provided as a gRNA molecule, the gRNA sequence may include multiple endoribonuclease recognition sites (e.g., Csy4) for multiplex processing. Alternatively, if provided as a DNA sequence to be transcribed, an endoribonuclease recognition site is encoded between neighboring gRNA sequences and more than one gRNA can be transcribed in a single expression cassette. Direct repeats can also serve as endoribonuclease recognition sites for multiplex processing. Guide RNAs are generally about 20 nucleotides to about 300 nucleotides in length and may contain a spacer sequence containing a plurality of bases and complementary to a protospacer sequence in the target sequence. The gRNA spacer sequence may be 50%, 60%, 75%, 80%, 85%, 90%, 95%, 97.5%, 98%, 99%, or more complementary to its intended target nucleic acid of interest.

(72) The gRNA of RNP1 is capable of complexing with the nucleic acid-guided nuclease to perform cis-cleavage of a target nucleic acid of interest (e.g., a DNA or RNA), which triggers non-sequence specific trans-cleavage of other molecules in the reaction mix. Guide RNAs include any polynucleotide sequence having sufficient complementarity with a target nucleic acid of interest (or

target sequences generated by unblocking blocked nucleic acid molecules or target sequences generated by synthesizing activating molecules as described below). Target sequences may include a protospacer-adjacent motif (PAM), and, following gRNA binding, the nucleic acid-guided nuclease induces a double-stranded break either inside or outside the protospacer region of the target sequence.

(73) In some embodiments, the gRNA (e.g., of RNP1) is an exo-resistant circular molecule that can include several DNA bases between the 5' end and the 3' end of a natural guide RNA and is capable of binding a target sequence. The length of the circularized guide for RNP1 can be such that the circular form of guide can be complexed with a nucleic acid-guided nuclease to form a modified RNP1 which can still retain its cis-cleavage (specific) and trans-cleavage (non-specific) nuclease activity.

(74) In any of the foregoing embodiments, the gRNA may be a modified or non-naturally occurring nucleic acid molecule. In some embodiments, the gRNAs of the disclosure may further contain a locked nucleic acid (LNA), a bridged nucleic acid (BNA), and/or a peptide nucleic acid (PNA). By way of further example, a modified nucleic acid molecule may contain a modified or non-naturally occurring nucleoside, nucleotide, and/or internucleoside linkage, such as a 2'-O-methyl (2'-O-Me) modified nucleoside, a 2'-fluoro (2'-F) modified nucleoside, and a phosphorothioate (PS) bond, or any other nucleic acid molecule modifications described herein.

(75) Ribonucleoprotein (RNP) Complex

(76) As described above, although the assay "reaction mix" may comprise separate nucleic acid-guided nucleases and gRNAs (or coding sequences therefor), the cascade assays preferably comprise preassembled ribonucleoprotein complexes (RNPs) in the reaction mix, allowing for faster detection kinetics. The present cascade assay employs at least two types of RNP complexes, RNP1 and RNP2, each type containing a nucleic acid-guided nuclease and a gRNA. RNP1 and RNP2 may comprise the same nucleic acid-guided nuclease or may comprise different nucleic acid-guided nucleases; however, the gRNAs in RNP1 and RNP2 are different and are configured to detect different nucleic acids. In some embodiments, the reaction mixture contains about 1 fM to about 10 μ M of a given RNP1, or about 1 pM to about 1 μ M of a given RNP1, or about 10 pM to about 500 pM of a given RNP1. In some embodiments the reaction mixture contains about $6 \times 10^{4.4}$ to about 6×10^{12} complexes per microliter (μ l) of a given RNP1, or about $6 \times 10^{5.6}$ to about 6×10^{10} complexes per microliter (μ l) of a given RNP1. In some embodiments, the reaction mixture contains about 1 fM to about 1 mM of a given RNP2, or about 1 pM to about 500 μ M of a given RNP2, or about 10 pM to about 100 μ M of a given RNP2. In some embodiments the reaction mixture contains about $6 \times 10^{4.4}$ to about 6×10^{14} complexes per microliter (μ l) of a given RNP2 or about $6 \times 10^{5.6}$ to about 6×10^{12} complexes per microliter (μ l) of a given RNP2. (See Example II below describing preassembling RNPs and Examples V-IX below describing various cascade assay conditions, including performing the cascade assay at room temperature.)

(77) In any of the embodiments of the disclosure, the reaction mixture includes 1 to about 1,000 different RNP1s (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, or 1,000 RNP1s), where different RNP1s comprise a different gRNA (or crRNA thereof) polynucleotide sequence. For example, a reaction mixture designed for syndromic testing by definition comprises more than one unique RNP1-gRNA (or RNP1-crRNA) ribonucleoprotein complex for the purpose of detecting more than one target nucleic acid of interest. More than one RNP1 may also be present for the purpose of targeting more than one target nucleic acid of interest from a single organism or condition.

(78) In any of the foregoing embodiments, the gRNA of RNP1 may be homologous or heterologous, relative to the gRNA of other RNP1 present in the reaction mixture. A homologous mixture of RNP1 gRNAs has a number of gRNAs with the same nucleotide sequence, whereas a

heterologous mixture of RNP1 gRNAs has multiple gRNAs with different nucleotide sequences (e.g., gRNAs targeting different loci, genes, variants, and/or microbial species). Therefore, the disclosed methods of identifying one or more target nucleic acids of interest may include a reaction mixture containing more than two heterologous gRNAs, more than three heterologous gRNAs, more than four heterologous gRNAs, more than five heterologous gRNAs, more than six heterologous gRNAs, more than seven heterologous gRNAs, more than eight heterologous gRNAs, more than nine heterologous gRNAs, more than ten heterologous gRNAs, more than eleven heterologous gRNAs, more than twelve heterologous gRNAs, more than thirteen heterologous gRNAs, more than fourteen heterologous gRNAs, more than fifteen heterologous gRNAs, more than sixteen heterologous gRNAs, more than seventeen heterologous gRNAs, more than eighteen heterologous gRNAs, more than nineteen heterologous gRNAs, more than twenty heterologous gRNAs, more than twenty-one heterologous gRNAs, more than twenty-three heterologous gRNAs, more than twenty-four heterologous gRNAs, or more than twenty-five heterologous gRNAs. Such a heterologous mixture of RNP1 gRNAs in a single reaction enables the capability of syndromic testing.

(79) As a first non-limiting example of a heterologous mixture of RNP1 gRNAs, the reaction mixture may contain: a number of RNP1s having a gRNA targeting parainfluenza virus 1; a number of RNP1s having a gRNA targeting human metapneumovirus; a number of RNP1s having a gRNA targeting human rhinovirus; a number of RNP1s having a gRNA targeting human enterovirus; and a number of RNP1s having a gRNA targeting coronavirus HKU1. As a second non-limiting example of a heterologous mixture of RNP1 gRNAs, the reaction mixture may contain: a number of RNP1s containing a gRNA targeting two or more SARS-Co-V-2 variants, e.g., B.1.1.7, B.1.351, P.1, B.1.617.2, BA.1, BA.2, BA.2.12.1, BA.4, and BA.5 and subvariants thereof.

(80) Reporter Moieties

(81) The cascade assay detects a target nucleic acid of interest via detection of a signal generated in the reaction mix by a reporter moiety. In some embodiments the detection of the target nucleic acid of interest occurs in about 10 minutes or less (e.g., 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 minute or less; e.g., FIGS. 6-9, and in some embodiments the detection of the target nucleic acid molecule is in about 5 minutes or less (e.g., 5, 4, 3, 2, or 1 minute or less; e.g., FIGS. 10-14). In some embodiments, the detection of the target nucleic acid molecule is in about 1 minute (e.g., FIGS. 10-13).

(82) Depending on the type of reporter moiety used, trans- and/or cis-cleavage by the nucleic acid-guided nuclease in RNP2 releases a signal. In some embodiments, trans-cleavage of stand-alone (e.g., not bound to any blocked nucleic acid molecules) reporter moieties may generate signal changes at rates that are proportional to the cleavage rate, as new RNP2s are activated over time (shown in FIG. 1B and at top of FIG. 4). Trans-cleavage by either an activated RNP1 or an activated RNP2 may release a signal. In alternative embodiments, the reporter moiety may be bound to the blocked nucleic acid molecule, where trans-cleavage of the blocked nucleic acid molecule and conversion to an unblocked nucleic acid molecule may generate signal changes at rates that are proportional to the cleavage rate, as new RNP2s are activated over time, thus allowing for real time reporting of results (shown at FIG. 4, center). In yet another embodiment, the reporter moiety may be bound to a blocked nucleic acid molecule such that cis-cleavage following the binding of the RNP2 to an unblocked nucleic acid molecule releases a PAM distal sequence, which in turn generates a signal at rates that are proportional to the cleavage rate (shown at FIG. 4, bottom). In this case, activation of RNP2 by cis- (target specific) cleavage of the unblocked nucleic acid molecule directly produces a signal, rather than producing a signal via indiscriminate trans-cleavage activity. Alternatively, or in addition, the reporter moiety may be bound to the gRNA.

(83) The reporter moiety may be a synthetic molecule linked or conjugated to a reporter and quencher such as, for example, a TAQMANTM probe with a dye label (e.g., FAM or FITC) on the 5' end and a quencher on the 3' end. The reporter and quencher may be about 20-30 bases apart or less

for effective quenching via fluorescence resonance energy transfer (FRET). Alternatively, signal generation may occur through different mechanisms. Other detectable moieties, labels, or reporters can also be used to detect a target nucleic acid of interest as described herein. Reporter moieties can be labeled in a variety of ways, including direct or indirect attachment of a detectable moiety such as a fluorescent moiety, hapten, or colorimetric moiety. Examples of detectable moieties include various radioactive moieties, enzymes, prosthetic groups, fluorescent markers, luminescent markers, bioluminescent markers, metal particles, and protein-protein binding pairs, e.g., protein-antibody binding pairs. Examples of fluorescent moieties include, but are not limited to, yellow fluorescent protein (YFP), green fluorescence protein (GFP), cyan fluorescence protein (CFP), umbelliferone, fluorescein, fluorescein isothiocyanate, rhodamine, dichlorotriazinylamine fluorescein, cyanines, dansyl chloride, phycocyanin, and phycoerythrin. Examples of bioluminescent markers include, but are not limited to, luciferase (e.g., bacterial, firefly, click beetle and the like), luciferin, and aequorin. Examples of enzyme systems having visually detectable signals include, but are not limited to, galactosidases, glucorinidases, phosphatases, peroxidases, and cholinesterases. Identifiable markers also include radioactive elements such as .sup.125I, .sup.35S, .sup.14C, or .sup.3H.

(84) The methods used to detect the generated signal will depend on the reporter moiety or moieties used. For example, a radioactive label can be detected using a scintillation counter, photographic film as in autoradiography, or storage phosphor imaging. Fluorescent labels can be detected by exciting the fluorochrome with the appropriate wavelength of light and detecting the resulting fluorescence. The fluorescence can be detected visually, by means of photographic film, by the use of electronic detectors such as charge coupled devices (CCDs) or photomultipliers and the like. Enzymatic labels can be detected by providing the appropriate substrates for the enzyme and detecting the resulting reaction product. Simple colorimetric labels can be detected by observing the color associated with the label. When pairs of fluorophores are used in an assay, fluorophores are chosen that have distinct emission patterns (wavelengths) so that they can be easily distinguished. In some embodiments, the signal can be detected by lateral flow assays (LFAs). Lateral flow tests are simple devices intended to detect the presence or absence of a target nucleic acid of interest in a sample. LFAs can use nucleic acid molecules conjugated nanoparticles (often gold, e.g., RNA-AuNPs or DNA-AuNPs) as a detection probe, which hybridizes to a complementary target sequence. (See FIGS. 5A and 5B and the description thereof below.) The classic example of an LFA is the home pregnancy test.

(85) Single-stranded nucleic acid reporter moieties such as ssDNA reporter moieties or RNA molecules can be introduced to show a signal change proportional to the cleavage rate, which increases with every new activated RNP2 complex over time. In some embodiments and as described in detail below, single-stranded nucleic acid reporter moieties can also be embedded into the blocked nucleic acid molecules for real time reporting of results.

(86) For example, the method of detecting a target nucleic acid molecule in a sample using a cascade assay as described herein can involve contacting the reaction mix with a labeled detection ssDNA containing a fluorescent resonance energy transfer (FRET) pair, a quencher/phosphor pair, or both. A FRET pair consists of a donor chromophore and an acceptor chromophore, where the acceptor chromophore may be a quencher molecule. FRET pairs (donor/acceptor) suitable for use include, but are not limited to, EDANS, IAEDANS/fluorescein, fluorescein/tetramethylrhodamine, fluorescein/Cy 5 (Cyanine 5), IEDANS/DABCYL, fluorescein/QSY-7™, fluorescein/LC Red 640, fluorescein/Cy 5.5, Texas Red (sulforhodamine 101 acid chloride)/DABCYL, BODIPY (boron-dipyrromethene)/DABCYL, Lucifer yellow/DABCYL, coumarin/DABCYL, and fluorescein/LC Red 705. In addition, a fluorophore/quantum dot donor/acceptor pair can be used. EDANS is (5-((2-Aminoethyl)amino)naphthalene-1-sulfonic acid); IAEDANS is 5-({2-[(iodoacetyl)amino]ethyl}amino)naphthalene-1-sulfonic acid); DABCYL is 4-(4-dimethylaminophenyl)diazanylbenzoic acid. Useful quenchers include, but are not limited to,

DABCYL, QSY-7-™ and QSY-33.

(87) In any of the foregoing embodiments, the reporter moiety may comprise one or more modified nucleic acid molecules, containing a modified nucleoside or nucleotide. In some embodiments the modified nucleoside or nucleotide is chosen from 2'-O-methyl (2'-O-Me) modified nucleoside, a 2'-fluoro (2'-F) modified nucleoside, and a phosphorothioate (PS) bond, or any other nucleic acid molecule modifications described below.

(88) Nucleic Acid Modifications

(89) For any of the nucleic acid molecules described herein (e.g., blocked nucleic acid molecules, blocked primer molecules, gRNAs, template molecules, synthesized activating molecules, and reporter moieties), the nucleic acid molecules may be used in a wholly or partially modified form. Typically, modifications to the blocked nucleic acids, gRNAs, template molecules, reporter moieties, and blocked primer molecules described herein are introduced to optimize the molecule's biophysical properties (e.g., increasing endonuclease resistance and/or increasing thermal stability). Modifications typically are achieved by the incorporation of, for example, one or more alternative nucleosides, alternative sugar moieties, and/or alternative internucleoside linkages.

(90) For example, one or more of the cascade assay components may include one or more of the following nucleoside modifications: 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl ($\text{—C}\equiv\text{C—CH.sub.3}$) uracil and cytosine and other alkynyl derivatives of pyrimidine bases, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 2-F-adenine, 2-amino-adenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-deazaadenine, and/or 3-deazaguanine and 3-deazaadenine. The nucleic acid molecules described herein (e.g., blocked nucleic acid molecules, blocked primer molecules, gRNAs, reporter molecules, synthesized activating molecules, and template molecules) may also include nucleobases in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine, and/or 2-pyridone. Further modification of the nucleic acid molecules described herein may include nucleobases disclosed in U.S. Pat. No. 3,687,808; Kroschwitz, ed. The Concise Encyclopedia of Polymer Science and Engineering, New York, John Wiley & Sons, 1990, pp. 858-859; Englisch, et al., Angewandte Chemie, 30:613 (1991); and Sanghvi, Chapter 16, Antisense Research and Applications, CRC Press, Gait, ed., 1993, pp. 289-302.

(91) In addition to or as an alternative to nucleoside modifications, the cascade assay components may comprise 2' sugar modifications, including 2'-O-methyl (2'-O-Me), 2'-methoxyethoxy (2'-O—CH.sub.2CH.sub.2OCH.sub.3, also known as 2'-O-(2-methoxyethyl) or 2'-MOE), 2'-dimethylaminoethoxy, i.e., a O(CH.sub.2).sub.2ON(CH.sub.3).sub.2 group, also known as 2'-DMAOE, and/or 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethylaminoethoxy-ethyl or 2'-DMAEOE), i.e., 2'-O—CH.sub.2OCH.sub.2N(CH.sub.3).sub.2. Other possible 2'-modifications that can modify the nucleic acid molecules described herein (i.e., blocked nucleic acids, gRNAs, synthesized activating molecules, reporter molecules, and blocked primer molecules) may include all possible orientations of OH; F; O-, S-, or N-alkyl (mono- or di-); O-, S-, or N-alkenyl (mono- or di-); O-, S- or N-alkynyl (mono- or di-); or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl may be substituted or unsubstituted C1 to C10 alkyl or C2 to C10 alkenyl and alkynyl. Other potential sugar substituent groups include, e.g., aminopropoxy ($\text{—OCH.sub.2CH.sub.2CH.sub.2NH.sub.2}$), allyl ($\text{—CH.sub.2—CH=CH.sub.2}$), —O-allyl ($\text{—O—CH.sub.2—CH=CH.sub.2}$) and fluoro (F). 2'-sugar substituent groups may be in the arabino (up) position or ribo (down) position. In some embodiments, the 2'-arabino modification is 2'-F. Similar

modifications may also be made at other positions on the interfering RNA molecule, particularly the 3' position of the sugar on the 3' terminal nucleoside or in 2'-5' linked oligonucleotides and the 5' position of 5' terminal nucleotide. Oligonucleotides may also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar.

(92) Finally, modifications to the cascade assay components may comprise internucleoside modifications such phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates, 5'-alkylene phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, selenophosphates, and boranophosphates having normal 3'-5' linkages, 2'-5' linked analogs of these, and those having inverted polarity wherein one or more internucleotide linkages is a 3' to 3', 5' to 5' or 2' to 2' linkage.

(93) The Cascade Assay Employing Blocked Nucleic Acids

(94) FIG. 1B depicts the cascade assay generally. A specific embodiment of the cascade assay utilizing blocked nucleic acids is depicted in FIG. 2A. In this embodiment, a blocked nucleic acid is used to prevent the activation of RNP2 in the absence of a target nucleic acid of interest. The method in FIG. 2A begins with providing the cascade assay components RNP1 (**201**), RNP2 (**202**) and blocked nucleic acid molecules (**203**). RNP1 (**201**) comprises a gRNA specific for a target nucleic acid of interest and a nucleic acid-guided nuclease (e.g., Cas 12a or Cas 14 for a DNA target nucleic acid of interest or a Cas 13a for an RNA target nucleic acid of interest) and RNP2 (**202**) comprises a gRNA specific for an unblocked nucleic acid molecule and a nucleic acid-guided nuclease (again, Cas 12a or Cas 14 for a DNA unblocked nucleic acid molecule or a Cas 13a for an RNA unblocked nucleic acid molecule). As described above, the nucleic acid-guided nucleases in RNP1 (**201**) and RNP2 (**202**) can be the same or different depending on the type of target nucleic acid of interest and unblocked nucleic acid molecule. What is key, however, is that the nucleic acid-guided nucleases in RNP1 and RNP2 may be activated to have trans-cleavage activity following initiation of cis-cleavage activity.

(95) In a first step, a sample comprising a target nucleic acid of interest (**204**) is added to the cascade assay reaction mix. The target nucleic acid of interest (**204**) combines with and activates RNP1 (**205**) but does not interact with or activate RNP2 (**202**). Once activated, RNP1 cuts the target nucleic acid of interest (**204**) via sequence-specific cis-cleavage, which then activates non-specific trans-cleavage of other nucleic acids present in the reaction mix, including the blocked nucleic acid molecules (**203**). At least one of the blocked nucleic acid molecules (**203**) becomes an unblocked nucleic acid molecule (**206**) when the blocking moiety (**207**) is removed. As described below, "blocking moiety" may refer to nucleoside modifications, topographical configurations such as secondary structures, and/or structural modifications.

(96) Once at least one of the blocked nucleic acid molecules (**203**) is unblocked, the unblocked nucleic acid molecule (**206**) can then interact with and activate an RNP2 (**208**) complex. Because the nucleic acid-guided nucleases in the RNP1x (**205**) and RNP2x (**208**) have both cis- and trans-cleavage activity, more blocked nucleic acid molecules (**203**) become unblocked nucleic acid molecules (**206**) triggering activation of more RNP2 (**208**) complexes and more trans-cleavage activity in a cascade. FIG. 2A at bottom depicts the concurrent activation of reporter moieties. Intact reporter moieties (**209**) comprise a quencher (**210**) and a fluorophore (**211**) linked by a nucleic acid sequence. As described above in relation to FIG. 1B, the reporter moieties are also subject to trans-cleavage by activated RNP1 (**205**) and RNP2 (**208**). The intact reporter moieties (**209**) become activated reporter moieties (**212**) when the quencher (**210**) is separated from the fluorophore (**211**), emitting a fluorescent signal (**213**). Signal strength increases rapidly as more blocked nucleic acid molecules (**203**) become unblocked nucleic acid molecules (**206**) triggering cis-cleavage activation of more RNP2s (**208**) and thus more trans-cleavage activity of the reporter moieties (**209**). Again, here the reporter moieties are shown as separate molecules from the blocked

nucleic acid molecules, but other configurations may be employed and are discussed in relation to FIG. 4. One particularly advantageous feature of the cascade assay is that, with the exception of the gRNA in the RNP1 (gRNA1), the cascade assay components are modular in the sense that the components stay the same no matter what target nucleic acid(s) of interest are being detected. (97) FIG. 2B is a diagram showing an exemplary blocked nucleic acid molecule (220) and an exemplary technique for unblocking the blocked nucleic acid molecules described herein. A blocked single-stranded or double-stranded, circular or linear, DNA or RNA molecule (220) comprising a target strand (222) may contain a partial hybridization with a complementary non-target strand nucleic acid molecule (224) containing unhybridized and cleavable secondary loop structures (226) (e.g., hairpin loops, tetraloops, pseudoknots, junctions, kissing hairpins, internal loops, bulges, and multibranch loops). Trans-cleavage of the loops by, e.g., activated RNP1s or RNP2s, generates short strand nucleotide sequences (228) which, because of the short length and low melting temperature $T_{sub.m}$, can dehybridize at room temperature (e.g., 15°-25° C.), thereby unblocking the blocked nucleic acid molecule (220) to create an unblocked nucleic acid molecule (230), enabling the internalization of the unblocked nucleic acid molecule (230) (target strand) into an RNP2, leading to RNP2 activation.

(98) A blocked nucleic acid molecule may be single-stranded or double-stranded, circular or linear, and may further contain a partially hybridized nucleic acid sequence containing cleavable secondary loop structures, as exemplified by “L” in FIGS. 2C-2E. Such blocked nucleic acids typically have a low binding affinity, or high dissociation constant ($K_{sub.d}$) in relation to binding to RNP2 and may be referred to herein as a high $K_{sub.d}$ nucleic acid molecule. In the context of the present disclosure, the binding of blocked or unblocked nucleic acid molecules or blocked or unblocked primer molecules to RNP2, low $K_{sub.d}$ values range from about 100 fM to about 1 aM or lower (e.g., 100 zM) and high $K_{sub.d}$ values are in the range of 100 nM to about 100 μ M (10 mM) and thus are about 10.^{sup.5}-, 10.^{sup.6}-, 10.^{sup.7}-, 10.^{sup.8}-, 10.^{sup.9}- to 10.^{sup.10}-fold or higher as compared to low $K_{sub.d}$ values.

(99) The blocked nucleic acid molecules (high $K_{sub.d}$ molecules) described herein can be converted into unblocked nucleic acid molecules (low $K_{sub.d}$ molecules—also in relation to binding to RNP2) via cleavage of nuclease-cleavable regions (e.g., via active RNP1s and RNP2s). The unblocked nucleic acid molecule has a higher binding affinity for the gRNA in the RNP2 than does the blocked nucleic acid molecule, although there may be some “leakiness” where some blocked nucleic acid molecules are able to interact with the gRNA in the RNP2. However, an unblocked nucleic acid molecule has a substantially higher likelihood than a blocked nucleic acid molecule to hybridize with the gRNA of RNP2.

(100) Once the unblocked nucleic acid molecule is bound to RNP2, the RNP2 activation triggers trans-cleavage activity, which in turn leads to more RNP2 activation by further cleaving blocked nucleic acid molecules, resulting in a positive feedback loop.

(101) In embodiments where blocked nucleic acid molecules are linear and/or form a secondary structure, the blocked nucleic acid molecules may be single-stranded (ss) or double-stranded (ds) and contain a first nucleotide sequence and a second nucleotide sequence. The first nucleotide sequence has sufficient complementarity to hybridize to a gRNA of RNP2, and the second nucleotide sequence does not. The first and second nucleotide sequences of a blocked nucleic acid molecule may be on the same nucleic acid molecule (e.g., for single-strand embodiments) or on separate nucleic acid molecules (e.g., for double strand embodiments). Trans-cleavage (e.g., via RNP1 or RNP2) of the second nucleotide sequence converts the blocked nucleic acid molecule to a single-strand unblocked nucleic acid molecule. The unblocked nucleic acid molecule contains only the first nucleotide sequence, which has sufficient complementarity to hybridize to the gRNA of RNP2, thereby activating the trans-endonuclease activity of RNP2.

(102) In some embodiments, the second nucleotide sequence at least partially hybridizes to the first nucleotide sequence, resulting in a secondary structure containing at least one loop (e.g., hairpin

loops, tetraloops, pseudoknots, junctions, kissing hairpins, internal loops, bulges, and multibranch loops). Such loops block the nucleic acid molecule from binding or incorporating into an RNP complex in a manner to initiate trans cleavage (see, e.g., the exemplary structures in FIGS. 2C-2E). (103) In some embodiments, the blocked nucleic acid molecule may contain a protospacer adjacent motif (PAM) sequence, or partial PAM sequence, positioned between the first and second nucleotide sequences, where the first sequence is 5' to the PAM sequence, or partial PAM sequence, (see FIG. 2G). Inclusion of a PAM sequence may increase the reaction kinetics internalizing the unblocked nucleic acid molecule into RNP2 and thus decrease the time to detection. In other embodiments, the blocked nucleic acid molecule does not contain a PAM sequence.

(104) In some embodiments, the blocked nucleic acid molecules (i.e., high K_{sub}d nucleic acid molecules—in relation to binding to RNP2) of the disclosure may include a structure represented by Formula I (e.g., FIG. 2C), Formula II (e.g., FIG. 2D), Formula III (e.g., FIG. 2E), or Formula IV (e.g., FIG. 2F) wherein Formulas I-IV are in the 5'-to-3' direction:

A-(B-L)J-C-M-T-D (Formula I); wherein A is 0-15 nucleotides in length; B is 4-12 nucleotides in length; L is 3-25 nucleotides in length; J is an integer between 1 and 10; C is 4-15 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then A-(B-L).sub.J-C and T-D are separate nucleic acid strands; T is 17-135 nucleotides in length (e.g., 17-100, 17-50, or 17-25) and comprises a sequence complementary to B and C; and D is 0-10 nucleotides in length and comprises a sequence complementary to A;

D-T-T'-C-(L-B)J-A (Formula II); wherein D is 0-10 nucleotides in length; T-T' is 17-135 nucleotides in length (e.g., 17-100, 17-50, or 17-25); T' is 1-10 nucleotides in length and does not hybridize with T; C is 4-15 nucleotides in length and comprises a sequence complementary to T; L is 3-25 nucleotides in length and does not hybridize with T; B is 4-12 nucleotides in length and comprises a sequence complementary to T; J is an integer between 1 and 10; A is 0-15 nucleotides in length and comprises a sequence complementary to D;

T-D-M-A-(B-L)J-C (Formula III); wherein T is 17-135 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-10 nucleotides in length; M is 1-25 nucleotides in length or is absent, wherein if M is absent then T-D and A-(B-L).sub.J-C are separate nucleic acid strands; A is 0-15 nucleotides in length and comprises a sequence complementary to D; B is 4-12 nucleotides in length and comprises a sequence complementary to T; L is 3-25 nucleotides in length; J is an integer between 1 and 10; and C is 4-15 nucleotides in length;

T-D-M-A-L_p-C (Formula IV); wherein T is 17-31 nucleotides in length (e.g., 17-100, 17-50, or 17-25); D is 0-15 nucleotides in length; M is 1-25 nucleotides in length; A is 0-15 nucleotides in length and comprises a sequence complementary to D; and L is 3-25 nucleotides in length; p is 0 or 1; C is 4-15 nucleotides in length and comprises a sequence complementary to T.

In alternative embodiments of any of these molecules, T (or T-T') can have a maximum length of 1000 nucleotides, e.g., at most 200, at most 135, at most 75, at most 50, or at most 25.

(105) Nucleotide mismatches can be introduced in any of the above structures containing double strand segments (for example, where M is absent in Formula I or Formula III) to reduce the melting temperature (T_m) of the segment such that once the loop (L) is cleaved, the double strand segment is unstable and dehybridizes rapidly. The percentage of nucleotide mismatches of a given segment may vary between 0% and 50%; however, the maximum number of nucleotide mismatches is limited to a number where the secondary loop structure still forms. "Segments" in the above statement refers to A, B, and C. In other words, the number of hybridized bases can be less than or equal to the length of each double strand segment and vary based on number of mismatches introduced.

(106) In any blocked nucleic acid molecule having the structure of Formula I, III, or IV, T will have sequence complementarity to a nucleotide sequence (e.g., a spacer sequence) within a gRNA of RNP2. The nucleotide sequence of T is to be designed such that hybridization of T to the gRNA of RNP2 activates the trans-nuclease activity of RNP2. In any blocked nucleic acid molecule having

structure of Formula II, T-T' will have sequence complementarity to a sequence (e.g., a spacer sequence) within the gRNA of RNP2. The nucleotide sequence of T-T' is to be designed such that hybridization of T-T' to the gRNA of RNP2 activates the trans-nuclease activity of RNP2. For T or T-T', full complementarity to the gRNA is not necessarily required, provided there is sufficient complementarity to cause hybridization and trans-cleavage activation of RNP2.

(107) Exemplary nucleotide sequences of blocked nucleic acid molecules (e.g., SEQ ID NOs: 14-1421) include those in Table 1.

(108) TABLE-US-00001 TABLE 1 Nucleotide sequences of blocked nucleic acid molecules. SEQ ID NO: Sequence SEQ ID NO: 14

GATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA TATATATATATAGTATC

SEQ ID NO: 15 GACACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA

TATATATATATAGTGTC SEQ ID NO: 16

GATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA TATATATATATATCGTATC

SEQ ID NO: 17 GGATCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA

TATATATATATAGATCC SEQ ID NO: 18

GACACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA TATATATATATATCGTGTC

SEQ ID NO: 19 GGATCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA

TATATATATATCGATCC SEQ ID NO: 20

GCGTCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA TATATATATATAGACGC

SEQ ID NO: 21 GCGTCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA

TATATATATATCGACGC SEQ ID NO: 22

GTATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTAT ATATATATATAGTATAC

SEQ ID NO: 23 GTGATCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTAT

ATATATATATAGATCAC SEQ ID NO: 24

GTATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTAT ATATATATATCGTATAC

SEQ ID NO: 25 GTATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTAT

ATACATATATCGTATAC SEQ ID NO: 26

GGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTAT ATATATATATAGTATCC

SEQ ID NO: 27 GTGATCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTAT

ATATATATATCGATCAC SEQ ID NO: 28

GTGATCTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTAT ATACATATATCGATCAC

SEQ ID NO: 29 GGATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTAT

ATATATATATCGTATCC SEQ ID NO: 30

GGATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTAT ATACATATATCGTATCC

SEQ ID NO: 31 GCGATCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTAT

ATATATATATAGATCGC SEQ ID NO: 32

GCGATCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTAT ATATATATATCGATCGC

SEQ ID NO: 33 GCGATCTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTAT

ATACATATATCGATCGC SEQ ID NO: 34

GATATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA TATATATATAGTATATC

SEQ ID NO: 35 GATATATTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA

TATATATATCATATATC SEQ ID NO: 36

GATATATTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA TACATATATCATATATC

SEQ ID NO: 37 GTGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA

TATATATATAGTATCAC SEQ ID NO: 38

GATATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA TATATATATCGTATATC

SEQ ID NO: 39 GATATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA

TACATATATCGTATATC SEQ ID NO: 40

GGTATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA TATATATATAGTATACC

SEQ ID NO: 41 GTGATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA

TATATATCGTATCAC SEQ ID NO: 42
GTGATACTTTTTATTTTTGATATATGTATATTTTTTATTTTTATA TACATATATCGTATCAC
SEQ ID NO: 43 GGTATACTTTTTATTTTTGATATATATATATTTTTTATTTTTATA
TATATATATCGTATACC SEQ ID NO: 44
GGTATACTTTTTATTTTTGATATATGTATATTTTTTATTTTTATA TACATATATCGTATACC
SEQ ID NO: 45 GGTGTACTTTTTATTTTTTATATATATATATTTTTTATTTTTATA
TATATATATAGTACACC SEQ ID NO: 46
GGTGTACTTTTTATTTTTGATATATATATATTTTTTATTTTTATA TATATATATCGTACACC
SEQ ID NO: 47 GGTGTACTTTTTATTTTTGATATATGTATATTTTTTATTTTTATA
TACATATATCGTACACC SEQ ID NO: 48
GTATATACTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT ATATATATAGTATATAC
SEQ ID NO: 49 GTATATACTTTTTATTTTTGATATATATATATTTTTTATTTTTTAT
ATATATATCGTATATAC SEQ ID NO: 50
GTATATACTTTTTATTTTTGATATATGTATATTTTTTATTTTTTAT ACATATATCGTATATAC
SEQ ID NO: 51 GTATATACTTTTTATTTTTGATCATGTATATTTTTTATTTTTATA
TACATGATCGTATATAC SEQ ID NO: 52
GTATATACTTTTTATTTTTGATCATATATGTTTTTTATTTTTTACA TATATGATCGTATATAC
SEQ ID NO: 53 GGATATACTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT
ATATATATAGTATATCC SEQ ID NO: 54
GGATATACTTTTTATTTTTGATATATATATATTTTTTATTTTTTAT ATATATATCGTATATCC
SEQ ID NO: 55 GGATATACTTTTTATTTTTGATATATGTATATTTTTTATTTTTTAT
ACATATATCGTATATCC SEQ ID NO: 56
GGATATACTTTTTATTTTTGATCATGTATATTTTTTATTTTTTATA TACATGATCGTATATCC
SEQ ID NO: 57 GGATATACTTTTTATTTTTGATCATATATGTTTTTTATTTTTTACA
TATATGATCGTATATCC SEQ ID NO: 58
GGTGATACTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT ATATATATAGTATCACC
SEQ ID NO: 59 GGTGATACTTTTTATTTTTGATATATATATATTTTTTATTTTTTAT
ATATATATCGTATCACC SEQ ID NO: 60
GGTGATACTTTTTATTTTTGATATATGTATATTTTTTATTTTTTAT ACATATATCGTATCACC
SEQ ID NO: 61 GGTGATACTTTTTATTTTTGATCATGTATATTTTTTATTTTTTATA
TACATGATCGTATCACC SEQ ID NO: 62
GGTGATACTTTTTATTTTTGATCATATATGTTTTTTATTTTTTACA TATATGATCGTATCACC
SEQ ID NO: 63 GGTGATCCTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT
ATATATATAGGATCACC SEQ ID NO: 64
GGTGATCCTTTTTATTTTTGATATATATATATTTTTTATTTTTTAT ATATATATCGGATCACC
SEQ ID NO: 65 GGTGATCCTTTTTATTTTTGATATATGTATATTTTTTATTTTTTAT
ACATATATCGGATCACC SEQ ID NO: 66
GGTGATCCTTTTTATTTTTGATCATGTATATTTTTTATTTTTTATA TACATGATCGGATCACC
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TATATGATCGGATCACC SEQ ID NO: 68
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ATATATCATGTATATAC SEQ ID NO: 70
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GGATATACATTTTTTATTTTTGATATATATATTTTTTATTTTTTAT ATATATCATGTATATCC

SEQ ID NO: 75 GGATATACATTTTTTATTTTTGATATATGTATTTTTTATTTTTTAC
ATATATCATGTATATCC SEQ ID NO: 76
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SEQ ID NO: 77 GGATATACATTTTTTATTTTTTGATCATATATTTTTTATTTTTTATA
TATGATCATGTATATCC SEQ ID NO: 78
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ATATGATCAATATATGTAC SEQ ID NO: 164
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SEQ ID NO: 165 GGTACATATATTTTTTTATTTTTGATATATATATTTTTATTTTTTA
TATATATCATATATGTACC SEQ ID NO: 166
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SEQ ID NO: 167 GGTACATATATTTTTTTATTTTTGATCATGTATTTTTATTTTTAT
ACATGATCATATATGTACC SEQ ID NO: 168
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ACATATATCAATATATGATCG SEQ ID NO: 171
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TACATGATCAATATATGATCG SEQ ID NO: 172
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TATATGATCAATATATGATCG SEQ ID NO: 173
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ATATATACATATATCGGATCACC SEQ ID NO: 564
GTATATACTTTTTATTTTTGATATAAGTAAATATTTTTATTTTT
ATATATACATATATCGTATATAC SEQ ID NO: 565
GGATATACTTTTTATTTTTGATATAAGTAAATATTTTTATTTTT
ATATATACATATATCGTATATCC SEQ ID NO: 566
GGTGATACTTTTTATTTTTGATATAAGTAAATATTTTTATTTTT
ATATATACATATATCGTATCACC SEQ ID NO: 567
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ATATATACATATATCGGATCACC SEQ ID NO: 568
GTATATACTTTTTATTTTTTATAAATATATATTTTTATTTTT ATATATATATATAGTATATAC
SEQ ID NO: 569 GTATATACTTTTTATTTTTGATATAAATATATTTTTATTTTT
ATATATATATATCGTATATAC SEQ ID NO: 570
GTATATACTTTTTATTTTTGATAAATGAATATATTTTTATTTTT
ATATACATATATCGTATATAC SEQ ID NO: 571
GGATATACTTTTTATTTTTTATAAATATATATTTTTATTTTT
ATATATATATATAGTATATCC SEQ ID NO: 572
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ATATATATATATCGTATATCC SEQ ID NO: 573
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ATATATATATATAGTATCACC SEQ ID NO: 575
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ATATATATATATCGTATCACC SEQ ID NO: 576
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ATATACATATATCGTATCACC SEQ ID NO: 577
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ATATATATATATAGGATCACC SEQ ID NO: 578
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ATATATATATATCGGATCACC SEQ ID NO: 579
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ATATACATATATCGGATCACC SEQ ID NO: 580
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ATATATATATATCGTATATAC SEQ ID NO: 581

GGATATATTTTTTTTATTTTATATAAATATTTTTTTTATTTTTT
ATATATATATATCGTATATCC SEQ ID NO: 582
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ATATATATATATCGTATCACC SEQ ID NO: 583
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ATATATATATATCGGATCACC SEQ ID NO: 584
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ATATATATATATATCGTGTC SEQ ID NO: 586
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SEQ ID NO: 587 GATATATAAAAAAAAAAAAAAGATATATGTATATAAAAAAAAAAAAA
ATATACATATATCATATATC SEQ ID NO: 588
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GATATATCACAAAAAAAAAAAAATATATATATAAAAAAAAAAAAA
TATATATATAGTGATATATC SEQ ID NO: 591
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TACATATATCATGTATATAC SEQ ID NO: 592
GGATATACATAAAAAAAAAAAAAAGATATATGTAAAAAAAAAAAA
ATACATATATCATGTATATCC SEQ ID NO: 593
GGATATACATAAAAAAAAAAAAAAGATCATGTATAAAAAAAAAAAAA
ATACATGATCATGTATATCC SEQ ID NO: 594
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TATATATATAGTATATACCC SEQ ID NO: 595
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AAAATATATATATATATATCGTATATAC SEQ ID NO: 597
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AAAATATATATATATATAGTATATCC SEQ ID NO: 598
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AAAATATATATATATATCGTATATCC SEQ ID NO: 599
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AATATATATATATATCGTATATCC SEQ ID NO: 603
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AATATATATATATAGTGATATATC SEQ ID NO: 604
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AATATATATATATCATGTATATCC SEQ ID NO: 605
GTACATATATTAAAAAAAAAAAAAGATATATATAAAAAAAAA
ATATATATATCAATATATGTAC SEQ ID NO: 606
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ATATATATATAGTATATACATC SEQ ID NO: 607
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AAATATATATCAATATATGATCG SEQ ID NO: 608
CGATCATATATTAAAAAAAAAAAAAGATATATGTAAAAAAAAAAAA
AATACATATATCAATATATGATCG SEQ ID NO: 609
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ATATATATATATATAGATCC SEQ ID NO: 611
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GACACAAAAAAAAAAAAAGATGATGTATATATAAAAAAAAAAAAA
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GCGTCAAAAAAAAAAAAAAGATAAATATATATATAAAAAAAAAAAAA
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SEQ ID NO: 615 GTATATACATAAAAAAAAAAAAAAGATAAATGTAAAAAAAAAAAA
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ATATATGATCATGTATATAC SEQ ID NO: 617
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ATATATATATCATGTATATCC SEQ ID NO: 618
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AATATATGATCATGTATATCC SEQ ID NO: 619
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AAAATATATATATATATAGTATATAC SEQ ID NO: 620
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AAAATATATATATATATATCGTATATCC SEQ ID NO: 623
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AAATATATACATATATCGTATATCC SEQ ID NO: 625
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AAATATATACATGATCGTATCACC SEQ ID NO: 626
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AAAATATATATATATAGTGATATATC SEQ ID NO: 627
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AACATATATGATCATGTATATAC SEQ ID NO: 629
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AAATATATATATATCATGTATATCC SEQ ID NO: 630
GTACATATATTAAAAAAAAAAAAAGATAAATATAAAAAAAAAAAAA
ATATATATATCAATATATGTAC SEQ ID NO: 631
GTACATATATTAAAAAAAAAAAAAGATAAATGTAAAAAAAAAAAA
ATACATATATCAATATATGTAC SEQ ID NO: 632
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AATATATGATCAATATATGTAC SEQ ID NO: 633
GGATATACATAAAAAAAAAAAAAAGATGATGAATAAAAAAAAAAAAA

AATACATGATGATGTATATCC SEQ ID NO: 634
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 TACATATATCATGTATATAC SEQ ID NO: 635
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 AATATATATATATATCGTATC SEQ ID NO: 636
 GATACAAAAAAAAAAAAAGATGATATAAGTACTAAAAAAAAAAAA
 AAGTACATATATGATCGTATC SEQ ID NO: 637
 GACACAAAAAAAAAAAAAGATAAATGAATATATAAAAAAAAAAAAA
 AATATATACATATATCGTGTC SEQ ID NO: 638
 GGATATACAAAAAAAAAAAAAGATATAAGTAAATATAAAAAAAAA
 AAAAATATATACATATATCGTATATCC SEQ ID NO: 639
 GGATATACAAAAAAAAAAAAAGATGATATAAGTACTAAAAAAAAAAAA
 AAAAGTACATATATGATCGTATATCC SEQ ID NO: 640
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 AAAATATATACATATATCGTATATAC SEQ ID NO: 641
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 AAAAGTACATATATGATCGTATATAC SEQ ID NO: 642
 GGATATACAAAAAAAAAAAAAGATAAATGAATATAAAAAAAAAAAAA
 AAATATATACATATATCGTATATCC SEQ ID NO: 643
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 AAATATATACATATATCGTATATAC SEQ ID NO: 644
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 AAGTACATATATGATCGTATATAC SEQ ID NO: 645
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 AATATATATATATATAGTATATAC SEQ ID NO: 646
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 AAAAATATATACATGATCATGTATATAC SEQ ID NO: 647
 GATATACAAAAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
 AATATATATATATCGTATATC SEQ ID NO: 648
 GTGATACAAAAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
 AATATATATATATCGTATCAC SEQ ID NO: 649
 GGTATACAAAAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
 AATATATATATATCGTATACC SEQ ID NO: 650
 GGATATACATAAAAAAAAAAAAAAGATAAATGAATAAAAAAAAAAAAA
 AAATATACATATATCATGTATATCC SEQ ID NO: 651
 GTATATACATAAAAAAAAAAAAAAGATAAATGTATTAAAAAAAAAAAAA
 AATATACATATATCATGTATATAC SEQ ID NO: 652
 GTATATACATAAAAAAAAAAAAAAGATGATAAATGTAAAAAAAAAAAA
 AAACATATATGATCATGTATATAC SEQ ID NO: 653
 GATACTTTTTTATTTTTTTATATATATATATATTTTTTTTATTTTTTATA
 T*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C SEQ ID NO: 654
 GACACTTTTTTATTTTTTTATATATATATATATTTTTTTTATTTTTTATA
 T*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*C SEQ ID NO: 655
 GATACTTTTTTATTTTTTGATATATATATATATTTTTTTTATTTTTTATA
 T*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 656
 GGATCTTTTTTATTTTTTTATATATATATATATTTTTTTTATTTTTTATA
 T*A*T*A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*C SEQ ID NO: 657
 GACACTTTTTTATTTTTTGATATATATATATATTTTTTTTATTTTTTATA
 T*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 658
 GGATCTTTTTTATTTTTTTGATATATATATATATTTTTTTTATTTTTTATA

T*A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*G SEQ ID NO: 659
GCGTCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*T*A*G*A*C*G*C SEQ ID NO: 660
GCGTCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 661
GTATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*C SEQ ID NO: 662
GTGATCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*A*C SEQ ID NO: 663
GTATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 664
GTATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 665
GGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*C SEQ ID NO: 666
GTGATCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 667
GTGATCTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 668
GGATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 669
GGATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 670
GCGATCTTTTTATTTTTTATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*G*C SEQ ID NO: 671
GCGATCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 672
GCGATCTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 673
GATATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*C SEQ ID NO: 674
GATATATTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 675
GATATATTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 676
GTGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*A*C SEQ ID NO: 677
GATATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 678
GATATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 679
GGTATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*C*C SEQ ID NO: 680
GTGATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 681
GTGATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 682
GGTATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 683
GGTATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA

T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C*SEQ ID NO: 684
GGTGTACTTTTTATTTTTTATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*C*A*C*SEQ ID NO: 685
GGTGTACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*C*A*C*SEQ ID NO: 686
GGTGTACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTATA
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GTATATACTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*A*C*SEQ ID NO: 688
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A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*SEQ ID NO: 689
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A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*SEQ ID NO: 690
GTATATACTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*C*SEQ ID NO: 691
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T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*A*C*SEQ ID NO: 692
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GGATATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTAT
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GGATATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C*SEQ ID NO: 695
GGATATACTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*C*SEQ ID NO: 696
GGATATACTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*C*SEQ ID NO: 697
GGTGATACTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*G*T*A*T*C*A*C*SEQ ID NO: 698
GGTGATACTTTTTATTTTTTGATATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C*SEQ ID NO: 699
GGTGATACTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C*SEQ ID NO: 700
GGTGATACTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*C*A*C*SEQ ID NO: 701
GGTGATACTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*C*A*C*SEQ ID NO: 702
GGTGATCCTTTTTATTTTTTATATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*G*G*A*T*C*A*C*SEQ ID NO: 703
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A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*C*A*C*SEQ ID NO: 704
GGTGATCCTTTTTATTTTTTGATATATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*A*T*C*G*G*A*T*C*A*C*SEQ ID NO: 705
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T*A*C*A*T*G*A*T*C*G*G*A*T*C*A*C*SEQ ID NO: 706
GGTGATCCTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*G*A*T*C*A*C*SEQ ID NO: 707
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A*T*A*T*A*T*A*G*T*G*A*T*A*T*A*T*C*SEQ ID NO: 708
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A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 709
GTATATACATTTTTTATTTTTTGATATATGTATTTTTTATTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 710
GTATATACATTTTTTATTTTTTGATCATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 711
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T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 712
GGATATACACTTTTTTATTTTTTATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*G*T*G*T*A*T*A*T*C*C SEQ ID NO: 713
GGATATACATTTTTTATTTTTTGATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 714
GGATATACATTTTTTATTTTTTGATATATGTATTTTTTATTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 715
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C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 716
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T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 717
GGGTATATACTTTTTTATTTTTTATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*G*T*A*T*A*T*A*C*C*C SEQ ID NO: 718
GGATATACACTTTTTTATTTTTTGATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*C SEQ ID NO: 719
GGATATACACTTTTTTATTTTTTGATATATGTATTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*C SEQ ID NO: 720
GGATATACACTTTTTTATTTTTTGATCATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*G*T*A*T*A*T*C*C SEQ ID NO: 721
GGATATACACTTTTTTATTTTTTGATCATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*G*T*A*T*A*T*C*C SEQ ID NO: 722
GGGTATATACTTTTTTATTTTTTGATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*C*C SEQ ID NO: 723
GGGTATATACTTTTTTATTTTTTGATATATGTATTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*C*C SEQ ID NO: 724
GGGTATATACTTTTTTATTTTTTGATCATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*A*T*A*T*A*C*C*C SEQ ID NO: 725
GGGTATATACTTTTTTATTTTTTGATCATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*A*T*A*T*A*C*C*C SEQ ID NO: 726
GGATGTACACTTTTTTATTTTTTATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*G*T*G*T*A*C*A*T*C*C SEQ ID NO: 727
GGATGTACACTTTTTTATTTTTTGATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*C SEQ ID NO: 728
GGATGTACACTTTTTTATTTTTTGATATATGTATTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*C SEQ ID NO: 729
GGATGTACACTTTTTTATTTTTTGATCATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*G*T*A*C*A*T*C*C SEQ ID NO: 730
GGATGTACACTTTTTTATTTTTTGATCATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*G*T*A*C*A*T*C*C SEQ ID NO: 731
GTATATACTTTTTTATTTTTTATATATATATATATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATAC SEQ ID NO: 732
GTATATACTTTTTTATTTTTTGATATATATATATATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 733
GGATATACTTTTTTATTTTTTATATATATATATATTTTTTATTTTTT

ATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*T*A*T*ATCC SEQ ID NO: 734
GGATATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 735
GGTGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*CACC SEQ ID NO: 736
GGTGATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ ID NO: 737
GGTGATCCTTTTTATTTTTTATATATATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*G*A*T*CACC SEQ ID NO: 738
GGTGATCCTTTTTATTTTTTGATATATATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ ID NO: 739
GATATATCACTTTTTATTTTTTATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*G*A*TATATC SEQ ID NO: 740
GTATATACATTTTTTATTTTTTGATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 741
GGATATACACTTTTTATTTTTTATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*ATATCC SEQ ID NO: 742
GGATATACATTTTTTATTTTTTGATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATCC SEQ ID NO: 743
GGGTATATACTTTTTATTTTTTATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATACCC SEQ ID NO: 744
GGATATACACTTTTTATTTTTTGATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 745
GGGTATATACTTTTTATTTTTTGATATATATATATATTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 746
GTATATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TAC SEQ ID NO: 747
GTATATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 748
GTATATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 749
GGATATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TCC SEQ ID NO: 750
GGATATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 751
GGATATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 752
GGTGATACTTTTTATTTTTTATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*ACC SEQ ID NO: 753
GGTGATACTTTTTATTTTTTGATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 754
GGTGATACTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 755
GGTGATCCTTTTTATTTTTTATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*A*G*G*A*T*C*ACC SEQ ID NO: 756
GGTGATCCTTTTTATTTTTTGATATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 757
GGTGATCCTTTTTATTTTTTGATATATGTATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 758
GATATATCACTTTTTATTTTTTATATATATATATATTTTTTATTTT

TTATA*T*A*T*A*T*A*T*A*T*A*T*A*T*A*T*A*T*G*A*T*G*A*T*ATAC SEQ ID NO: 759
GTATATACATTTTTTATTTTTTGATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 760
GTATATACATTTTTTATTTTTTGATATATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 761
GGATATACACTTTTTTATTTTTTATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*A*TATCC SEQ ID NO: 762
GGATATACATTTTTTATTTTTTGATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 763
GGATATACATTTTTTATTTTTTGATATATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 764
GGGTATATACTTTTTTATTTTTTATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TACCC SEQ ID NO: 765
GGATATACACTTTTTTATTTTTTGATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 766
GGATATACACTTTTTTATTTTTTGATATATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 767
GGGTATATACTTTTTTATTTTTTGATATATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 768
GGGTATATACTTTTTTATTTTTTGATATATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 769
GATATATCACTTTTTTATTTTTTATATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*A*T*A*TATC SEQ ID NO: 770
GTATATACATTTTTTATTTTTTGATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 771
GTATATACATTTTTTATTTTTTGATATATGTATATTTTTTATTTT
ATAT*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 772
GGATATACACTTTTTTATTTTTTATATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*A*T*ATCC SEQ ID NO: 773
GGATATACATTTTTTATTTTTTGATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 774
GGATATACATTTTTTATTTTTTGATATATGTATATTTTTTATTTT
ATAT*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 775
GGGTATATACTTTTTTATTTTTTATATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*ACCC SEQ ID NO: 776
GGATATACACTTTTTTATTTTTTGATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 777
GGATATACACTTTTTTATTTTTTGATATATGTATATTTTTTATTTT
ATAT*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 778
GGGTATATACTTTTTTATTTTTTGATATATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 779
GGGTATATACTTTTTTATTTTTTGATATATGTATATTTTTTATTTT
ATAT*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 780
GATATATCACTTTTTTATTTTTTATATATATATATTTTTTATTTT
ATA*T*A*T*A*T*A*T*A*T*A*G*T*G*A*T*A*T*ATC SEQ ID NO: 781
GTATATACATTTTTTATTTTTTGATATATATATATTTTTTATTTT
ATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 782
GTATATACATTTTTTATTTTTTGATATATGTATATTTTTTATTTT
ATA*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 783
GTATATACATTTTTTATTTTTTGATCATGTATATTTTTTATTTTA

TAT*A*C*A*T*T*G*A*T*C*A*T*G*T*A*T*A*T*TAC SEQ ID NO: 784
GTATATACATTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTA
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 785
GGATATACACTTTTTTATTTTTTATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*G*T*G*T*A*T*A*TCC SEQ ID NO: 786
GGATATACATTTTTTATTTTTTGATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 787
GGATATACATTTTTTATTTTTTGATATATGTATATTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 788
GGATATACATTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTA
TAT*A*C*A*T*G*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 789
GGATATACATTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTA
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 790
GGGTATATACTTTTTTATTTTTTATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*G*T*A*T*A*T*A*CCC SEQ ID NO: 791
GGATATACACTTTTTTATTTTTTGATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 792
GGATATACACTTTTTTATTTTTTGATATATGTATATTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 793
GGATATACACTTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTA
TAT*A*C*A*T*G*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 794
GGATATACACTTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTA
CAT*A*T*A*T*G*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 795
GGGTATATACTTTTTTATTTTTTGATATATATATATTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 796
GGGTATATACTTTTTTATTTTTTGATATATGTATATTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 797
GGGTATATACTTTTTTATTTTTTGATCATGTATATTTTTTATTTTTTA
TAT*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 798
GGGTATATACTTTTTTATTTTTTGATCATATATGTTTTTTATTTTTTA
CAT*A*T*A*T*G*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 799
GTACATATATTTTTTTATTTTTTGATATATATATTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 800
GTACATATATTTTTTTATTTTTTGATATATGTATTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 801
GTACATATATTTTTTTATTTTTTGATCATGTATTTTTTATTTTTTAT
AC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 802
GTACATATATTTTTTTATTTTTTGATCATATATTTTTTATTTTTTAT
AT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 803
GATGTATATACTTTTTTATTTTTTATATATATATTTTTTATTTTTTA
TA*T*A*T*A*T*A*G*T*A*T*A*T*A*C*A*TC SEQ ID NO: 804
GGTACATATATTTTTTTATTTTTTGATATATATATTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 805
GGTACATATATTTTTTTATTTTTTGATATATGTATTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 806
GGTACATATATTTTTTTATTTTTTGATCATGTATTTTTTATTTTTTAT
AC*A*T*G*A*T*C*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 807
GGTACATATATTTTTTTATTTTTTGATCATATATTTTTTATTTTTTAT
AT*A*T*G*A*T*C*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 808
CGATCATATATTTTTTTATTTTTTGATATATATATTTTTTATTTTTT

ATA*T*A*T*A*T*T*C*A*A*T*A*T*T*A*T*G*A*TCG SEQ ID NO: 809
CGATCATATAATTTTTTTATTTTTTGATATATGTATTTTTATTTTTT
ACA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 810
CGATCATATAATTTTTTTATTTTTTGATCATGTATTTTTATTTTTTA
TAC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 811
CGATCATATAATTTTTTTATTTTTTGATCATATAATTTTTATTTTTTA
TAT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 812
GATACTTTTTATTTTTTTATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C SEQ ID NO: 813
GACACTTTTTATTTTTTTATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*C SEQ ID NO: 814
GATACTTTTTATTTTTTGATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 815
GATACTTTTTATTTTTTGATAAATGTATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 816
GATACTTTTTATTTTTTGATGATGTATATATATTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C SEQ ID NO: 817
GATACTTTTTATTTTTTGATGATATATGTACTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C SEQ ID NO: 818
GGATCTTTTTATTTTTTTATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*C SEQ ID NO: 819
GACACTTTTTATTTTTTGATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 820
GACACTTTTTATTTTTTGATAAATGTATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 821
GACACTTTTTATTTTTTGATGATGTATATATATTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*T*G*T*C SEQ ID NO: 822
GACACTTTTTATTTTTTGATGATATATGTACTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*T*G*T*C SEQ ID NO: 823
GGATCTTTTTATTTTTTGATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*C SEQ ID NO: 824
GGATCTTTTTATTTTTTGATAAATGTATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*C SEQ ID NO: 825
GGATCTTTTTATTTTTTGATGATGTATATATATTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*C SEQ ID NO: 826
GGATCTTTTTATTTTTTGATGATATATGTACTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*C SEQ ID NO: 827
GCGTCTTTTTATTTTTTTATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*A*G*A*C*G*C SEQ ID NO: 828
GCGTCTTTTTATTTTTTGATAAATATATATATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 829
GCGTCTTTTTATTTTTTGATAAATGTATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 830
GCGTCTTTTTATTTTTTGATGATGTATATATATTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*A*C*G*C SEQ ID NO: 831
GCGTCTTTTTATTTTTTGATGATATATGTACTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*A*C*G*C SEQ ID NO: 832
GTATACTTTTTATTTTTTGATAAATATATATATTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 833
GTATACTTTTTATTTTTTGATAAATGTATATATTTTTATTTTTTAT

A*T*A*C*A*T*A*T*A*T*T*C*G*T*A*A*T*A*C SEQ ID NO: 834
GTATACTTTTTATTTTTGATGATGTATATATTTTTTATTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*C SEQ ID NO: 835
GTATACTTTTTATTTTTGATGATATATGTACTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*C SEQ ID NO: 836
GTGATCTTTTTATTTTTGATAAATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 837
GTGATCTTTTTATTTTTGATAAATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 838
GTGATCTTTTTATTTTTGATGATGTATATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*A*C SEQ ID NO: 839
GTGATCTTTTTATTTTTGATGATATATGTACTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*A*C SEQ ID NO: 840
GGATACTTTTTATTTTTGATAAATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 841
GGATACTTTTTATTTTTGATAAATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 842
GGATACTTTTTATTTTTGATGATGTATATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C*C SEQ ID NO: 843
GGATACTTTTTATTTTTGATGATATATGTACTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C*C SEQ ID NO: 844
GCGATCTTTTTATTTTTGATAAATATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 845
GCGATCTTTTTATTTTTGATAAATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 846
GCGATCTTTTTATTTTTGATGATGTATATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*G*C SEQ ID NO: 847
GCGATCTTTTTATTTTTGATGATATATGTACTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*G*C SEQ ID NO: 848
GATATACTTTTTATTTTTTATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*C SEQ ID NO: 849
GATATATTTTTTATTTTTGATAAATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 850
GATATATTTTTTATTTTTGATGATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*G*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 851
GATATATTTTTTATTTTTGATGATATATGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 852
GTGATACTTTTTATTTTTTATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*A*C SEQ ID NO: 853
GATATACTTTTTATTTTTGATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 854
GATATACTTTTTATTTTTGATGATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 855
GATATACTTTTTATTTTTGATGATATATGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 856
GGTATACTTTTTATTTTTTATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*C*C SEQ ID NO: 857
GTGATACTTTTTATTTTTGATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 858
GTGATACTTTTTATTTTTGATAAATGTATATTTTTTATTTTTTATA

T*A*C*A*T*A*T*A*T*A*T*T*C*G*T*A*T*C*A*T*C*A*C SEQ ID NO: 859
GTGATACTTTTTATTTTTTGATGATGTATATATTTTTATTTTTTAT
A*T*A*C*A*T*G*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 860
GTGATACTTTTTATTTTTTGATGATATATGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 861
GGTATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*T*C*G*T*A*T*A*C*C SEQ ID NO: 862
GGTATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*T*C*G*T*A*T*A*C*C SEQ ID NO: 863
GGTATACTTTTTATTTTTTGATGATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 864
GGTATACTTTTTATTTTTTGATGATATATGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 865
GGTGTACTTTTTATTTTTTATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*G*T*A*C*A*C*C SEQ ID NO: 866
GGTGTACTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*T*C*G*T*A*C*A*C*C SEQ ID NO: 867
GGTGTACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*T*C*G*T*A*C*A*C*C SEQ ID NO: 868
GGTGTACTTTTTATTTTTTGATGATGTATATATTTTTTATTTTTTAT
A*T*A*C*A*T*G*A*T*C*G*T*A*C*A*C*C SEQ ID NO: 869
GGTGTACTTTTTATTTTTTGATGATATATGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*C*A*C*C SEQ ID NO: 870
GTATATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 871
GTATATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 872
GTATATACTTTTTATTTTTTGATGATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 873
GTATATACTTTTTATTTTTTGATGATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 874
GGATATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*T*C*G*T*A*T*A*T*C*C SEQ ID NO: 875
GGATATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*T*C*G*T*A*T*A*T*C*C SEQ ID NO: 876
GGATATACTTTTTATTTTTTGATGATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*C*C SEQ ID NO: 877
GGATATACTTTTTATTTTTTGATGATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*C*C SEQ ID NO: 878
GGTGATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*T*C*G*T*A*T*C*A*C*C SEQ ID NO: 879
GGTGATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*T*C*G*T*A*T*C*A*C*C SEQ ID NO: 880
GGTGATACTTTTTATTTTTTGATGATGTATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*C*A*C*C SEQ ID NO: 881
GGTGATACTTTTTATTTTTTGATGATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*C*A*C*C SEQ ID NO: 882
GGTGATCCTTTTTATTTTTTGATAAATATATATTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*T*C*G*G*A*T*C*A*C*C SEQ ID NO: 883
GGTGATCCTTTTTATTTTTTGATAAATGTATATTTTTTATTTTTTAT

A*C*A*T*A*T*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 884
GGTGATCCTTTTTATTTTTGATGATGATATATTTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 885
GGTGATCCTTTTTATTTTTGATGATATATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 886
GTATATACATTTTTTATTTTTTGATAAATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 887
GTATATACATTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 888
GTATATACATTTTTTATTTTTTGATGATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 889
GTATATACATTTTTTATTTTTTGATGATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 890
GGATATACATTTTTTATTTTTTGATAAATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 891
GGATATACATTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 892
GGATATACATTTTTTATTTTTTGATGATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 893
GGATATACATTTTTTATTTTTTGATGATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 894
GGATATACACTTTTTTATTTTTTGATAAATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 895
GGATATACACTTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 896
GGATATACACTTTTTTATTTTTTGATGATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 897
GGATATACACTTTTTTATTTTTTGATGATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 898
GGGTATATACTTTTTTATTTTTTGATAAATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*G*G SEQ ID NO: 899
GGGTATATACTTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTTAC
A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*G*G SEQ ID NO: 900
GGGTATATACTTTTTTATTTTTTGATGATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*A*T*A*T*A*C*G*G SEQ ID NO: 901
GGGTATATACTTTTTTATTTTTTGATGATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*A*T*A*T*A*C*G*G SEQ ID NO: 902
GGATGTACACTTTTTTATTTTTTGATAAATATATTTTTTATTTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 903
GGATGTACACTTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 904
GGATGTACACTTTTTTATTTTTTGATGATGTATTTTTTATTTTTTATA
C*A*T*G*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 905
GGATGTACACTTTTTTATTTTTTGATGATATATTTTTTATTTTTTATA
T*A*T*G*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 906
GTATATACTTTTTTATTTTTTTATAAATATATATATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATAC SEQ ID NO: 907
GTATATACTTTTTTATTTTTTGATAAATATATATATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 908
GTATATACTTTTTTATTTTTTGATAAATGTATATATTTTTTATTTTTT

ATAT*A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ	ID	NO:	909
GTATATACTTTTTATTTTTGATGATGTATATATATTTTTATTTTT			
TATA*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*ATAC SEQ	ID	NO:	910
GTATATACTTTTTATTTTTGATGATATATGTACTTTTTTATTTTT			
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATAC SEQ	ID	NO:	911
GGATATACTTTTTATTTTTTATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATCC SEQ	ID	NO:	912
GGATATACTTTTTATTTTTTGATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ	ID	NO:	913
GGATATACTTTTTATTTTTTGATAAATGTATATATTTTTTATTTTT			
ATAT*A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ	ID	NO:	914
GGATATACTTTTTATTTTTGATGATGTATATATATTTTTTATTTTT			
TATA*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*ATCC SEQ	ID	NO:	915
GGATATACTTTTTATTTTTGATGATATATGTACTTTTTTATTTTT			
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATCC SEQ	ID	NO:	916
GGTGATACTTTTTATTTTTTATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*CACC SEQ	ID	NO:	917
GGTGATACTTTTTATTTTTTGATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ	ID	NO:	918
GGTGATACTTTTTATTTTTTGATAAATGTATATATTTTTTATTTTT			
ATAT*A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ	ID	NO:	919
GGTGATACTTTTTATTTTTGATGATGTATATATATTTTTTATTTTT			
TATA*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*CACC SEQ	ID	NO:	920
GGTGATACTTTTTATTTTTGATGATATATGTACTTTTTTATTTTT			
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*CACC SEQ	ID	NO:	921
GGTGATCCTTTTTATTTTTTATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*G*A*T*CACC SEQ	ID	NO:	922
GGTGATCCTTTTTATTTTTTGATAAATATATATATTTTTTATTTTT			
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ	ID	NO:	923
GGTGATCCTTTTTATTTTTTGATAAATGTATATATTTTTTATTTTT			
ATAT*A*T*A*C*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ	ID	NO:	924
GGTGATCCTTTTTATTTTTGATGATGTATATATATTTTTTATTTTT			
TATA*T*A*T*A*C*A*T*G*A*T*C*G*G*A*T*CACC SEQ	ID	NO:	925
GGTGATCCTTTTTATTTTTGATGATATATGTACTTTTTTATTTTT			
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*G*A*T*CACC SEQ	ID	NO:	926
GATATATCACTTTTTATTTTTTATAAATATATATATTTTTTATTT			
TTATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*A*TATATC SEQ	ID	NO:	927
GTATATACATTTTTTATTTTTTGATAAATATATATATTTTTTATTT			
TTATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ	ID	NO:	928
GTATATACATTTTTTATTTTTTGATAAATGTATATATTTTTTATTT			
TTATAT*A*T*A*C*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ	ID	NO:	929
GTATATACATTTTTTATTTTTTGATGATGTATATATATTTTTTATTT			
TTTATA*T*A*T*A*C*A*T*G*A*T*C*A*T*G*T*ATATAC SEQ	ID	NO:	930
GTATATACATTTTTTATTTTTTGATGATATATGTACTTTTTTATTT			
TTAGTA*C*A*T*A*T*A*T*G*A*T*C*A*T*G*T*ATATAC SEQ	ID	NO:	931
GGATATACACTTTTTATTTTTTATAAATATATATATTTTTTATTT			
TTATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*ATATCC SEQ	ID	NO:	932
GGATATACATTTTTTATTTTTTGATAAATATATATATTTTTTATTT			
TTATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATCC SEQ	ID	NO:	933
GGATATACATTTTTTATTTTTTGATAAATGTATATATTTTTTATTT			

TTATAT*A*T*A*C*A*T*A*T*A*T*A*T*G*T*ATATCC SEQ ID NO: 934
GGATATACATTTTTTATTTTTTGATGATGTATATATATTTTTTATTT
TTTATA*T*A*T*A*C*A*T*G*A*T*C*A*T*G*T*ATATCC SEQ ID NO: 935
GGATATACATTTTTTATTTTTTGATGATATATGTACTTTTTTTATTT
TTAGTA*C*A*T*A*T*A*T*G*A*T*C*A*T*G*T*ATATCC SEQ ID NO: 936
GGGTATATACTTTTTATTTTTTATAAATATATATATTTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATACCC SEQ ID NO: 937
GGATATACACTTTTTATTTTTTGATAAATATATATATTTTTTTATT
TTTATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 938
GGATATACACTTTTTATTTTTTGATAAATGTATATATTTTTTTATT
TTTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 939
GGATATACACTTTTTATTTTTTGATGATGTATATATATTTTTTTATT
TTTTTATA*T*A*T*A*C*A*T*G*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 940
GGATATACACTTTTTATTTTTTGATGATATATGTACTTTTTTTATTT
TTAGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 941
GGGTATATACTTTTTATTTTTTGATAAATATATATATTTTTTTATTT
TTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 942
GGGTATATACTTTTTATTTTTTGATAAATGTATATATTTTTTTATTT
TTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 943
GGGTATATACTTTTTATTTTTTGATGATGTATATATATTTTTTTATTT
TTTTATA*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 944
GGGTATATACTTTTTATTTTTTGATGATATATGTACTTTTTTTATTT
TTAGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 945
GTATATACTTTTTATTTTTTGATAAATATATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 946
GTATATACTTTTTATTTTTTGATAAATGTATATATTTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 947
GTATATACTTTTTATTTTTTGATGATGTATATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 948
GTATATACTTTTTATTTTTTGATGATATATGTACTTTTTTATTTTTTG
TAC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 949
GGATATACTTTTTATTTTTTGATAAATATATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 950
GGATATACTTTTTATTTTTTGATAAATGTATATATTTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 951
GGATATACTTTTTATTTTTTGATGATGTATATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 952
GGATATACTTTTTATTTTTTGATGATATATGTACTTTTTTATTTTTTG
TAC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 953
GGTGATACTTTTTATTTTTTGATAAATATATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 954
GGTGATACTTTTTATTTTTTGATAAATGTATATATTTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 955
GGTGATACTTTTTATTTTTTGATGATGTATATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 956
GGTGATACTTTTTATTTTTTGATGATATATGTACTTTTTTATTTTTTG
TAC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 957
GGTGATCCTTTTTATTTTTTGATAAATATATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 958
GGTGATCCTTTTTATTTTTTGATAAATGTATATATTTTTTTATTTTTT

ATA*T*A*C*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 959
GGTGATCCTTTTTATTTTTGATGATGATATATATTTTTTATTTTTA
TAT*A*T*A*C*A*T*G*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 960
GGTGATCCTTTTTATTTTTGATGATATATGTACTTTTTATTTTTG
TAC*A*T*A*T*A*T*G*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 961
GTATATACATTTTTTATTTTTGATAAATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 962
GTATATACATTTTTTATTTTTGATGATGATATATATTTTTTATTTT
TATAT*A*T*A*C*A*T*G*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 963
GTATATACATTTTTTATTTTTGATGATATATGTACTTTTTTATTTT
TGTAC*A*T*A*T*A*T*G*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 964
GGATATACATTTTTTATTTTTGATAAATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 965
GGATATACATTTTTTATTTTTGATAAATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 966
GGATATACATTTTTTATTTTTGATGATGATATATATTTTTTATTTT
TATAT*A*T*A*C*A*T*G*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 967
GGATATACATTTTTTATTTTTGATGATATATGTACTTTTTTATTTT
TGTAC*A*T*A*T*A*T*G*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 968
GGATATACACTTTTTATTTTTGATAAATATATATATTTTTTATTT
TTTATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 969
GGATATACACTTTTTATTTTTGATAAATGTATATATTTTTTATTT
TTTATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 970
GGATATACACTTTTTATTTTTGATGATGATATATATTTTTTATTTT
TATAT*A*T*A*C*A*T*G*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 971
GGATATACACTTTTTATTTTTGATGATATATGTACTTTTTTATTTT
TGTAC*A*T*A*T*A*T*G*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 972
GGGTATATACTTTTTATTTTTGATAAATATATATATTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 973
GGGTATATACTTTTTATTTTTGATAAATGTATATATTTTTTATTTT
TTATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 974
GGGTATATACTTTTTATTTTTGATGATGATATATATTTTTTATTTT
TATAT*A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 975
GGGTATATACTTTTTATTTTTGATGATATATGTACTTTTTTATTTT
TGTAC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 976
GATATATCACTTTTTATTTTTTATAAATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*A*T*A*TATC SEQ ID NO: 977
GTATATACATTTTTTATTTTTGATAAATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 978
GTATATACATTTTTTATTTTTGATGATGATATATATTTTTTATTTT
TATA*T*A*C*A*T*G*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 979
GTATATACATTTTTTATTTTTGATGATATATGATTTTTTATTTTT
TACA*T*A*T*A*T*G*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 980
GGATATACACTTTTTATTTTTTATAAATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*T*A*T*ATCC SEQ ID NO: 981
GGATATACATTTTTTATTTTTGATAAATATATATTTTTTATTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 982
GGATATACATTTTTTATTTTTGATAAATGTATATTTTTTATTTT
ATAT*A*C*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 983
GGATATACATTTTTTATTTTTGATGATGATATATATTTTTTATTTT

TATA*T*A*C*A*T*T*G*A*T*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 984
GGATATACATTTTTTATTTTTTGATGATATATGTATTTTTATTTTT
TACA*T*A*T*A*T*G*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 985
GGGTATATACTTTTTATTTTTTATAAATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*ACCC SEQ ID NO: 986
GGATATACACTTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 987
GGATATACACTTTTTTATTTTTTGATAAATGTATATTTTTTATTTTT
ATAT*A*C*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 988
GGATATACACTTTTTTATTTTTTGATGATGTATATATTTTTTATTTTT
TATA*T*A*C*A*T*G*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 989
GGATATACACTTTTTTATTTTTTGATGATATATGTATTTTTATTTTT
TACA*T*A*T*A*T*G*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 990
GGGTATATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 991
GGGTATATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTT
ATAT*A*C*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 992
GGGTATATACTTTTTATTTTTTGATGATGTATATATTTTTTATTTTT
TATA*T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 993
GGGTATATACTTTTTATTTTTTGATGATATATGTATTTTTATTTTT
TACA*T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 994
GTATATACATTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*TAC SEQ ID NO: 995
GTATATACATTTTTTATTTTTTGATGATGTATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*TAC SEQ ID NO: 996
GTATATACATTTTTTATTTTTTGATGATATATGTTTTTTATTTTTA
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*TAC SEQ ID NO: 997
GGATATACATTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*TCC SEQ ID NO: 998
GGATATACATTTTTTATTTTTTGATAAATGTATATTTTTTATTTTT
ATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*TCC SEQ ID NO: 999
GGATATACATTTTTTATTTTTTGATGATGTATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*TCC SEQ ID NO: 1000
GGATATACATTTTTTATTTTTTGATGATATATGTTTTTTATTTTTA
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*T*TCC SEQ ID NO: 1001
GGATATACACTTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATA*T*A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*TCC SEQ ID NO: 1002
GGATATACACTTTTTTATTTTTTGATAAATGTATATTTTTTATTTTT
ATA*C*A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*TCC SEQ ID NO: 1003
GGATATACACTTTTTTATTTTTTGATGATGTATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*G*T*G*T*A*T*A*T*TCC SEQ ID NO: 1004
GGATATACACTTTTTTATTTTTTGATGATATATGTTTTTTATTTTTA
CAT*A*T*A*T*G*A*T*C*G*T*G*T*A*T*A*T*TCC SEQ ID NO: 1005
GGGTATATACTTTTTATTTTTTGATAAATATATATTTTTTATTTTT
ATA*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1006
GGGTATATACTTTTTATTTTTTGATAAATGTATATTTTTTATTTTT
ATA*C*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1007
GGGTATATACTTTTTATTTTTTGATGATGTATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1008
GGGTATATACTTTTTATTTTTTGATGATATATGTTTTTTATTTTTA

CAT*A*T*A*T*G*A*T*C*G*T*A*T*A*T*GCC SEQ ID NO: 1009
GTACATATATTTTTTATTTTTTGATAAATATATTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1010
GTACATATATTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1011
GTACATATATTTTTTATTTTTTGATGATGTATTTTTTATTTTTAT
AC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1012
GTACATATATTTTTTATTTTTTGATGATATATTTTTTATTTTTAT
AT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1013
GGTACATATATTTTTTATTTTTTGATAAATATATTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1014
GGTACATATATTTTTTATTTTTTGATAAATGTATTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1015
GGTACATATATTTTTTATTTTTTGATGATGTATTTTTTATTTTTAT
AC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1016
GGTACATATATTTTTTATTTTTTGATGATATATTTTTTATTTTTAT
AT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1017
CGATCATATATTTTTTATTTTTTGATAAATATATTTTTTATTTTTT
ATA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1018
CGATCATATATTTTTTATTTTTTGATAAATGTATTTTTTATTTTTT
ACA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1019
CGATCATATATTTTTTATTTTTTGATGATGTATTTTTTATTTTTA
TAC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1020
CGATCATATATTTTTTATTTTTTGATGATATATTTTTTATTTTTA
TAT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1021
GTATATACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTT
TATA*T*A*T*A*C*A*A*T*G*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1022
GTATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTT
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1023
GGATATACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTT
TATA*T*A*T*A*C*A*A*T*G*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1024
GGATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTT
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1025
GGTGATACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTT
TATA*T*A*T*A*C*A*A*T*G*A*T*C*G*T*A*T*CACC SEQ ID NO: 1026
GGTGATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTT
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*CACC SEQ ID NO: 1027
GGTGATCCTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTT
TATA*T*A*T*A*C*A*A*T*G*A*T*C*G*G*A*T*CACC SEQ ID NO: 1028
GGTGATCCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTT
AGTA*C*A*T*A*T*A*T*G*A*T*C*G*G*A*T*CACC SEQ ID NO: 1029
GTATATACATTTTTTATTTTTTGATGATGTAAATATATTTTTTATTT
TTTATA*T*A*T*A*C*A*A*T*G*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 1030
GTATATACATTTTTTATTTTTTGATGATATAAGTACTTTTTTATTT
TTAGTA*C*A*T*A*T*A*T*G*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 1031
GGATATACATTTTTTATTTTTTGATAAATGTAAATATTTTTTATT
TTTATAT*A*T*A*C*A*A*T*A*T*A*T*C*A*A*T*G*T*ATATCC SEQ ID NO: 1032
GGATATACATTTTTTATTTTTTGATGATGTAAATATATTTTTATT
TTTTATA*T*A*T*A*C*A*A*T*G*A*T*C*A*A*T*G*T*ATATCC SEQ ID NO: 1033
GGATATACATTTTTTATTTTTTGATGATATAAGTACTTTTTTATT

TTTAGTA*C*A*T*A*T*A*T*G*A*A*T*C*A*A*T*G*T*ATATCC SEQ ID NO: 1034
GGATATACACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATT
TTTTATA*T*A*T*A*C*A*A*T*G*A*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 1035
GGATATACACTTTTTTATTTTTTGATGATATAAGTACTTTTTTTATT
TTTAGTA*C*A*T*A*T*A*T*G*A*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 1036
GGGTATATACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATT
TTTTATA*T*A*T*A*C*A*A*T*G*A*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 1037
GGGTATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTTATT
TTTAGTA*C*A*T*A*T*A*T*G*A*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 1038
GTATATACTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*A*T*G*A*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1039
GTATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTTG
TAC*A*T*A*T*A*T*G*A*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1040
GGATATACTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*A*T*G*A*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1041
GGATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTT
GTAC*A*T*A*T*A*T*G*A*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1042
GGTGATACTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*A*T*G*A*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1043
GGTGATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTT
GTAC*A*T*A*T*A*T*G*A*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1044
GGTGATCCTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTTTA
TAT*A*T*A*C*A*A*T*G*A*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 1045
GGTGATCCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTTG
TAC*A*T*A*T*A*T*G*A*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 1046
GTATATACATTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTT
TATAT*A*T*A*C*A*A*T*G*A*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 1047
GTATATACATTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTT
TGTAC*A*T*A*T*A*T*G*A*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 1048
GGATATACATTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTT
TATAT*A*T*A*C*A*A*T*G*A*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 1049
GGATATACATTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTT
TTGTAC*A*T*A*T*A*T*G*A*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 1050
GGATATACACTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTT
TTATAT*A*T*A*C*A*A*T*G*A*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 1051
GGATATACACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTT
TTGTAC*A*T*A*T*A*T*G*A*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 1052
GGGTATATACTTTTTTATTTTTTGATGATGTAAATATTTTTTTATTTTT
TATAT*A*T*A*C*A*A*T*G*A*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 1053
GGGTATATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTT
TTGTAC*A*T*A*T*A*T*G*A*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 1054
GTATATACATTTTTTTATTTTTTGATGATATAAGTATTTTTTATTTTT
TACA*T*A*T*A*T*G*A*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 1055
GGATATACATTTTTTTATTTTTTGATAAATGAATATTTTTTTATTTTT
ATAT*A*C*A*A*T*A*T*A*T*C*A*A*T*G*T*A*T*ATCC SEQ ID NO: 1056
GGATATACATTTTTTTATTTTTTGATGATATAAGTATTTTTTATTTTT
TACA*T*A*T*A*T*G*A*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 1057
GGATATACACTTTTTTATTTTTTGATAAATGAATATTTTTTTATTTTT
TATAT*A*C*A*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 1058
GGATATACACTTTTTTATTTTTTGATGATATAAGTATTTTTTATTTTT

TTACA*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1059
GGGTATATACTTTTTATTTTTTGATAAATGAATATTTTTTATTTTT
ATAT*A*C*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 1060
GGGTATATACTTTTTATTTTTTGATGATATAAGTATTTTTTATTTTT
TACA*T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 1061
GTATATACATTTTTTATTTTTTGATGATGAATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1062
GGATATACATTTTTTATTTTTTGATAAATGAATATTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1063
GGATATACATTTTTTATTTTTTGATGATGAATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1064
GGATATACATTTTTTATTTTTTGATGATAAATGTTTTTTATTTTTA
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1065
GGATATACACTTTTTTATTTTTTGATGATGAATATTTTTTATTTTTT
ATAT*A*C*A*T*G*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 1066
GGGTATATACTTTTTATTTTTTGATGATGAATATTTTTTATTTTTA
TAT*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1067
GATACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C SEQ ID NO: 1068
GATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C SEQ ID NO: 1069
GACACTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*T*G*T*C SEQ ID NO: 1070
GACACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*T*G*T*C SEQ ID NO: 1071
GGATCTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*C SEQ ID NO: 1072
GGATCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*C SEQ ID NO: 1073
GCGTCTTTTTTATTTTTTGATGATGTAAATATATTTTTTATTTTTTAT
A*T*A*T*A*C*A*T*G*A*T*C*G*A*C*G*C SEQ ID NO: 1074
GCGTCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTAGT
A*C*A*T*A*T*A*T*G*A*T*C*G*A*C*G*C SEQ ID NO: 1075
GTATACTTTTTTATTTTTTGATGATGTAAATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*A*C SEQ ID NO: 1076
GTATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*C SEQ ID NO: 1077
GTGATCTTTTTTATTTTTTGATGATGTAAATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*A*C SEQ ID NO: 1078
GTGATCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*A*C SEQ ID NO: 1079
GGATACTTTTTTATTTTTTGATGATGTAAATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C*C SEQ ID NO: 1080
GGATACTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTGT
AC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C*C SEQ ID NO: 1081
GCGATCTTTTTTATTTTTTGATGATGTAAATATTTTTTATTTTTTATA
T*A*T*A*C*A*T*G*A*T*C*G*A*T*C*G*C SEQ ID NO: 1082
GCGATCTTTTTTATTTTTTGATGATATAAGTACTTTTTTATTTTTGTA
C*A*T*A*T*A*T*G*A*T*C*G*A*T*C*G*C SEQ ID NO: 1083
GATATATTTTTTATTTTTTGATGATATAAGTATTTTTTATTTTTTAC

A*T*A*T*A*T*G*A*T*T*C*A*T*T*A*T*T*C SEQ ID NO: 1084
GATATACTTTTTATTTTTGATGATATAAGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*T*C SEQ ID NO: 1085
GTGATACTTTTTATTTTTGATAAATGAATATTTTTATTTTTATA
T*A*C*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 1086
GTGATACTTTTTATTTTTGATGATATAAGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 1087
GGTATACTTTTTATTTTTGATAAATGAATATTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 1088
GGTATACTTTTTATTTTTGATGATATAAGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 1089
GGTGTACTTTTTATTTTTGATAAATGAATATTTTTATTTTTTATA
T*A*C*A*T*A*T*A*T*C*G*T*A*C*A*A*C*C SEQ ID NO: 1090
GGTGTACTTTTTATTTTTGATGATATAAGTATTTTTATTTTTTAC
A*T*A*T*A*T*G*A*T*C*G*T*A*C*A*A*C*C SEQ ID NO: 1091
GTATACTTTTTATTTTTGATGATGAATATTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 1092
GTATACTTTTTATTTTTGATGATAAATGTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*T*A*T*A*T*A*C SEQ ID NO: 1093
GGATACTTTTTATTTTTGATGATGAATATTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*A*T*C*C SEQ ID NO: 1094
GGTGATACTTTTTATTTTTGATGATGAATATTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*T*A*T*C*A*A*C*C SEQ ID NO: 1095
GGTGATACTTTTTATTTTTGATGATAAATGTTTTTATTTTTTAC
AT*A*T*A*T*G*A*T*C*G*T*A*T*C*A*A*C*C SEQ ID NO: 1096
GGTGATCCTTTTTATTTTTGATGATGAATATTTTTATTTTTTATA
T*A*C*A*T*G*A*T*C*G*G*A*T*C*A*A*C*C SEQ ID NO: 1097
GATACTTTTTATTTTTGATATAAATATATAATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 1098
GATACTTTTTATTTTTGATAAATGAATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 1099
GACACTTTTTATTTTTGATATAAATATATAATTTTTATTTTTTAT
AT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1100
GACACTTTTTATTTTTGATAAATGAATATATTTTTATTTTTTAT
AT*A*T*A*C*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1101
GACACTTTTTATTTTTGATATAAGTAAATATTTTTATTTTTTAT
AT*A*T*A*C*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1102
GGATCTTTTTATTTTTGATATAAATATATAATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*C SEQ ID NO: 1103
GGATCTTTTTATTTTTGATAAATGAATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*C SEQ ID NO: 1104
GGATCTTTTTATTTTTGATATAAGTAAATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*C SEQ ID NO: 1105
GCGTCTTTTTATTTTTGATATAAATATATAATTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 1106
GCGTCTTTTTATTTTTGATAAATGAATATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 1107
GCGTCTTTTTATTTTTGATATAAGTAAATATTTTTATTTTTTATA
T*A*T*A*C*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 1108
GTATATACATTTTTTATTTTTGATATAAATATATAATTTTTATTT

TTATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 1109
GTATATACATTTTTTATTTTTTGATAAATGAATATATTTTTTATTT
TTATAT*A*T*A*C*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 1110
GTATATACATTTTTTATTTTTTGATATAAGTAAATATTTTTTATTT
TTATAT*A*T*A*C*A*T*A*T*A*T*C*A*T*G*T*ATATAC SEQ ID NO: 1111
GGATATACATTTTTTATTTTTTGATATAAATATATAATTTTTTATT
TTTATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*ATATCC SEQ ID NO: 1112
GGATATACACTTTTTTATTTTTTGATATAAATATATAATTTTTTATT
TTTATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 1113
GGATATACACTTTTTTATTTTTTGATAAATGAATATATTTTTTTATT
TTTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 1114
GGATATACACTTTTTTATTTTTTGATATAAGTAAATATTTTTTTATT
TTTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*ATATCC SEQ ID NO: 1115
GGGTATATACTTTTTTATTTTTTGATATAAATATATAATTTTTTATT
TTTATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 1116
GGGTATATACTTTTTTATTTTTTGATAAATGAATATATTTTTTTATT
TTTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 1117
GGGTATATACTTTTTTATTTTTTGATATAAGTAAATATTTTTTTATT
TTTATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATACCC SEQ ID NO: 1118
GTATACTTTTTTATTTTTTATAAATATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*C SEQ ID NO: 1119
GTGATCTTTTTTATTTTTTATAAATATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*A*C SEQ ID NO: 1120
GTATACTTTTTTATTTTTTGATATAAATATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 1121
GTATACTTTTTTATTTTTTGATATATAAATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 1122
GTATACTTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C SEQ ID NO: 1123
GGATACTTTTTTATTTTTTATAAATATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*C SEQ ID NO: 1124
GTGATCTTTTTTATTTTTTGATATAAATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 1125
GTGATCTTTTTTATTTTTTGATATATAAATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 1126
GTGATCTTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*A*T*C*G*A*T*C*A*C SEQ ID NO: 1127
GGATACTTTTTTATTTTTTGATATAAATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 1128
GGATACTTTTTTATTTTTTGATATATAAATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 1129
GGATACTTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTTTAT
A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*C*C SEQ ID NO: 1130
GCGATCTTTTTTATTTTTTATAAATATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*G*C SEQ ID NO: 1131
GCGATCTTTTTTATTTTTTGATATAAATATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 1132
GCGATCTTTTTTATTTTTTGATATATAAATATTTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 1133
GCGATCTTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTTTAT

A*T*A*C*A*T*A*T*A*T*C*G*A*T*C*G*C SEQ ID NO: 1134
GATATATCACTTTTTATTTTTTATAAATATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*G*T*G*A*T*ATATC SEQ ID NO: 1135
GTATATACATTTTTTATTTTTTGATATAAATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 1136
GTATATACATTTTTTATTTTTTGATATATAAATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 1137
GGATATACACTTTTTTATTTTTTATAAATATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*G*T*G*T*A*TATCC SEQ ID NO: 1138
GGATATACATTTTTTATTTTTTGATATAAATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 1139
GGATATACATTTTTTATTTTTTGATATATAAATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 1140
GGATATACATTTTTTATTTTTTGATAAATGAATATATTTTTTATTT
TTTATA*T*A*C*A*T*A*T*A*T*C*A*T*G*T*A*TATCC SEQ ID NO: 1141
GGGTATATACTTTTTTATTTTTTATAAATATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TACCC SEQ ID NO: 1142
GGATATACACTTTTTTATTTTTTGATATAAATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 1143
GGATATACACTTTTTTATTTTTTGATATATAAATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 1144
GGATATACACTTTTTTATTTTTTGATAAATGAATATATTTTTTATTT
TTTATA*T*A*C*A*T*A*T*A*T*C*G*T*G*T*A*TATCC SEQ ID NO: 1145
GGGTATATACTTTTTTATTTTTTGATATAAATATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 1146
GGGTATATACTTTTTTATTTTTTGATATATAAATATTTTTTTTATTTT
TTATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 1147
GGGTATATACTTTTTTATTTTTTGATAAATGAATATATTTTTTATTT
TTTATA*T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*TACCC SEQ ID NO: 1148
GATATACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 1149
GTGATACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 1150
GGTATACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 1151
GGTGTACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTTATA
T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*C*A*C*C SEQ ID NO: 1152
GTATATACATTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATAC SEQ ID NO: 1153
GGATATACATTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*ATCC SEQ ID NO: 1154
GGATATACACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTT
TATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*A*T*ATCC SEQ ID NO: 1155
GGGTATATACTTTTTTATTTTTTGATAAATATATAATTTTTTATTTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*ACCC SEQ ID NO: 1156
GTATATACTTTTTTATTTTTTGATAAATATATTTTTTTTATTTTTTTAT
A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*T*A*C SEQ ID NO: 1157
GTATATACTTTTTTATTTTTTGATAAATGTATTTTTTTTATTTTTTTAT
A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*T*A*C SEQ ID NO: 1158
GGATATACTTTTTTATTTTTTGATAAATATATTTTTTTTATTTTTTTAT

A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C*G SEQ ID NO: 1159
GGATATACTTTTTATTTTTGATAAATGTATTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*C*G*T*A*T*A*T*C*G SEQ ID NO: 1160
GGATATACTTTTTATTTTTGATGATAAATGTTTTTTATTTTTTAC
AT*A*T*A*T*G*A*T*C*G*T*A*T*A*T*C*G SEQ ID NO: 1161
GGTGATACTTTTTATTTTTGATAAATATATTTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C*G SEQ ID NO: 1162
GGTGATACTTTTTATTTTTGATAAATGTATTTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*C*G*T*A*T*C*A*C*G SEQ ID NO: 1163
GGTGATCCTTTTTATTTTTGATAAATATATTTTTTTATTTTTTAT
A*T*A*T*A*T*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 1164
GGTGATCCTTTTTATTTTTGATAAATGTATTTTTTTATTTTTTAT
A*C*A*T*A*T*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 1165
GGTGATCCTTTTTATTTTTGATGATAAATGTTTTTTATTTTTTACA
T*A*T*A*T*G*A*T*C*G*G*A*T*C*A*C*G SEQ ID NO: 1166
GTATATACATTTTTTATTTTTGATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1167
GTATATACATTTTTTATTTTTGATAAATGTATTTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1168
GTATATACATTTTTTATTTTTGATGATAAATGTTTTTTATTTTTT
CAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1169
GGATATACATTTTTTATTTTTGATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1170
GGATATACACTTTTTATTTTTGATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 1171
GGATATACACTTTTTATTTTTGATAAATGTATTTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 1172
GGATATACACTTTTTATTTTTGATGATAAATGTTTTTTATTTTTT
ACAT*A*T*A*T*G*A*T*C*G*T*G*T*A*T*A*TCC SEQ ID NO: 1173
GGGTATATACTTTTTATTTTTGATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1174
GGGTATATACTTTTTATTTTTGATAAATGTATTTTTTTATTTTTT
ATA*C*A*T*A*T*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1175
GGGTATATACTTTTTATTTTTGATGATAAATGTTTTTTATTTTTT
CAT*A*T*A*T*G*A*T*C*G*T*A*T*A*T*A*CCC SEQ ID NO: 1176
GTATATACATTTTTTATTTTTGATAAATATTTTTTTATTTTTTAT
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 1177
GTATATACATTTTTTATTTTTGATAAATGTTTTTTTATTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 1178
GGATATACATTTTTTATTTTTGATAAATATTTTTTTATTTTTTAT
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 1179
GGATATACATTTTTTATTTTTGATAAATGTTTTTTTATTTTTTAC
A*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 1180
GGATATACATTTTTTATTTTTGATGATGAATTTTTTATTTTTTATA
C*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*G SEQ ID NO: 1181
GGATATACACTTTTTATTTTTGATAAATATTTTTTTATTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 1182
GGATATACACTTTTTATTTTTGATAAATGTTTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*T*A*T*C*G SEQ ID NO: 1183
GGGTATATACTTTTTATTTTTGATAAATATTTTTTTATTTTTTAT

A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*C*G SEQ ID NO: 1184
GGGTATATACTTTTTATTTTTTGATAAATGTTTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*A*T*A*T*A*C*C*G SEQ ID NO: 1185
GGATGTACACTTTTTATTTTTTGATAAATATTTTTTTATTTTTTAT
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 1186
GGATGTACACTTTTTATTTTTTGATAAATGTTTTTTTATTTTTTAC
A*T*A*T*A*T*C*G*T*G*T*A*C*A*T*C*G SEQ ID NO: 1187
GTACATATATTTTTTTATTTTTTGATAAATATTTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1188
GTACATATATTTTTTTATTTTTTGATAAATGTTTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1189
GGTACATATATTTTTTTATTTTTTGATAAATATTTTTTTATTTTTTA
TA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1190
GGTACATATATTTTTTTATTTTTTGATAAATGTTTTTTTATTTTTTA
CA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*A*CC SEQ ID NO: 1191
CGATCATATATTTTTTTATTTTTTGATAAATATTTTTTTATTTTTT
ATA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1192
CGATCATATATTTTTTTATTTTTTGATAAATGTTTTTTTATTTTTT
ACA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1193
CGATCATATATTTTTTTATTTTTTGATGATGAATTTTTTATTTTTTA
TAC*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1194
CGATCATATATTTTTTTATTTTTTGATGATAAATTTTTTATTTTTTA
TAT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1195
GTATATACTTTTTATTTTTTGATATAAATATATAATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1196
GTATATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1197
GGATATACTTTTTATTTTTTGATATAAATATATAATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1198
GGATATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1199
GGTGATACTTTTTATTTTTTGATATAAATATATAATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ ID NO: 1200
GGTGATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ ID NO: 1201
GGTGATCCTTTTTATTTTTTGATATAAATATATAATTTTTTATTTTT
ATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ ID NO: 1202
GGTGATCCTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ ID NO: 1203
GTATATACTTTTTATTTTTTGATATAAGTAAATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1204
GGATATACTTTTTATTTTTTGATATAAGTAAATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1205
GGTGATACTTTTTATTTTTTGATATAAGTAAATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*CACC SEQ ID NO: 1206
GGTGATCCTTTTTATTTTTTGATATAAGTAAATATTTTTTATTTTT
ATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*G*A*T*CACC SEQ ID NO: 1207
GTATATACTTTTTATTTTTTATAAATATATAATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TAC SEQ ID NO: 1208
GTATATACTTTTTATTTTTTGATATAAATATATTTTTTTATTTTTT

ATA*T*A*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*TAC SEQ ID NO: 1209
GTATATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1210
GGATATACTTTTTATTTTTTATAAATATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TCC SEQ ID NO: 1211
GGATATACTTTTTATTTTTTGATATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1212
GGATATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1213
GGTGATACTTTTTATTTTTTATAAATATATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C*ACC SEQ ID NO: 1214
GGTGATACTTTTTATTTTTTGATATAAATATATTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1215
GGTGATACTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1216
GGTGATCCTTTTTATTTTTTATAAATATATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*A*G*G*A*T*C*ACC SEQ ID NO: 1217
GGTGATCCTTTTTATTTTTTGATATAAATATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 1218
GGTGATCCTTTTTATTTTTTGATAAATGAATATATTTTTTATTTTTT
ATA*T*A*C*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 1219
GTATATACTTTTTATTTTTTGATATATAAATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1220
GGATATACTTTTTATTTTTTGATATATAAATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1221
GGTGATACTTTTTATTTTTTGATATATAAATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1222
GGTGATCCTTTTTATTTTTTGATATATAAATATTTTTTTTATTTTTT
ATA*T*A*T*A*T*A*T*A*T*C*G*G*A*T*C*ACC SEQ ID NO: 1223
GATACAAAAAAAAAAAAATATATATATATATAAAAAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C SEQ ID NO: 1224
GACACAAAAAAAAAAAAAGATATATATATATAAAAAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1225
GATATACAAAAAAAAAAAAATATATATATATATAAAAAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*C SEQ ID NO: 1226
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ATAT*A*C*A*T*A*T*A*T*C*A*T*A*T*A*T*C SEQ ID NO: 1227
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ATAT*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 1228
GGTATACAAAAAAAAAAAAATATATATATATATAAAAAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*G*T*A*T*A*C*C SEQ ID NO: 1229
GATATATCACAAAAAAAAAAAAATATATATATAAAAAAAAAAAAAA
TATA*T*A*T*A*T*A*G*T*G*A*T*A*T*A*T*C SEQ ID NO: 1230
GTATATACATAAAAAAAAAAAAAAGATATATGTAAAAAAAAAAAAA
TACA*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*T*C SEQ ID NO: 1231
GGATATACATAAAAAAAAAAAAAAGATATATGTAAAAAAAAAAAAA
ATACA*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 1232
GGATATACATAAAAAAAAAAAAAAGATCATGTATAAAAAAAAAAAAAA
ATAC*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 1233
GGGTATATACAAAAAAAAAAAAATATATATATAAAAAAAAAAAAAA

TATA*T*A*T*A*T*A*G*T*A*T*A*T*A*C*C*SEQ ID NO: 1234
GTATATACAAAAAAAAAATATATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*G*T*A*T*ATAC SEQ ID NO: 1235
GTATATACAAAAAAAAAAGATATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1236
GGATATACAAAAAAAAAATATATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATCC SEQ ID NO: 1237
GGATATACAAAAAAAAAAGATATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1238
GTATATACAAAAAAAAAATATATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TAC SEQ ID NO: 1239
GTATATACAAAAAAAAAAGATATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1240
GGATATACAAAAAAAAAATATATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TCC SEQ ID NO: 1241
GGATATACAAAAAAAAAAGATATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1242
GATATATCACAAAAAAAAAATATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*G*T*G*A*T*A*T*ATC SEQ ID NO: 1243
GGATATACATAAAAAAAAAAAGATATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1244
GTACATATATTAAAAAAAAAAGATATATATAAAAAAAAAA
ATATA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1245
GATGTATATACAAAAAAAAAATATATATATAAAAAAAAAA
ATATA*T*A*T*A*T*A*G*T*A*T*A*T*A*C*A*TC SEQ ID NO: 1246
CGATCATATATTAAAAAAAAAAGATATATATAAAAAAAAAA
AATATA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1247
CGATCATATATTAAAAAAAAAAGATATATGTAAAAAAAAA
AATACA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*A*TCG SEQ ID NO: 1248
GATACAAAAAAAAAATATAAATATATATATAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*C SEQ ID NO: 1249
GGATCAAAAAAAAAAATATAAATATATATATAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*T*A*G*A*T*C*C SEQ ID NO: 1250
GACACAAAAAAAAAAGATAAATATATATATAAAAAAAAAA
AATAT*A*T*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1251
GACACAAAAAAAAAAGATGATGTATATATAAAAAAAAAA
ATATA*T*A*T*A*C*A*T*G*A*T*C*G*T*G*T*C SEQ ID NO: 1252
GCGTCAAAAAAAAAAAGATAAATATATATATAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*T*C*G*A*C*G*C SEQ ID NO: 1253
GATATACAAAAAAAAAATATAAATATATATATAAAAAAAAAA
ATAT*A*T*A*T*A*T*A*T*A*G*T*A*T*A*T*C SEQ ID NO: 1254
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ATACA*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 1255
GTATATACATAAAAAAAAAAAGATGATATATAAAAAAAAAA
ATAT*A*T*G*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 1256
GGATATACATAAAAAAAAAAAGATAAATATAAAAAAAAAA
ATATA*T*A*T*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 1257
GGATATACATAAAAAAAAAAAGATGATATATAAAAAAAAAA
AATAT*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 1258
GTATATACAAAAAAAAAATATAAATATATATATAAAAAAAAAA

AAATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*T*A*T*ATAC SEQ ID NO: 1259
GTATATACAAAAAAAAAAAAAGATAAATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1260
GGATATACAAAAAAAAAAAAATATAAATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*A*T*A*G*T*A*T*ATCC SEQ ID NO: 1261
GGATATACAAAAAAAAAAAAAGATAAATATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1262
GTATATACAAAAAAAAAAAAAGATAAATGTATATAAAAAAAAAA
AATATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1263
GGATATACAAAAAAAAAAAAAGATAAATGTATATAAAAAAAAAA
AAATATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1264
GGTGATACAAAAAAAAAAAAAGATGATGTATATATAAAAAAAAAA
AAATAT*A*T*A*C*A*T*G*A*T*C*G*T*A*T*C*ACC SEQ ID NO: 1265
GATATATCACAAAAAAAAAAAAATATAAATATATATAAAAAAAAAA
AAAATAT*A*T*A*T*A*T*A*T*A*T*A*G*T*G*A*T*A*TATC SEQ ID NO: 1266
GTATATACATAAAAAAAAAAAAAAGATAAATATATAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1267
GTATATACATAAAAAAAAAAAAAAGATGATATATGTAAAAAAAAA
AAACAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1268
GGATATACATAAAAAAAAAAAAAAGATAAATATATAAAAAAAAAA
AAATATA*T*A*T*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1269
GTACATATATTAAAAAAAAAAAAAGATAAATATAAAAAAAAAA
ATATA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1270
GTACATATATTAAAAAAAAAAAAAGATAAATGTAAAAAAAAA
ATACA*T*A*T*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1271
GTACATATATTAAAAAAAAAAAAAGATGATATATAAAAAAAAAA
AATAT*A*T*G*A*T*C*A*A*T*A*T*A*T*G*T*AC SEQ ID NO: 1272
GGATATACATAAAAAAAAAAAAAAGATGATGAATAAAAAAAAAA
AATAC*A*T*G*A*T*C*A*T*G*T*A*T*A*T*C*C SEQ ID NO: 1273
GTATATACATAAAAAAAAAAAAAAGATAAATGTTAAAAAAAAA
TACA*T*A*T*A*T*C*A*T*G*T*A*T*A*T*A*C SEQ ID NO: 1274
GATACAAAAAAAAAAAAAGATATAAATATATAAAAAAAAAA
AATAT*A*T*A*T*A*T*A*T*A*T*A*T*C*G*T*A*T*C SEQ ID NO: 1275
GATACAAAAAAAAAAAAAGATGATATAAGTACTAAAAAAAAA
AAGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*C SEQ ID NO: 1276
GACACAAAAAAAAAAAAAGATAAATGAATATATAAAAAAAAAA
AATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*G*T*C SEQ ID NO: 1277
GGATATACAAAAAAAAAAAAAGATATAAGTAAATATAAAAAA
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GGATATACAAAAAAAAAAAAAGATGATATAAGTACTAAAAAAAAA
AAAAGTA*C*A*T*A*T*A*T*G*A*T*C*G*T*A*T*ATCC SEQ ID NO: 1279
GTATATACAAAAAAAAAAAAAGATATAAGTAAATATAAAAAA
AAAATAT*A*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*ATAC SEQ ID NO: 1280
GTATATACAAAAAAAAAAAAAGATGATATAAGTACTAAAAAAAAA
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GGATATACAAAAAAAAAAAAAGATAAATGAATATAAAAAA
AAATATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TCC SEQ ID NO: 1282
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AAATATA*T*A*C*A*T*A*T*A*T*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1283
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AGTAC*A*T*A*T*A*T*G*A*T*C*G*T*A*T*A*TAC SEQ ID NO: 1284
GTATATACAAAAAAAAAATATAAATATATATTAAAAAAAAAA
AATATA*T*A*T*A*T*A*T*A*T*A*G*T*A*T*A*TAC SEQ ID NO: 1285
GTATATACATAAAAAAAAAAAGATGATGTAAATATAAAAAAAAA
AAAAATAT*A*T*A*C*A*T*G*A*T*C*A*T*G*T*A*TATAC SEQ ID NO: 1286
GATATACAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
AATAT*A*T*A*T*A*T*A*T*C*G*T*A*T*A*T*C SEQ ID NO: 1287
GTGATACAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
AATAT*A*T*A*T*A*T*A*T*C*G*T*A*T*C*A*C SEQ ID NO: 1288
GGTATACAAAAAAAAAAGATAAATATATAAAAAAAAAAAAA
AATAT*A*T*A*T*A*T*A*T*C*G*T*A*T*A*C*C SEQ ID NO: 1289
GGATATACATAAAAAAAAAAAGATAAATGAATAAAAAAAAA
AAATATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TCC SEQ ID NO: 1290
GTATATACATAAAAAAAAAAAGATAAATGTATTAAAAAAAAA
AATATA*C*A*T*A*T*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1291
GTATATACATAAAAAAAAAAAGATGATAAATGTAAAAAAAA
AAACAT*A*T*A*T*G*A*T*C*A*T*G*T*A*T*A*TAC SEQ ID NO: 1292
TATATATTATTTTATTTTAATCGAGTCTTTTTGACTCGATATAC AATATATA SEQ ID
NO: 1293 GATATATTATTTTATTTTAATCGAGTCTTTTTGACTCGATATAC AATATATC
SEQ ID NO: 1294 GATATATCATTTTATTTTAATCGAGTCTTTTTGACTCGATATAC
GATATATC SEQ ID NO: 1295
GATATGTCATTTTATTTTAATCGAGTCTTTTTGACTCGATATAC GACATATC SEQ ID
NO: 1296 GTGATGTCATTTTATTTTAATCGAGTCTTTTTGACTCGATATAC GACATCAC
SEQ ID NO: 1297 TATATATTATTTTATTTTATGCGAGTCTTTTTGACTCGCAGCCC
AATATATA SEQ ID NO: 1298
GATATATTATTTTATTTTATGCGAGTCTTTTTGACTCGCAGCCC AATATATC SEQ ID
NO: 1299 GATATATCATTTTATTTTATGCGAGTCTTTTTGACTCGCAGCCC GATATATC
SEQ ID NO: 1300 GATATGTCATTTTATTTTATGCGAGTCTTTTTGACTCGCAGCCC
GACATATC SEQ ID NO: 1301
GTGATGTCATTTTATTTTATGCGAGTCTTTTTGACTCGCAGCCC GACATCAC SEQ ID
NO: 1302 TATATATTATTTTATTTTAGTATATCGGACTCGATATACAATAT ATA SEQ ID
NO: 1303 GATATATTATTTTATTTTAGTATATCGGACTCGATATACAATAT ATC SEQ ID
NO: 1304 GATATATCATTTTATTTTAGTATATCGGACTCGATATACGATAT ATC SEQ ID
NO: 1305 GATATGTCATTTTATTTTAGTATATCGGACTCGATATACGACAT ATC SEQ
ID NO: 1306 GTGATGTCATTTTATTTTAGTATATCGGACTCGATATACGACAT CAC
SEQ ID NO: 1307 TATATATTATTTTATTTTAGGGCTGCGGACTCGCAGCCCAATAT
ATA SEQ ID NO: 1308 GATATATTATTTTATTTTAGGGCTGCGGACTCGCAGCCCAATA
TATC SEQ ID NO: 1309
GATATATCATTTTATTTTAGGGCTGCGGACTCGCAGCCCGATA TATC SEQ ID NO:
1310 GATATGTCATTTTATTTTAGGGCTGCGGACTCGCAGCCCGACA TATC SEQ ID
NO: 1311 GTGATGTCATTTTATTTTAGGGCTGCGGACTCGCAGCCCGACA TCAC SEQ
ID NO: 1312 TATATATATTATTACTATATGGACTCGCATATAGATATATA SEQ ID
NO: 1313 GATATATATTATTACTATATGGACTCGCATATAGATATATC SEQ ID NO:
1314 GATATACATTATTACTATATGGACTCGCATATAGGTATATC SEQ ID NO: 1315
GATATCCATTATTACTATATGGACTCGCATATAGGGATATC SEQ ID NO: 1316
GTGATACATTATTACTATATGGACTCGCATATAGGTATCAC SEQ ID NO: 1317
TATATATTTTATTTTCGGGCTGGACTCGCAGCCCGATATATA SEQ ID NO: 1318
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GATATACATTATTACGGGCTGGACTCGCAGCCCGGTATATC SEQ ID NO: 1320
GATATCCATTATTACGGGCTGGACTCGCAGCCCGGGATATC SEQ ID NO: 1321

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TATATATTTTATTTCTATATGTTTATTTTCGAGTCTTTTGACTCGC ATATAGATATATA SEQ
ID NO: 1323 GATATATATTATTACTATATGATTATTACGAGTCTTTTGACTCG
CATATAGATATATC SEQ ID NO: 1324
GATATACATTATTACTATATGATTATTACGAGTCTTTTGACTCG CATATAGGTATATC SEQ
ID NO: 1325 GATATCCATTATTACTATATGATTATTACGAGTCTTTTGACTCG
CATATAGGGATATC SEQ ID NO: 1326
GTGATACATTATTACTATATGATTATTACGAGTCTTTTGACTCG CATATAGGTATCAC
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CAGCCCGATATATA SEQ ID NO: 1328
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SEQ ID NO: 1329 GATATACATTATTACGGGCTGATTATTACGAGTCTTTTGACTC
GCAGCCCGGTATATC SEQ ID NO: 1330
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GCAGCCCGGTATCAC SEQ ID NO: 1332
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GTGTGCATTATTACCGGGCGACTCGCAGCCCGGGCACAC SEQ ID NO: 1339
TATATATTTTATTTTCATATCTTTATTTTGCGAGTCTTTTGACTCGC AGATATGTATATA SEQ
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CAGATATGGATATC SEQ ID NO: 1341
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GACTCGCAGCCCGGGATATCGCGCGCGCAATAAGCGCGCATT
ATTAGCGATATTATTAATCCCATTATTAGGGCTATTATTAGCG AG SEQ ID NO: 1420
GACTCGCAGCCCGGGCATAACGCGCGCGCAATAAGCGCGCATT
ATTAGCGTAATTATTATGCCCATTATTAGGGCTATTATTAGCG AG SEQ ID NO: 1421
GACTCGCAGCCCGGGCACACGCGCGCGCAATAAGCGCGCATT
ATTAGCGTGATTATTATGCCCATTATTAGGGCTATTATTAGCG AG *indicate the bonds that are phosphorothioate (PS) modified. These sequences may include nuclease resistant modifications such as PS modifications in all bases except the Loop sequences, where Loop sequences are the unhybridized bases. The number of modifications, e.g., PS, can vary from “0” to “max = total number of bases – number of bases in loops.

(109) In any of the foregoing embodiments, the blocked nucleic acid molecules of the disclosure may further contain a reporter moiety attached thereto such that cleavage of the blocked nucleic acid releases a signal from the reporter moiety. (See FIG. 4, mechanisms depicted at center and bottom.)

(110) Also, in any of the foregoing embodiments, the blocked nucleic acid molecule may be a modified or non-naturally occurring nucleic acid molecule. In some embodiments, the blocked nucleic acid molecules of the disclosure may further contain a locked nucleic acid (LNA), a bridged nucleic acid (BNA), and/or a peptide nucleic acid (PNA). The blocked nucleic acid molecule may contain a modified or non-naturally occurring nucleoside, nucleotide, and/or internucleoside linkage, such as a 2'-O-methyl (2'-O-Me) modified nucleoside, a 2'-fluoro (2'-F) modified nucleoside, and a phosphorothioate (PS) bond, any other nucleic acid molecule modifications described above, and any combination thereof.

(111) FIG. 2G at left shows an exemplary single-strand blocked nucleic acid molecule and how the configuration of this blocked nucleic acid molecule is able to block R-loop formation with an RNP complex, thereby blocking activation of the trans-cleavage activity of RNP2. The single-strand

blocked nucleic acid molecule is self-hybridized and comprises: a target strand (TS) sequence complementary to the gRNA (e.g., crRNA) of RNP2; a cleavable non-target strand (NTS) sequence that is partially hybridized (e.g., it contains secondary loop structures) to the TS sequence; and a protospacer adjacent motif (PAM) sequence (e.g., 5' NAAA 3') that is specifically located at the 3' end of the TS sequence. An RNP complex with 3'.fwdarw.5' diffusion (e.g., 1D diffusion) initiates R-loop formation upon PAM recognition. R-loop formation is completed upon a stabilizing ≥ 17 base hybridization of the TS to the gRNA of RNP2; however, because of the orientation of the PAM sequence relative to the secondary loop structure(s), the blocked nucleic acid molecule sterically prevents the TS sequence from hybridizing with the gRNA of RNP2, thereby blocking the stable R-loop formation required for the cascade reaction.

(112) FIG. 2G at right shows the blocked nucleic acid molecule being unblocked via trans-cleavage (e.g., by RNP1) and subsequent dehybridization of the NTS's secondary loop structures, followed by binding of the TS sequence to the gRNA of RNP2, thereby completing stable R-loop formation and activating the trans-cleavage activity of the RNP2 complex.

(113) In some embodiments, the blocked nucleic acid molecules provided herein are circular DNAs, RNAs or chimeric (DNA-RNA) molecules (FIG. 2H), and the blocked nucleic acid molecules may include different base compositions depending on the Cas enzyme used for RNP1 and RNP2. For the circular design of blocked nucleic acid molecules, the 5' and 3' ends are covalently linked together. This configuration makes internalization of the blocked nucleic acid molecule into RNP2—and subsequent RNP2 activation—sterically unfavorable, thereby blocking the progression of a CRISPR Cascade reaction. Thus, RNP2 activation (e.g., trans-cleavage activity) happens after cleavage of a portion of the blocked nucleic acid molecule followed by linearization and internalization of unblocked nucleic acid molecule into RNP2.

(114) In some embodiments, the blocked nucleic acid molecules are topologically circular molecules with 5' and 3' portions hybridized to each other using DNA, RNA, LNA, BNA, or PNA bases which have a very high melting temperature (T_m). The high T_m causes the structure to effectively behave as a circular molecule even though the 5' and 3' ends are not covalently linked. The 5' and 3' ends can also have base non-naturally occurring modifications such as phosphorothioate bonds to provide increased stability.

(115) In embodiments where the blocked nucleic acid molecules are circularized (e.g., circular or topologically circular), as illustrated in FIG. 2H, each blocked nucleic acid molecule includes a first region, which is a target sequence specific to the gRNA of RNP2, and a second region, which is a sequence that can be cleaved by nuclease enzymes of activated RNP1 and/or RNP2. The first region may include a nuclease-resistant nucleic acid sequence such as, for example, a phosphorothioate group or other non-naturally occurring nuclease-resistant base modifications, for protection from trans-endonuclease activity. In some embodiments, when the Cas enzyme in both RNP1 and RNP2 is Cas12a, the first region of the blocked nucleic acid molecule includes a nuclease-resistant DNA sequence, and the second region of the blocked nucleic acid molecule includes a cleavable DNA sequence. In other embodiments, when the Cas enzyme in RNP1 is Cas12a and the Cas enzyme in RNP2 is Cas13a, the first region of the blocked nucleic acid molecule includes a nuclease-resistant RNA sequence, and the second region of the blocked nucleic acid molecule includes a cleavable DNA sequence and a cleavable RNA sequence. In yet other embodiments, when the Cas enzyme in RNP1 is Cas13a and the Cas enzyme in RNP2 is Cas12a, the first region of the blocked nucleic acid molecule includes a nuclease-resistant DNA sequence, and the second region of the blocked nucleic acid molecule includes a cleavable DNA sequence and a cleavable RNA sequence. In some other embodiments, when the Cas enzyme in both RNP1 and RNP2 is Cas13a, the first region of the blocked nucleic acid molecule includes a nuclease-resistant RNA sequence, and the second region of the blocked nucleic acid molecule includes a cleavable RNA sequence.

(116) The Cascade Assay Employing Blocked Primer Molecules

(117) The blocked nucleic acids described above may also be blocked primer molecules. Blocked primer molecules include a sequence complementary to a primer binding domain (PBD) on a template molecule (see description below in reference to FIGS. 3A and 3B) and can have the same general structures as the blocked nucleic acid molecules described above. A PBD serves as a nucleotide sequence for primer hybridization followed by primer polymerization by a polymerase. In any of Formulas I, II, or III described above, the blocked primer nucleic acid molecule may include a sequence complementary to the PBD on the 5' end of T. The unblocked primer nucleic acid molecule can bind to a template molecule at the PBD and copy the template molecule via polymerization by a polymerase.

(118) Other specific embodiments of the cascade assay that utilize blocked primer molecules and are depicted in FIGS. 3A and 3B. In the embodiments using blocked nucleic acid molecules described above, activation of RNP1 and trans-cleavage of the blocked nucleic acid molecules were used to activate RNP2—that is, the unblocked nucleic acid molecules are a target sequence for the gRNA in RNP2. In contrast, in the embodiments using blocked primers, activation of RNP1 and trans-cleavage unblocks a blocked primer molecule that is then used to prime a template molecule for extension by a polymerase, thereby synthesizing activating molecules that are the target sequence for the gRNA in RNP2.

(119) FIG. 3A is a diagram showing the sequence of steps in an exemplary cascade assay involving circular blocked primer molecules and linear template molecules. At left of FIG. 3A is a cascade assay reaction mix comprising 1) RNP1s (301) (only one RNP1 is shown); 2) RNP2s (302); 3) linear template molecules (330) (which is the non-target strand); 4) a circular blocked primer molecule (334) (i.e., a high K_{sub.d} molecule); and 5) a polymerase (338), such as a Φ 29 polymerase. The linear template molecule (330) (non-target strand) comprises a PAM sequence (331), a primer binding domain (PBD) (332) and, optionally, a nucleoside modification (333) to protect the linear template molecule (330) from 3'.fwdarw.5' exonuclease activity. Blocked primer molecule (334) comprises a cleavable region (335) and a complement to the PBD (332) on the linear template molecule (330).

(120) Upon addition of a sample comprising a target nucleic acid of interest (304) (capable of complexing with the gRNA in RNP1 (301)), the target nucleic acid of interest (304) combines with and activates RNP1 (305) but does not interact with or activate RNP2 (302). Once activated, RNP1 cuts the target nucleic acid of interest (304) via sequence specific cis-cleavage, which activates non-specific trans-cleavage of other nucleic acids present in the reaction mix, including at least one of the blocked primer molecules (334). The circular blocked primer molecule (334) (i.e., a high K_{sub.d} molecule, where high K_{sub.d} relates to binding to RNP2) upon cleavage becomes an unblocked linear primer molecule (344) (a low K_{sub.d} molecule, where low K_{sub.d} related to binding to RNP2), which has a region (336) complementary to the PBD (332) on the linear template molecule (330) and can bind to the linear template molecule (330).

(121) Once the unblocked linear primer molecule (344) and the linear template molecule (330) are hybridized (i.e., hybridized at the PBD (332) of the linear template molecule (330) and the PBD complement (336) on the unblocked linear primer molecule (344)), 3'.fwdarw.5' exonuclease activity of the polymerase (338) removes the unhybridized single-stranded DNA at the end of the unblocked primer molecule (344) and the polymerase (338) can copy the linear template molecule (330) to produce a synthesized activating molecule (346) (a complement of the non-target strand, which is a target strand). The synthesized activating molecule (346) is capable of activating RNP2 (302.fwdarw.308). As described above, because the nucleic acid-guided nuclease in the RNP2 (308) complex exhibits (that is, possesses) both cis- and trans-cleavage activity, more blocked primer molecules (334) become unblocked primer molecules (344) triggering activation of more RNP2s (308) and more trans-cleavage activity in a cascade. As stated above in relation to blocked and unblocked nucleic acid molecules (both linear and circular), the unblocked primer molecule has a higher binding affinity for the gRNA in RNP2 than does the blocked primer molecule,

although there may be some “leakiness” where some blocked primer molecules are able to interact with the gRNA in RNP2. However, an unblocked primer molecule has a substantially higher likelihood than a blocked primer molecule to hybridize with the gRNA of RNP2.

(122) FIG. 3A at bottom depicts the concurrent activation of reporter moieties. Intact reporter moieties (309) comprise a quencher (310) and a fluorophore (311). As described above in relation to FIG. 1B, the reporter moieties are also subject to trans-cleavage by activated RNP1 (305) and RNP2 (308). The intact reporter moieties (309) become activated reporter moieties (312) when the quencher (310) is separated from the fluorophore (311), and the fluorophore emits a fluorescent signal (313). Signal strength increases rapidly as more blocked primer molecules (334) become unblocked primer molecules (344) generating synthesized activating molecules (346) and triggering activation of more RNP2 (308) complexes and more trans-cleavage activity of the reporter moieties (309). Again, here the reporter moieties are shown as separate molecules from the blocked nucleic acid molecules, but other configurations may be employed and are discussed in relation to FIG. 4. Also, as with the cascade assay embodiment utilizing blocked nucleic acid molecules that are not blocked primers, with the exception of the gRNA in RNP1, the cascade assay components stay the same no matter what target nucleic acid(s) of interest are being detected.

(123) FIG. 3B is a diagram showing the sequence of steps in an exemplary cascade assay involving blocked primer molecules and circular template molecules. The cascade assay of FIG. 3B differs from that depicted in FIG. 3A by the configuration of the template molecule. Where the template molecule in FIG. 3A was linear, in FIG. 3B the template molecule is circular. At left of FIG. 3B is a cascade assay reaction mix comprising 1) RNP1s (301) (only one RNP1 is shown); 2) RNP2s (302); 3) a circular template molecule (352) (non-target strand); 4) a circular blocked primer molecule (334); and 5) a polymerase (338), such as a Φ 29 polymerase. The circular template molecule (352) (non-target strand) comprises a PAM sequence (331) and a primer binding domain (PBD) (332). Blocked primer molecule (334) comprises a cleavable region (335) and a complement to the PBD (332) on the circular template molecule (352).

(124) Upon addition of a sample comprising a target nucleic acid of interest (304) (capable of complexing with the gRNA in RNP1 (301)), the target nucleic acid of interest (304) combines with and activates RNP1 (305) but does not interact with or activate RNP2 (302). Once activated, RNP1 cuts the target nucleic acid of interest (304) via sequence specific cis-cleavage, which activates non-specific trans-cleavage of other nucleic acids present in the reaction mix, including at least one of the blocked primer molecules (334). The circular blocked primer molecule (334), upon cleavage, becomes an unblocked linear primer molecule (344), which has a region (336) complementary to the PBD (332) on the circular template molecule (352) and can hybridize with the circular template molecule (352).

(125) Once the unblocked linear primer molecule (344) and the circular template molecule (352) are hybridized (i.e., hybridized at the PBD (332) of the circular template molecule (352) and the PBD complement (336) on the unblocked linear primer molecule (344)), 3'.fwdarw.5' exonuclease activity of the polymerase (338) removes the unhybridized single-stranded DNA at the 3' end of the unblocked primer molecule (344). The polymerase (338) can now use the circular template molecule (352) (non-target strand) to produce concatenated activating nucleic acid molecules (360) (which are concatenated target strands), which will be cleaved by the trans-cleavage activity of activated RNP1. The cleaved regions of the concatenated synthesized activating molecules (360) (target strand) are capable of activating the RNP2 (302.fwdarw.308) complex.

(126) As described above, because the nucleic acid-guided nuclease in RNP2 (308) comprises both cis- and trans-cleavage activity, more blocked primer molecules (334) become unblocked primer molecules (344) triggering activation of more RNP2s (308) and more trans-cleavage activity in a cascade. FIG. 3B at bottom depicts the concurrent activation of reporter moieties. Intact reporter moieties (309) comprise a quencher (310) and a fluorophore (311). As described above in relation to FIG. 1B, the reporter moieties are also subject to trans-cleavage by activated RNP1 (305) and

RNP2 (308). The intact reporter moieties (309) become activated reporter moieties (312) when the quencher (310) is separated from the fluorophore (311), and the fluorescent signal (313) is unquenched and can be detected. Signal strength increases rapidly as more blocked primer molecules (334) become unblocked primer molecules (344) generating synthesized activating nucleic acid molecules and triggering activation of more RNP2s (308) and more trans-cleavage activity of the reporter moieties (309). Again, here the reporter moieties are shown as separate molecules from the blocked nucleic acid molecules, but other configurations may be employed and are discussed in relation to FIG. 4. Also note that as with the other embodiments of the cascade assay, in this embodiment, with the exception of the gRNA in RNP1, the cascade assay components stay the same no matter what target nucleic acid(s) of interest are being detected.

(127) The polymerases used in the “blocked primer molecule” embodiments serve to polymerize a reverse complement strand of the template molecule (non-target strand) to generate a synthesized activating molecule (target strand) as described above. In some embodiments, the polymerase is a DNA polymerase, such as a BST, T4, or Terminator polymerase (New England BioLabs Inc., Ipswich MA., USA). In some embodiments, the polymerase is a Klenow fragment of a DNA polymerase. In some embodiments the polymerase is a DNA polymerase with 5'.fwdarw.3' DNA polymerase activity and 3'.fwdarw.5' exonuclease activity, such as a Type I, Type II, or Type III DNA polymerase. In some embodiments, the DNA polymerase, including the Phi29, T7, Q5®, Q5U®, Phusion®, OneTaq®, LongAmp®, Vent®, or Deep Vent® DNA polymerases (New England BioLabs Inc., Ipswich MA., USA), or any active portion or variant thereof. Also, a 3' to 5' exonuclease can be separately used if the polymerase lacks this activity.

(128) FIG. 4 depicts three mechanisms in which a cascade assay reaction can release a signal from a reporter moiety. FIG. 4 at top shows the mechanism discussed in relation to FIGS. 2A, 3A and 3B. In this embodiment, a reporter moiety 409 is a separate molecule from the blocked nucleic acid molecules present in the reaction mix. Reporter moiety (409) comprises a quencher (410) and a fluorophore (411). An activated reporter moiety (412) emits a signal from the fluorophore (411) once it has been physically separated from the quencher (410).

(129) FIG. 4 at center shows a blocked nucleic acid molecule (403), which is also a reporter moiety. In addition to quencher (410) and fluorophore (411), a blocking moiety (407) can be seen (see also blocked nucleic acid molecules 203 in FIG. 2A). Blocked nucleic acid molecule/reporter moiety (403) comprises a quencher (410) and a fluorophore (411). In this embodiment of the cascade assay, when the blocked nucleic acid molecule (403) is unblocked due to trans-cleavage initiated by the target nucleic acid of interest binding to RNP1, the unblocked nucleic acid molecule (406) also becomes an activated reporter moiety with fluorophore (411) separated from quencher (410). Note both the blocking moiety (407) and the quencher (410) are removed. In this embodiment, reporter signal is directly generated as the blocked nucleic acid molecules become unblocked.

(130) FIG. 4 at the bottom shows that cis-cleavage of an unblocked nucleic acid or a synthesized activation molecule at a PAM distal sequence by RNP2 generates a signal. Shown are activated RNP2 (408), unblocked nucleic acid molecule (461), quencher (410), and fluorophore (411) forming an activated RNP2 with the unblocked nucleic acid/reporter moiety intact (460). Cis-cleavage of the unblocked nucleic acid/reporter moiety (461) results in an activated RNP2 with the reporter moiety activated (462), comprising the activated RNP2 (408), the unblocked nucleic acid molecule with the reporter moiety activated (463), quencher (410) and fluorophore (411).

(131) Applications of the Cascade Assay

(132) The present disclosure describes cascade assays for detecting a target nucleic acid of interest in a sample. As described above, the various embodiments of the cascade assay are notable in that, with the exception of the gRNA in RNP1, the cascade assay components stay the same no matter what target nucleic acid(s) of interest are being detected.

(133) Target nucleic acids of interest are derived from samples. Suitable samples for testing

include, but are not limited to, any environmental sample, such as air, water, soil, surface, food, clinical sites and products, industrial sites and products, pharmaceuticals, medical devices, nutraceuticals, cosmetics, personal care products, agricultural equipment and sites, and commercial samples, and any biological sample obtained from an organism or a part thereof, such as a plant, animal, or bacteria. In some embodiments, the biological sample is obtained from an animal subject, such as a human subject. A biological sample is any solid or fluid sample obtained from, excreted by or secreted by any living organism, including, without limitation, single celled organisms, such as bacteria, yeast, protozoans, and amoebas among others, multicellular organisms including plants or animals, including samples from a healthy or apparently healthy human subject or a human patient affected by a condition or disease to be diagnosed or investigated, such as an infection with a pathogenic microorganism, such as a pathogenic bacteria or virus. For example, a biological sample can be a biological fluid obtained from, for example, blood, plasma, serum, urine, stool, sputum, mucous, lymph fluid, synovial fluid, bile, ascites, pleural effusion, seroma, saliva, cerebrospinal fluid, aqueous or vitreous humor, or any bodily secretion, a transudate, an exudate (for example, fluid obtained from an abscess or any other site of infection or inflammation), or fluid obtained from a joint (for example, a normal joint or a joint affected by disease, such as rheumatoid arthritis, osteoarthritis, gout or septic arthritis), or a swab of skin or mucosal membrane surface (e.g., a nasal or buccal swab).

(134) In some embodiments, the sample can be a viral or bacterial sample or a biological sample that has been minimally processed, e.g., only treated with a brief lysis step prior to detection. In some embodiments, minimal processing can include thermal lysis at an elevated temperature to release nucleic acids. Suitable methods are contemplated in U.S. Pat. No. 9,493,736, among other references. Common methods for cell lysis involve thermal, chemical, enzymatic, or mechanical treatment of the sample or a combination of those. In some embodiments, minimal processing can include treating the sample with chaotropic salts such as guanidine isothiocyanate or guanidine HCl. Suitable methods are contemplated in U.S. Pat. Nos. 8,809,519, 7,893,251, among other references. In some embodiments, minimal processing may include contacting the sample with reducing agents such as DTT or TCEP and EDTA to inactivate inhibitors and/or other nucleases present in the crude samples. In other embodiments, minimal processing for biofluids may include centrifuging the samples to obtain cell-debris free supernatant before applying the reagents. Suitable methods are contemplated in U.S. Pat. No. 8,809,519, among other references. In still other embodiments, minimal processing may include performing DNA/RNA extraction to get purified nucleic acids before applying CRISPR Cascade reagents.

(135) FIG. 5A shows a lateral flow assay (LFA) device that can be used to detect the cleavage and separation of a signal from a reporter moiety. For example, the reporter moiety may be a single-stranded or double-stranded oligonucleotide with terminal biotin and fluorescein amidite (FAM) modifications; and, as described above, the reporter moiety may also be part of a blocked nucleic acid. The LFA device may include a pad with binding particles, such as gold nanoparticles functionalized with anti-FAM antibodies; a control line with a first binding moiety attached, such as avidin or streptavidin; a test line with a second binding moiety attached, such as antibodies; and an absorption pad. After completion of a cascade assay (see FIGS. 2A, 3A, and 3B), the assay reaction mix is added to the pad containing the binding particles, (e.g., antibody labeled gold nanoparticles). When the target nucleic acid of interest is present, a reporter moiety is cleaved, and when the target nucleic acid of interest is absent, the reporter is not cleaved.

(136) A moiety on the reporter binds to the binding particles and is transported to the control line. When the target nucleic acid of interest is absent, the reporter moiety is not cleaved, and the first binding moiety binds to the reporter moiety, with the binding particles attached. When the target nucleic acid of interest is present, one portion of the cleaved reporter moiety binds to the first binding moiety, and another portion of the cleaved reporter moiety bound to the binding particles via the moiety binds to the second binding moiety. In one example, anti-FAM gold nanoparticles

bind to a FAM terminus of a reporter moiety and flow sequentially towards the control line and then to the test line. For reporters that are not trans-cleaved, gold nanoparticles attach to the control line via biotin-streptavidin and result in a dark control line. In a negative test, since the reporter has not been cleaved, all gold conjugates are trapped on control line due to attachment via biotin-streptavidin. A negative test will result in a dark control line with a blank test line. In a positive test, reporter moieties have been trans-cleaved by the cascade assay, thereby separating the biotin terminus from the FAM terminus. For cleaved reporter moieties, nanoparticles are captured at the test line due to anti-FAM antibodies. This positive test results in a dark test line in addition to a dark control line.

(137) In some embodiments, the LFA device is designed for syndromic testing. For example, multiple strips with pooled RNP1s targeting different target nucleic acids of interest may be employed, either as separate devices or in a combined device. As a non-limiting example, a syndromic testing device could include four lateral flow strips, with each strip indicating the presence of at least one out of several generally related (e.g., by genetics or by treatment) pathogens (FIG. 5B). One example of a use for syndromic testing is in respiratory illness. For example, the first lateral flow strip could indicate the presence of at least one of the several strains of influenza that cause the common flu (e.g., influenza A, influenza A/H1, influenza A/H3, influenza A/H1-2009, and influenza B); the second lateral flow strip could indicate the presence of at least one of the multiple strains of respiratory syncytial virus (RSV), such as RSV-A and RSV-B; the third lateral flow strip could indicate the presence of at least one variant of SARS-CoV-2 (e.g., B.1.1.7, B.1.351, P.1, B.1.617.2, BA.1, BA.2, BA.2.12.1, BA.4, and BA.5); and the fourth lateral flow strip could indicate the presence of at least one of other pathogens of interest (e.g., parainfluenza virus 1-4, human metapneumovirus, human rhinovirus, human enterovirus, adenovirus, coronavirus HKU1, coronavirus NL63, coronavirus 229E, coronavirus OC43, MERS, and many more). The results shown in FIG. 5B indicate a positive test for the presence of RSV A and/or RSV B nucleic acid molecules. Also as seen in FIG. 5B, the syndromic testing device could further include a lateral flow strip for a negative control and a lateral flow strip for a positive control.

(138) The components of the cascade assay may be provided in various kits. In one aspect, the kit for detecting a target nucleic acid of interest in a sample includes: first ribonucleoprotein complexes (RNP1s), second ribonucleoprotein complexes (RNP2s), blocked nucleic acid molecules, and reporter moieties. The first complex (RNP1) comprises a first nucleic acid-guided nuclease and a first gRNA, where the first gRNA includes a sequence complementary to the target nucleic acid(s) of interest. Binding of the first complex (RNP1) to the target nucleic acid(s) of interest activates trans-cleavage activity of the first nucleic acid-guided nuclease. The second complex (RNP2) comprises a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest. The blocked nucleic acid molecule comprises a sequence complementary to the second gRNA, where trans-cleavage of the blocked nucleic acid molecule results in an unblocked nucleic acid molecule and the unblocked nucleic acid molecule can bind to the second complex (RNP2), thereby activating the trans-cleavage activity of the second nucleic acid-guided nuclease. Activating trans-cleavage activity in RNP2 results in an exponential increase in unblocked nucleic acid molecules and in active reporter moieties, where reporter moieties are nucleic acid molecules and/or are operably linked to the blocked nucleic acid molecules and produce a detectable signal upon cleavage by RNP2.

(139) In a second aspect, the kit for detecting a target nucleic acid molecule in sample includes: first ribonucleoprotein complexes (RNP1s), second ribonucleoprotein complexes (RNP2s), template molecules, blocked primer molecules, a polymerase, NTPs, and reporter moieties. The first ribonucleoprotein complex (RNP1) comprises a first nucleic acid-guided nuclease and a first gRNA, where the first gRNA includes a sequence complementary to the target nucleic acid of interest and where binding of RNP1 to the target nucleic acid(s) of interest activates trans-cleavage

activity of the first nucleic acid-guided nuclease. The second complex (RNP2) comprises a second nucleic acid-guided nuclease and a second gRNA that is not complementary to the target nucleic acid of interest. The template molecules comprise a primer binding domain (PBD) sequence as well as a sequence corresponding to a spacer sequence of the second gRNA. The blocked primer molecules comprise a sequence that is complementary to the PBD on the template nucleic acid molecule and a blocking moiety.

(140) Upon binding to the target nucleic acid of interest, RNP1 becomes active triggering trans-cleavage activity that cuts at least one of the blocked primer molecules to produce at least one unblocked primer molecule. The unblocked primer molecule hybridizes to the PBD of one of the template nucleic acid molecules, is trimmed of excess nucleotides by the 3'-to-5' exonuclease activity of the polymerase and is then extended by the polymerase and NTPs to form a synthesized activating molecule with a sequence that is complementary to the second gRNA of RNP2. Upon activating RNP2, additional trans-cleavage activity is initiated, cleaving at least one additional blocked primer molecule. Continued cleavage of blocked primer molecules and subsequent activation of more RNP2s proceeds at an exponential rate. A signal may be generated upon cleavage of a reporter molecule by active RNP2 complexes; therefore, a change in signal production indicates the presence of the target nucleic acid molecule.

(141) Any of the kits described herein may further include a sample collection device, e.g., a syringe, lancet, nasal swab, or buccal swab for collecting a biological sample from a subject, and/or a sample preparation reagent, e.g., a lysis reagent. Each component of the kit may be in separate container or two or more components may be in the same container. The kit may further include a lateral flow device used for contacting the biological sample with the reaction mixture, where a signal is generated to indicate the presence or absence of the target nucleic acid molecule of interest. In addition, the kit may further include instructions for use and other information.

EXAMPLES

(142) The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the present invention and are not intended to limit the scope of what the inventors regard as their invention, nor are they intended to represent or imply that the experiments below are all of or the only experiments performed. It will be appreciated by persons skilled in the art that numerous variations and/or modifications may be made to the invention as shown in the specific aspects without departing from the spirit or scope of the invention as broadly described. The present aspects are, therefore, to be considered in all respects as illustrative and not restrictive.

Example 1: Preparation of Nucleic Acids of Interest

(143) Mechanical lysis: Nucleic acids of interest may be isolated by various methods depending on the cell type and source (e.g., tissue, blood, saliva, environmental sample, etc.). Mechanical lysis is a widely-used cell lysis method and may be used to extract nucleic acids from bacterial, yeast, plant and mammalian cells. Cells are disrupted by agitating a cell suspension with “beads” at high speeds (beads for disrupting various types of cells can be sourced from, e.g., OPS Diagnostics (Lebanon NJ, US) and MP Biomedicals (Irvine, CA, USA)). Mechanical lysis via beads begins with harvesting cells in a tissue or liquid, where the cells are first centrifuged and pelleted. The supernatant is removed and replaced with a buffer containing detergents as well as lysozyme and protease. The cell suspension is mixed to promote breakdown of the proteins in the cells and the cell suspension then is combined with small beads (e.g., glass, steel, or ceramic beads) that are mixed (e.g., vortexed) with the cell suspension at high speeds. The beads collide with the cells, breaking open the cell membrane with shear forces. After “bead beating”, the cell suspension is centrifuged to pellet the cellular debris and beads, and the supernatant may be purified via a nucleic acid binding column (such as the MagMAX™ Viral/Pathogen Nucleic Acid Isolation Kit from ThermoFisher (Waltham, MA, USA) and others from Qiagen (Hilden Germany), TakaraBio (San Jose, CA, USA), and Biocomma (Shenzhen, China)) to collect the nucleic acids (see the discussion

of solid phase extraction below).

(144) Solid phase extraction (SPE): Another method for capturing nucleic acids is through solid phase extraction. SPE involves a liquid and stationary phase, which selectively separate the target analyte (here, nucleic acids) from the liquid in which the cells are suspended based on specific hydrophobic, polar, and/or ionic properties of the target analyte in the liquid and the stationary solid matrix. Silica binding columns and their derivatives are the most commonly used SPE techniques, having a high binding affinity for DNA under alkaline conditions and increased salt concentration; thus, a highly alkaline and concentrated salt buffer is used. The nucleic acid sample is centrifuged through a column with a highly porous and high surface area silica matrix, where binding occurs via the affinity between negatively charged nucleic acids and positively charged silica material. The nucleic acids bind to the silica matrices, while the other cell components and chemicals pass through the matrix without binding. One or more wash steps typically are performed after the initial sample binding (i.e., the nucleic acids to the matrix), to further purify the bound nucleic acids, removing excess chemicals and cellular components non-specifically bound to the silica matrix. Alternative versions of SPE include reverse SPE and ion exchange SPE, and use of glass particles, cellulose matrices, and magnetic beads.

(145) Thermal lysis: Thermal lysis involves heating a sample of mammalian cells, virions, or bacterial cells at high temperatures thereby damaging the cellular membranes by denaturizing the membrane proteins. Denaturizing the membrane proteins results in the release of intracellular DNA. Cells are generally heated above 90° C., however time and temperature may vary depending on sample volume and sample type. Once lysed, typically one or more downstream methods, such as use of nucleic acid binding columns for solid phase extraction as described above, are required to further purify the nucleic acids.

(146) Physical lysis: Common physical lysis methods include sonication and osmotic shock. Sonication involves creating and rupturing of cavities or bubbles to release shockwaves, thereby disintegrating the cellular membranes of the cells. In the sonication process, cells are added into lysis buffer, often containing phenylmethylsulfonyl fluoride, to inhibit proteases. The cell samples are then placed in a water bath and a sonication wand is placed directly into the sample solution. Sonication typically occurs between 20-50 kHz, causing cavities to be formed throughout the solution as a result of the ultrasonic vibrations; subsequent reduction of pressure then causes the collapse of the cavity or bubble resulting in a large amount of mechanical energy being released in the form of a shockwave that propagates through the solution and disintegrates the cellular membrane. The duration of the sonication pulses and number of pulses performed varies depending on cell type and the downstream application. After sonication, the cell suspension typically is centrifuged to pellet the cellular debris and the supernatant containing the nucleic acids may be further purified by solid phase extraction as described above.

(147) Another form of physical lysis is osmotic shock, which is most typically used with mammalian cells. Osmotic shock involves placing cells in DI/distilled water with no salt added. Because the salt concentration is lower in the solution than in the cells, water is forced into the cell causing the cell to burst, thereby rupturing the cellular membrane. The sample is typically purified and extracted by techniques such as e.g., solid phase extraction or other techniques known to those of skill in the art.

(148) Chemical lysis: Chemical lysis involves rupturing cellular and nuclear membranes by disrupting the hydrophobic-hydrophilic interactions in the membrane bilayers via detergents. Salts and buffers (such as, e.g., Tris-HCl pH8) are used to stabilize pH during extraction, and chelating agents (such as ethylenediaminetetraacetic acid (EDTA)) and inhibitors (e.g., Proteinase K) are also added to preserve the integrity of the nucleic acids and protect against degradation. Often, chemical lysis is used with enzymatic disruption methods (see below) for lysing bacterial cell walls. In addition, detergents are used to lyse and break down cellular membranes by solubilizing the lipids and membrane proteins on the surface of cells. The contents of the cells include, in addition to the

desired nucleic acids, inner cellular proteins and cellular debris. Enzymes and other inhibitors are added after lysis to inactivate nucleases that may degrade the nucleic acids. Proteinase K is commonly added after lysis, destroying DNase and RNase enzymes capable of degrading the nucleic acids. After treatment with enzymes, the sample is centrifuged, pelleting cellular debris, while the nucleic acids remain in the solution. The nucleic acids may be further purified as described above.

(149) Another form of chemical lysis is the widely-used procedure of phenol-chloroform extraction. Phenol-chloroform extraction involves the ability for nucleic acids to remain soluble in an aqueous solution in an acidic environment, while the proteins and cellular debris can be pelleted down via centrifugation. Phenol and chloroform ensure a clear separation of the aqueous and organic (debris) phases. For DNA, a pH of 7-8 is used, and for RNA, a more acidic pH of 4.5 is used.

(150) Enzymatic lysis: Enzymatic disruption methods are commonly combined with other lysis methods such as those described above to disrupt cellular walls (bacteria and plants) and membranes. Enzymes such as lysozyme, lysostaphin, zymolase, and protease are often used in combination with other techniques such as physical and chemical lysis. For example, one can use cellulase to disrupt plant cell walls, lysosomes to disrupt bacterial cell walls and zymolase to disrupt yeast cell walls.

Example II: RNP Formation

(151) For RNP complex formation, 250 nM of LbCas12a nuclease protein was incubated with 375 nM of a target specific gRNA in 1× Buffer (10 mM Tris-HCl, 100 µg/mL BSA) with 2-15 mM MgCl.sub.2 at 25° C. for 20 minutes. The total reaction volume was 2 µL. Other ratios of LbCas12a nuclease to gRNAs were tested, including 1:1, 1:2 and 1:5. The incubation temperature can range from 20° C.-37° C., and the incubation time can range from 10 minutes to 4 hours.

Example III: Blocked Nucleic Acid Molecule Formation

(152) Ramp cooling: For formation of the secondary structure of blocked nucleic acids, 2.5 µM of a blocked nucleic acid molecule (any of Formulas I-IV) was mixed in a T50 buffer (20 mM Tris HCl, 50 mM NaCl) with 10 mM MgCl.sub.2 for a total volume of 50 µL. The reaction was heated to 95° C. at 1.6° C./second and incubated at 95° C. for 5 minutes to dehybridize any secondary structures. Thereafter, the reaction was cooled to 37° C. at 0.015° C./second to form the desired secondary structure.

(153) Snap cooling: For formation of the secondary structure of blocked nucleic acids, 2.5 µM of a blocked nucleic acid molecule (any of Formulas I-IV) was mixed in a T50 buffer (20 mM Tris HCl, 50 mM NaCl) with 10 mM MgCl.sub.2 for a total volume of 50 µL. The reaction was heated to 95° C. at 1.6° C./second and incubated at 95° C. for 5 minutes to dehybridize any secondary structures. Thereafter, the reaction was cooled to room temperature by removing the heat source to form the desired secondary structure.

(154) Snap cooling on ice: For formation of the secondary structure of blocked nucleic acids, 2.5 µM of a blocked nucleic acid molecule (any of Formulas I-IV) was mixed in a T50 buffer (20 mM Tris HCl, 50 mM NaCl) with 10 mM MgCl.sub.2 for a total volume of 50 µL. The reaction was heated to 95° C. at 1.6° C./second and incubated at 95° C. for 5 minutes to dehybridize any secondary structures. Thereafter, the reaction was cooled to room temperature by placing the reaction tube on ice to form the desired secondary structure.

Example IV: Reporter Moiety Formation

(155) The reporter moieties used in the reactions herein were single-stranded DNA oligonucleotides 5-10 bases in length (e.g., with sequences of TTATT, TTTATTT, ATTAT, ATTTATTTA, AAAAA, or AAAAAAAAAA) with a fluorophore and a quencher attached on the 5' and 3' ends, respectively. In one example using a Cas12a cascade, the fluorophore was FAM-6, and the quencher was IOWA BLACK® (Integrated DNA Technologies, Coralville, IA). In another example using a Cas13 cascade, the reporter moieties were single stranded RNA oligonucleotides

5-10 bases in length (e.g., r(U)n, r(UUAUU)n, r(A)n).

Example V: Cascade Assay

(156) 9+1 Format (final reaction mix components added at the same time): RNP1 was assembled using the LbCas12a nuclease and a gRNA for the Methicillin resistant *Staphylococcus aureus* (MRSA) DNA according to the RNP complex formation protocol described in Example II (for this sequence, see Example VIII). Briefly, 250 nM LbCas12a nuclease was assembled with 375 nM of the MRSA-target specific gRNA. Next, RNP2 was formed using the LbCas12a nuclease and a gRNA specific for a selected blocked nucleic acid molecule (Formula I-IV) using 500 nM LbCas12a nuclease assembled with 750 nM of the blocked nucleic acid-specific gRNA incubated in 1×NEB 2.1 Buffer (New England Biolabs, Ipswich, MA) with 5 mM MgCl.sub.2 at 25° C. for 20-40 minutes. Following incubation, RNP1s were diluted to a concentration of 75 nM LbCas12a: 112.5 nM gRNA. Thereafter, the final reaction was carried out in 1× Buffer, with 500 nM of the ssDNA reporter moiety, 1×ROX dye (Thermo Fisher Scientific, Waltham, MA) for passive reference, 2.5 mM MgCl.sub.2, 4 mM NaCl, 15 nM LbCas12a: 22.5 nM gRNA RNP1, 20 nM LbCas12a: 35 nM gRNA RNP2, and 50 nM blocked nucleic acid molecule (any one of Formula I-IV) in a total volume of 9 µL. 1 µL of MRSA DNA target (with samples having as low as three copies and as many as 30000 copies—see FIGS. 6-14) was added to make a final volume of 10 µL. The final reaction was incubated in a thermocycler at 25° C. with fluorescence measurements taken every 1 minute.

(157) 2+1+7 Format (RNP1 and MRSA target pre-incubated before addition to final reaction mix): RNP1 was assembled using the LbCas12a nuclease and a gRNA for the MRSA DNA according to RNP formation protocol described in Example II (for this sequence, see Example VIII). Briefly, 250 nM LbCas12a nuclease was assembled with 375 nM of the MRSA-target specific gRNA. Next, RNP2 was formed using the LbCas12a nuclease and a gRNA specific for a selected blocked nucleic acid molecule (Formula I-IV) using 500 nM LbCas12a nuclease assembled with 750 nM of the blocked nucleic acid-specific gRNA incubated in 1×NEB 2.1 Buffer (New England Biolabs, Ipswich, MA) with 5 mM MgCl.sub.2 at 25° C. for 20-40 minutes. Following incubation, RNP1s were diluted to a concentration of 75 nM LbCas12a: 112.5 nM gRNA. After dilution, the formed RNP1 was mixed with 1 µL of MRSA DNA target and incubated at 20° C.-37° C. for up to 10 minutes to activate RNP1. The final reaction was carried out in 1× Buffer, with 500 nM of the ssDNA reporter moiety, 1×ROX dye (Thermo Fisher Scientific, Waltham, MA) for passive reference, 2.5 mM MgCl.sub.2, 4 mM NaCl, the pre-incubated and activated RNP1, 20 nM LbCas12a: 35 nM gRNA RNP2, and 50 nM blocked nucleic acid molecule (any one of Formula I-IV) in a total volume of 9 µL. The final reaction was incubated in a thermocycler at 25° C. with fluorescence measurements taken every 1 minute.

(158) 2+1+6+1 Format (RNP1 and MRSA target pre-incubated before addition to final reaction mix and blocked nucleic acid molecule added to final reaction mix last): RNP1 was assembled using the LbCas12a nuclease and a gRNA for the MRSA DNA according to the RNP complex formation protocol described in Example II (for this sequence, see Example VIII). Briefly, 250 nM LbCas12a nuclease was assembled with 375 nM of the MRSA-target specific gRNA. Next, RNP2 was formed using the LbCas12a nuclease and a gRNA specific for a selected blocked nucleic acid molecule (Formula I-IV) using 500 nM LbCas12a nuclease assembled with 750 nM of the blocked nucleic acid-specific gRNA incubated in 1×NEB 2.1 Buffer (New England Biolabs, Ipswich, MA) with 5 mM MgCl.sub.2 at 25° C. for 20-40 minutes. Following incubation, RNP1s were diluted to a concentration of 75 nM LbCas12a: 112.5 nM gRNA. After dilution, the formed RNP1 was mixed with 1 µL of MRSA DNA target and incubated at 20° C.-37° C. for up to 10 minutes to activate RNP1. The final reaction was carried out in 1× Buffer, with 500 nM of the ssDNA reporter moiety, 1×ROX dye (Thermo Fisher Scientific, Waltham, MA) for passive reference, 2.5 mM MgCl.sub.2, 4 mM NaCl, the pre-incubated and activated RNP1, and 20 nM LbCas12a: 35 nM gRNA RNP2 in a total volume of 9 µL. Once the reaction mix was made, 1 µL (50 nM) blocked nucleic acid

molecule (any one of Formula I-IV) was added for a total volume of 10 μ L. The final reaction was incubated in a thermocycler at 25° C. with fluorescence measurements taken every 1 minute.

Example VI: Detection of SARS-CoV-2 with the Cascade Assay in Under 10 Minutes

(159) To detect the presence of SARS-CoV-2 in a sample and determine the sensitivity of detection with the cascade assay, titration experiments were performed using a SARS-CoV-2 gamma-inactivated virus and a synthesized positive control. To serve as the positive control for the detection system, a plasmid containing a 316 bp SARS-CoV-2 nucleocapsid gene (N-gene) was synthesized by IDT (Integrated DNA Technologies, Coralville, IA). The N-gene sequence was as follows.

(160) TABLE-US-00002 SARS-CoV-2 N-gene Target Sequence (Positive Control; SEQ ID NO: 3):

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CTCAAGGAACAACATTGCCAAAAGGCTTCTACGCAGAAGGGAGCAGAGGC
GGCAGTCAAGCCTCTTCTCGTTCCTCATCACGTAGTCGCAACAGTTCAAG
AAATTCAACTCCAGGCAGCAGTAGGGGAAGTTCTCCTGCTAGAATGGCTG
GCAATGGCGGTGATGCTGCTCTTGCTTTGCTGCTGCTTGACAGATTGAAC
CAGCTTGAGAGCAAAATGTCTGGTAAAGGCCAACAACAAGGCCAAAC
TGTCATAAGAAATCTGCTGCTGAGGCTTCTAAGAAGCCTCGGCAAAAAC
GTACTGCCACTAAAGC
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(161) For the detection of SARS-CoV-2, a gamma-inactivated virus was incubated in a buffer at 95° C. for 1 minute in order to lyse and release viral RNA, followed by reverse transcription to convert the viral RNA to cDNA. The reverse transcription primer is designed to reverse transcribe the SARS-CoV-2 N-gene. The reverse transcription primer is as follows.

(162) TABLE-US-00003 Reverse Transcription Primer (SEQ ID NO: 4):
GTTTGGCCTTGTTGTTGTT

RNP1 was preassembled with a guide RNA (gRNA) sequence designed to target the N-gene of SARS-CoV-2. The guide sequence is as follows.

(163) TABLE-US-00004 Guide Sequence (SEQ ID NO: 5):
UAAUUUCUACUAAGUGUAGAUUUGAACUGUUGCGACUACGUGAU

(164) RNP2 was preassembled with a gRNA sequence designed to target an unblocked nucleic acid molecule that results from unblocking (i.e., linearizing) a circularized blocked nucleic acid molecule. A circularized blocked nucleic acid molecule was designed and synthesized. The blocked nucleic acid molecule was as follows.

(165) TABLE-US-00005 Blocked nucleic acid molecule (SEQ ID NO: 6):
GTT*AT*TA*AA*TG*AC*TT*CT*CATT

where the * indicate bonds that are phosphorothioate modified. The 5' and 3' ends were covalently linked to form a circularized molecule. The SARS-CoV-2 gamma-inactivated virus or positive control with 1700, 170, 17, or 5 total copies of N-gene DNA, or a negative control (0 copies of N-gene), were added to a reaction mixture to begin the cascade assay. The reaction mix contained the preassembled RNP1, preassembled RNP2, a blocked nucleic acid molecule in a buffer (~ pH 8) containing 4 mM MgCl₂ and 101 mM NaCl. The buffering conditions were optimized to reduce non-specific nickase activity by the RNP complexes.

(166) The cascade assay reaction proceeded for 20 minutes at 37° C. and fluorescence from the reporter molecule was measured. In all the SARS-CoV-2 gamma-inactivated virus and positive control titrations, a significant change in fluorescence was observed after 10 and 5 minutes, relative to the negative control (see the results in FIGS. 6 and 7). For the results shown in FIG. 6, the presence of the N-gene was detected in 10 minutes or less at 37° C. The data represent 3 independent biological replicates. Data is presented as mean $\pm$ s.d. ****= $p < 0.0001$ (student t-test). For the results shown in FIG. 7, the presence of SARS-CoV-2 was detected in 10 minutes or 5 minutes at 37° C. The data represent 3 independent biological replicates. Data is presented as mean $\pm$ s.d. ****= $p < 0.0001$ (student t-test). The results indicate that the cascade assay can detect as

few as 5 SARS-CoV-2 target molecules in 10 minutes or less at room temperature.

Example VII: Detection of MRSA in 5 Minutes with Cascade Assay at 37° C.

(167) To detect the presence of Methicillin resistant *Staphylococcus aureus* (MRSA) and determine the sensitivity of detection with the cascade assay, titration experiments with a MRSA DNA target nucleic acid of interest were performed. The MRSA DNA sequence (NCBI Reference Sequence NC: 007793.1) is as follows.

(168) TABLE-US-00006 SEQ ID NO: 7:

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ATGAAAAAGATAAAAATTGTTCCACTTATTTTAATAGTTGTAGTTGTCGG
GTTTGGTATATATTTTTATGCTTCAAAAGATAAAGAAATTAATAATACTA
TTGATGCAATTGAAGATAAAAATTTCAAACAAGTTTATAAAGATAGCAGT
TATATTTCTAAAAGCGATAATGGTGAAGTAGAAATGACTGAACGTCCGAT
AAAAATATATAATAGTTTtaggcgttaaagatataaacattcaggatcgtA
AAATAAAAAAAGTATCTAAAAATAAAAAACGAGTAGATGCTCAATATAAA
ATTAAACAAACTACGGTAACATTGATCGCAACGTTCAATTTAATTTTGT
TAAAGAAGATGGTATGTGGAAGTTAGATTGGGATCATAGCGTCATTATTC
CAGGAATGCAGAAAGACCAAAGCATAACATTTGAAAATTTAAAATCAGAA
CGTGGTAAAATTTTAGACCGAAACAATGTGGAATTGGCCAATACAGGAAC
AGCATATGAGATAGGCATCGTTCCAAAGAATGTATCTAAAAAAGATTATA
AAGCAATCGCTAAAGAACTAAGTATTTCTGAAGACTATATCAAACAACAA
ATGGATCAAATTTGGGTACAAGATGATACCTTCGTTCCACTTAAAACCGT
TAAAAAATGGATGAATATTTAAGTGATTTCGCAAAAAAATTCATCTTA
CACTAATGAAACAGAAAGTCGTAACCTATCCTCTAGGAAAAGCGACTTCA
CATCTATTAGGTTATGTTGGTCCCATTAACCTCTGAAGAATTAACAACAAA
AGAATATAAAGGCTATAAAGATGATGCAGTTATTGGTAAAAAGGGACTCG
AAAAACTTTACGATAAAAAGCTCCAACATGAAGATGGCTATCGTGTCACA
ATCGTTGACGATAATAGCAATACAATCGCACATACATTAATAGAGAAAAA
GAAAAAAGATGGCAAAGATATTCAACTAACTATTGATGCTAAAGTTCAAA
AGAGTATTTATAACAACATGAAAAATGATTATGGCTCAGGTACTGCTATC
CACCTCAAACAGGTGAATTATTAGCACTTGTAAGCACACCTTCATATGA
CGTCTATCCATTTATGTATGGCATGAGTAACGAAGAATATAATAAATTAA
CCGAAGATAAAAAAGAACCTCTGCTCAACAAGTTCCAGATTACAACCTTCA
CCAGGTTCAACTCAAAAAATATTAACAGCAATGATTGGGTAAATAACAA
AACATTAGACGATAAAACAAGTTATAAAATCGATGGTAAAGGTTGGCAAA
AAGATAAATCTTGGGGTGGTTACAACGTTACAAGATATGAAGTGGTAAAT
GGTAATATCGACTTAAACAAGCAATAGAATCATCAGATAACATTTTCTT
TGCTAGAGTAGCACTCGAATTAGGCAGTAAGAAATTTGAAAAAGGCATGA
AAAAACTAGGTGTTGGTGAAGATATACCAAGTGATTATCCATTTTATAAT
GCTCAAATTTCAAACAAAAATTTAGATAATGAAATATTATTAGCTGATTC
AGGTTACGGACAAGGTGAAATACTGATTAACCCAGTACAGATCCTTTCAA
TCTATAGCGCATTAGAAAATAATGGCAATATTAACGCACCTCACTTATTA
AAAGACACGAAAAACAAAGTTTGGAAGAAAAATATTATTTCCAAAGAAAA
TATCAATCTATTAACCTGATGGTATGCAACAAGTCGTAAATAAAACACATA
AAGAAGATATTTATAGATCTTATGCAAACCTTAATTGGCAAATCCGGTACT
GCAGAACTCAAATGAAACAAGGAGAACTGGCAGACAAATTGGGTGGTT
TATATCATATGATAAAGATAATCCAAACATGATGATGGCTATTAATGTTA
AAGATGTACAAGATAAAGGAATGGCTAGCTACAATGCCAAAATCTCAGGT
AAAGTGTATGATGAGCTATATGAGAACGGTAATAAAAAATACGATATAGA TGAATAA
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(169) Briefly, an RNP1 was preassembled with a gRNA sequence designed to target MRSA DNA. Specifically, RNP1 was designed to target a 20 bp region of the *mecA* gene of MRSA:

TGTATGGCATGAGTAACGAA (SEQ ID NO: 8). An RNP2 was preassembled with a gRNA

sequence designed to target an unblocked nucleic acid molecule that results from unblocking (i.e., linearizing) a circularized blocked nucleic acid molecule. The circularized blocked nucleic acid molecule was designed and synthesized (SEQ ID NO: 6):

GTT*AT*TA*AA*TG*AC*TT*CT*CATT, where the * indicate bonds that are phosphorothioate modified. The 5' and 3' ends were covalently linked to form a circularized molecule. MRSA DNA (SEQ ID NO: 7) with 3000, 300, 30, or 3 total copies, or a negative control (e.g., 0 copies), were added to a reaction mixture to begin the cascade assay. The reaction mix contained the preassembled RNP1, preassembled RNP2, and a circularized blocked nucleic acid molecule, in a buffer (pH of about 8) containing 4 mM MgCl₂ and 101 mM NaCl. The buffering conditions were optimized to reduce non-specific nickase activity by the RNP complexes. The cascade assay proceeded for 10 minutes at 37° C., and fluorescence from the reporter moiety was measured. In all titrations, a significant change in fluorescence was observed after 10 and 5 minutes, relative to the negative control (see the results in FIG. 8). The cascade assay was initiated to identify the presence of MRSA in 10 minutes or 5 minutes at 37° C. Data represent 3 independent biological replicates. Data is presented as mean±s.d. ****= $p < 0.0001$ (student t-test). The results indicate that the cascade assay can detect as few as 3 MRSA target molecules in only 5 minutes when at 37° C.

Example VIII: Detection of MRSA in Under 10 Minutes with a Cascade Assay at 25° C.

(170) To detect the presence of MRSA and determine the sensitivity of detection with the cascade assay, titration experiments with MRSA DNA (SEQ ID NO: 7) were performed.

(171) Briefly, an RNP1 was preassembled with a guide RNA (gRNA) sequence designed to target MRSA DNA. Specifically, RNP1 was designed to target the following 20 bp sequence in the *mecA* gene of MRSA: TGTATGGCATGAGTAACGAA (SEQ ID NO: 8). An RNP2 was preassembled with a gRNA sequence designed to target an unblocked nucleic acid molecule that results from unblocking (i.e., linearizing) a circularized blocked nucleic acid molecule. A circularized blocked nucleic acid molecule was designed and (SEQ synthesized ID NO: 6):

GTT*AT*TA*AA*TG*AC*TT*CT*CATT, where the * indicate bonds that are phosphorothioate modified. The 5' and 3' ends were covalently linked to form a circularized molecule.

(172) MRSA DNA (SEQ ID NO: 7) with 30000, 3000, 300, 30, or 3 total copies, or a negative control (e.g., 0 copies), was added to a reaction mixture to begin the cascade assay. The reaction mix contained the preassembled RNP1, preassembled RNP2, the circularized blocked nucleic acid molecule in a buffer (pH 8) containing 4 mM MgCl₂ and 101 mM NaCl. The buffering conditions were optimized to reduce non-specific nickase activity by the RNP complexes. The cascade reaction proceeded for 20 minutes at 25° C., and fluorescence by the reporter molecule was measured. In all titrations, a significant change in fluorescence was observed after 10 and 5 minutes, relative to the negative control (see the results in FIG. 9), indicating that the cascade assay can detect as few as 3 MRSA target molecules in 10 minutes or less while at room temperature. The data represent 3 independent biological replicates and is presented as mean±s.d. ****= $p < 0.0001$ (student t-test).

Example IX: Optimized Detection of MRSA in 1 Minute with the Cascade Assay at 25° C.

(173) RNP1 was preassembled with a gRNA sequence designed to target MRSA DNA (SEQ ID NO: 7). Specifically, RNP1 was designed to target the following 20 bp sequence in the *mecA* gene of MRSA: TGTATGGCATGAGTAACGAA (SEQ ID NO: 8). RNP2 was preassembled with a gRNA sequence designed to target an unblocked nucleic acid molecule that results from unblocking a blocked nucleic acid molecule. Five different double stranded and linear blocked nucleic acid molecules were designed, synthesized, and tested: molecule C5, molecule C6, molecule C7, molecule C8, and molecule C9. The nucleotide sequences of molecules C5-C9 are as follows.

(174) TABLE-US-00007 C5 (SEQ ID NO: 9):

GTTATTGAGAATTATTGTCATATTATTCTAATATTATTAAGGCTTATTCA

CTGTTATTATTATAATTATTAAGCTTATT C6 (SEQ ID NO: 10):

GTTATTGAGAAGTTATTATCTATTATTATAAAGTTATTGCCACTATT
ATTGTTATAATTATTAAAGCTTATT C7 (SEQ ID NO: 11):
GTTATTGAGAAGTATTATTCATCTAATTATTATAAGGCCTTATTACTGTT
ATTATTAATAAGCTTATT C8 (SEQ ID NO: 12):
GTTATTGAGAAGTCTTATTATCTAATATTATTAGGCCACTGTTATTATTA
TAATAAGCTTATT C9 (SEQ ID NO: 13):
GTTATTGAGAAGTCATTATTATCTAATAAGTTATTGCCACTGTTATTATT
ATAATAAGCTTATT

(175) Three copies of MRSA DNA (SEQ ID NO: 7) or a negative control (e.g., 0 copies) were added to a reaction mix to begin the cascade assay. The reaction mix contained the preassembled RNP1, preassembled RNP2, and one of the five blocked nucleic acid molecules in a buffer (~pH 8) containing 4 mM MgCl₂ and 71 mM NaCl. These buffering conditions were optimized to reduce non-specific nickase activity by the RNP complexes. Each cascade assay proceeded for 10-20 minutes at 25° C., and fluorescence by the reporter molecule was measured for each cascade assay containing C5 (see the results shown in FIG. 10, where the presence of just 3 MRSA targets was detected in 5 minutes or less at 25° C. The data represent 9 independent biological replicates and is presented as mean±s.d. ****=p<0.0001 (student t-test)), molecule C6 (see the results shown in FIG. 11, where the presence of just 3 MRSA targets was detected in 5 minutes or less at 25° C. The data represent 6 independent biological replicates and is presented as mean±s.d. ****=p<0.0001 (student t-test)), molecule C7 (see the results shown in FIG. 12, where the presence of just 3 MRSA targets was detected in 5 minutes or less at 25° C. Data represent 6 independent biological replicates and is presented as mean±s.d. ****=p<0.0001 (student t-test)), molecule C8 (see the results shown in FIG. 13, where the presence of just 3 MRSA targets was detected in 5 minutes or less at 25° C. Data represent 6 independent biological replicates and is presented as mean±s.d. *=p<0.0001 (student t-test)), and molecule C9 (see the results shown in FIG. 14, where the presence of just 3 MRSA targets was detected in 10 minutes or less at 25° C. Data represent 6 independent biological replicates and data is presented as mean±s.d. ****=p<0.0001 (student t-test)). A significant change in fluorescence is observed after 1 minute and after 5 minutes, relative to the negative control, indicating that the cascade assay can be optimized to detect as few as 3 MRSA target molecules in as little as 1 minute while at room temperature.

(176) While certain embodiments have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the present disclosures. Indeed, the novel methods, apparatuses, modules, instruments and systems described herein can be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods, apparatuses, modules, instruments and systems described herein can be made without departing from the spirit of the present disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the present disclosures.

PARTIES TO A JOINT RESEARCH AGREEMENT

(177) The presently claimed invention was made by or on behalf of the below-listed parties to a joint research agreement. The joint research agreement was in effect on or before the date the claimed invention was made, and the claimed invention was part of the joint research agreement and made as a result of activities undertaken within the scope of the joint research agreement. The parties to the joint research agreement are The Board of Trustees of the University of Illinois and LabSimply, Inc. (now VedaBio, Inc.).

Claims

1. A cascade assay method for detecting a target nucleic acid of interest in a sample comprising the steps of: (a) providing a reaction mixture comprising: i. a first ribonucleoprotein (RNP) complex (RNP1) comprising a first Cas12a nucleic acid-guided nuclease and a first guide RNA (gRNA); wherein the first gRNA comprises a sequence complementary to a target nucleic acid of interest, and wherein the first nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; ii. a second ribonucleoprotein complex (RNP2) comprising a second Cas12a nucleic acid-guided nuclease and a second gRNA; wherein the second gRNA is complementary to synthesized activating molecules and the second nucleic acid-guided nuclease exhibits both cis-cleavage activity and trans-cleavage activity; iii. a plurality of template molecules comprising a sequence having homology to the second gRNA and a primer binding domain; iv. a plurality of blocked primer molecules comprising a sequence complementary to the primer binding domain of the plurality of template molecules and wherein the blocked primer molecules cannot be extended by a polymerase; v. a Phi29 or T7 DNA polymerase and dNTPs; wherein when blocked, the blocked primer molecules cannot bind to the template molecules and be extended by the polymerase and wherein when unblocked, the blocked primer molecules can bind to the primer binding domain of the template molecules and be extended by the polymerase to produce synthesized activating molecules; (b) contacting the reaction mixture with the sample under conditions that allow target nucleic acids of interest in the sample to bind to the first gRNA; wherein: upon binding of the target nucleic acid of interest, the RNP1 becomes active cleaving at least one of the blocked primer molecules, thereby producing at least one unblocked primer molecule that can be extended by the polymerase; at least one unblocked primer molecule binds to one of the template molecules and is extended by the polymerase and nucleotides to form at least one synthesized activating molecule having a sequence complementary to the second gRNA; at least one synthesized activating molecule binds to the second gRNA, and RNP2 becomes active cleaving at least one further blocked primer molecule; and (c) detecting the cleavage products of step (a), thereby detecting the target nucleic acid of interest in the sample.
2. The method of claim 1, wherein the polymerase exhibits 3'.5' exonuclease activity.
3. The method of claim 1, wherein the blocked primer molecule comprises a modified nucleoside or nucleotide.
4. The method of claim 3, wherein the modified nucleoside or nucleotide comprises a locked nucleic acid (LNA), peptide nucleic acid (PNA), 2'-O-methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bond.
5. The method of claim 1, wherein the template molecules do not comprise a PAM sequence complement.
6. The method of claim 1, wherein the template molecules comprise a PAM sequence complement.
7. The method of claim 1, further comprising a reporter moiety: wherein the reporter moiety comprises a DNA, RNA or chimeric nucleic acid molecule and produces a detectable signal upon cleavage by the ribonucleoprotein complex.
8. The method of claim 7, wherein the reporter moiety comprises a modified nucleoside or nucleotide.
9. The method of claim 8, wherein the modified nucleoside or nucleotide comprises a locked nucleic acid (LNA), peptide nucleic acid (PNA), 2'-O-methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, and/or a phosphorothioate (PS) bond.
10. The method of claim 7, wherein the detectable signal is a fluorescent, chemiluminescent, radioactive, colorimetric or other optical signal.
11. The method of claim 10, wherein the detectable signal is the fluorescent signal.
12. The method of claim 10, wherein the detectable signal is the chemiluminescent signal.
13. The method of claim 10, wherein the detectable signal is the colorimetric signal.
14. The method of claim 1, comprising about 1 fM to about 1 mM of the RNP2.

15. The method of claim 1, wherein a K.sub.d of the blocked primer molecules to the template molecules is about 10.sup.6-fold greater or more than the K.sub.d of an unblocked primer molecule resulting from unblocking of the blocked primer molecules.
 16. The method of claim 15, wherein a K.sub.d of the blocked primer molecules to the template molecules is about 10.sup.7-fold greater or more than the K.sub.d of an unblocked primer molecule resulting from unblocking of the blocked primer molecules.
 17. The method of claim 16, wherein a K.sub.d of the blocked primer molecules to the template molecules is about 10.sup.8-fold greater or more than the K.sub.d of an unblocked primer molecule resulting from unblocking of the blocked primer molecules.
 18. The method of claim 17, wherein a K.sub.d of the blocked primer molecules to the template molecules is about 10.sup.10-fold greater or more than the K.sub.d of an unblocked primer molecule resulting from unblocking of the blocked primer molecules.
 19. The method of claim 1, wherein there are at least two different first ribonucleoprotein complexes, wherein said different first ribonucleoprotein complexes comprise a first gRNA complementary to a different target nucleic acid of interest.
 20. The method of claim 19, wherein there are at least ten different first ribonucleoprotein complexes, wherein said different first ribonucleoprotein complexes comprise a first gRNA complementary to a different target nucleic acid of interest.
 21. The method of claim 20, wherein there are at least 100 different first ribonucleoprotein complexes wherein different first ribonucleoprotein complexes comprise first gRNAs complementary to different target nucleic acids of interest.
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